

Final Technical Report

Component 4 - Site Technology Design and Development



For Serious Shea – Shea Butter Eco-Processing in Burkina Faso

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1. Executive summary

The Shea tree, (*Vitellaria paradoxa*) belongs to the family of Sapotaceae. Shea trees grow in the wild. Sheanut (*Butyrospermum paradoxum*) is an oil rich tropical tree crop, which is indigenous to the West African savannah zone. The Sheanut, fruits and seeds obtained from the Shea tree plays an important role in the nutritional and economy of the people; it is the most important source of fatty acids, glycerol in the diet. Shea processing is a major industry in Burkina Faso, and conducted almost exclusively by women's groups; however, it still involves outdated methods: potentially degrading habitats including Shea trees, emitting significant greenhouse gas emissions, causing a number of health and hygiene risks, and not realising its potential to support social and -economic development.

TARA has been assigned to assess and design the optimal process in order to support the value chain, using renewable energies. The goal of the assignment is to outline potential technological options for collecting, storing, processing and transporting of Shea butter on a large-scale across Burkina Faso. TARA provided potential solutions in collaboration with the key stakeholders (including Serious Shea and the women's groups), rank/prioritize preferred solutions and agree preferred technical design. TARA, based on inputs from the local stakeholders prepare detailed designs, costings, and implementation plan for collection sites and processing centres with specification for better storage facility, and energy efficient technologies.

Understanding the production process from collection and processing was stipe learning because of the challenges at both level. To design technological solutions, is fundamental to understand every step of the value chain. Listing correct processes, its timings, gaps and causes, appropriate solutions, resources required at each stage, expertise to operate machines, seasonality and sourcing of the crops, processing done at village level, quality requirement, procedure for transport, human resources for quality monitoring and ensuring the quality these were questions which were discussed at length before providing any technological solutions.

To maintain the Shea butter quality, it was complex to produce the Shea butter in rural areas. Therefore, we have a designed a model with two levels of processing, item one – collection centre - location rural area close to village will process the fruits into high quality dry pulp and high quality shea almonds. Second item – processing centre – it will be located in the bigger city. Five collection centres plus one processing centre will work as a cluster. In each processing centre is designed to process 3000 tons of kernels into 1000 tons of Shea butter.

The production is often conducted by women's groups and helps to improve the living conditions of rural households. In traditional methods, non-renewable biomass is inefficiently consumed in boiling nuts denatures & de-husking, roasting, heating water for kneading, and boiling fat to dry and clean, and accounts for ~25kgCO₂e/kg butter at farm. Both the deforestation and carbon emissions could be reversed and negated respectively through the implementation of the Eco-Processing Centres, as they will minimise the need for fuelwood, and enable the protection and reforestation of forest populations.

The concept of collection centre and processing centre evolved after understanding that to make it more environment and energy conservative, it is important to segregate the pre processing of fruits at collection centre and Shea butter production using environment friendly technologies at processing centre. Each centre is expected to save >32,000tCO₂e/year of GHG emissions and drive widespread reforestation.

The Serious Shea project aims to decarbonise the Shea butter value chain in Burkina Faso through the development of 50 x Collection Sites and 10 x Eco-Processing Centres:

- Each set of 5 collection sites will be fully equipped to process 7,500mt of fruit into 3,750mt of almonds

- Each processing centre will be fully equipped to process 3,750mt of almonds into 1,000 mt of Shea butter per year, with further fractionization and refining potential

Why we need collection sites and processing centres?

- To organize Shea Butter production system
- To ensure better quality production
- To enhance efficiency and effectiveness in entire Shea butter production process
- To build the capacity of women and strengthen their engagement in new structure of the production process
- To increase profit share of women
- To control the production quality
- To ensure sustainability in production system across the units

Key features of collection site

- Designates space to store kernels
- Professionally designed space for pre processing of fruits and kernels
- Separate space to process pulp
- Provides better working environment to the women
- Women groups can use it as ware house for pre processed kernels
- To maintain the quality of kernels

The women groups can use the services of the collection site on the following conditions

- They will sign an official contract with the group and Serious Shea jointly for the supply of fruits and kernels to the Serious Shea
- Serious Shea team will provide orientation to all the women on the procedure of the collection site and how to use them for shared understanding
- They will provide 50 % of their total fruits/kernels to Serious Shea as fees to use the services of the collection site.
- If the women want to sell the 50% of the kernel to Serious Shea for butter making then they Serious Shea will purchase those seeds on the market rates of that day and make payment to the women

Key features of processing centre

- One stop solution approach
- Well equipped with high quality equipment
- Well trained human resources to operate the equipment and produce quality butter
- Eco friendly
- Production of renewable energy by [bio gas](#) using the fruit waste, [syngas](#) using the nut shell and cake and solar energy to reduce carbon foot prints
- Scientifically storage facility for processed butter and other products

Composition of the clusters

- **Collection centre**
Serious is driving an effort to transform the lives of rural women Shea workers in Burkina Faso through the development of 50 x Collection Sites:
 - a. Each set of collection sites will be fully equipped to process Shea fruit into Shea kernels
 - b. Each processing centre will be fully equipped to process Shea kernels into Shea butter, with further fractionization and refining potentially
 - c. The facilities will draw on renewable energy sources (e.g., biogas and solar PV) and include all the equipment necessary for clean, efficient, and environmentally-friendly Shea butter production

▪ **Processing centre**

Serious is driving an effort to transform the lives of rural women Shea workers in Burkina Faso through the development of 50 x Collection Sites and 10 x Eco-Processing Centres:

More specifically, each centre to be equipped with an optimal mix of the following:

- a. Solar cookers and steam boilers to dry and boil the nuts cleanly and efficiently
- b. Solar-powered motors, which enable the women's groups to produce the Shea paste quickly without the strenuous physical labour of traditional manual methods
- c. Large-scale improved cookstoves to boil the mixture and extract the butter
- d. All the necessary apparatuses and equipment to store, transport, package and export the final Shea butter product

The report talks about proposed layouts for collection and processing centres, resource flow at each centre, type of machines required and its specification, storage, spacing required for different processes at both the centres, area required for each centre and how much quantum can be managed at each centre. The high definition layouts give clarity on all these aspects.

To understand the actual production results testing is essential. The report talks about the testing protocols at each stage and how to conduct these tests. A total 14 tests protocols were developed and test were conducted with fresh fruits and kernels. In the fruits season tests were conducted with different size and shape fruits and kernels to understand the actual production of Shea butter and other products. This has helped in defining the processes, timing, seasonality, how much quantum can be managed at each centre, what types of machines required etc.

There are different types of machines which are needed at each stages. A review was conducted to understand the production quality, quantity, resource required and its duration. This was helpful in selecting the machines and technologies. The report provides details of different types of machines, its quality and resource requirements.

To run any production unit, it is essential to calculate its operational cost and capital cost. The report provides detailed calculations required to operate a collection unit and processing unit. The calculation of operation and capital cost was done on the basis of actual requirements at each centre. These calculations were done for a 70 tons per annum capacity unit. The calculations were done on the basis of detailed analysis and requirements to increase efficiency and productivity at each centre.



To enhance the production quality and quantity energy efficient machine are essential. The report also provides details about new machines and its costs along with quotations from the supplier. The processing of the whole fruit can provide significant benefits for food and energy security. The fruit can be processed into valuable foodstuff which can expand the value chain and significantly increase the return to the stakeholders. To process the fruit pulp into other by product is our idea and this will provide additional income to the woman and also the waste from the fruits will give the additional source of energy to be used in production processes. Some Foodstuff which could be obtained from the Shea pulp do not require sophisticate equipment of processes. Fruit waste will go to bio digester and shell will be used as energy source to produce Syn Gas. This will provide additional energy at the collection centre. Processing centre we are going to use mostly the cake to produce syn gas and solar energy will be used.

Significant quantity of biomass can be collected from the fruit process to contribute to the decarbonisation of the value chain. The pulp wastes which can be estimated to 20% in fruit processing can produce large amount of biogas.

Developing layout to increase efficiency and productivity was a challenge because designing a layout considering woman should have major say in entire production process was not easy. The team worked hard to understand the entire production processes and accordingly designed the layout and also selected the technologies. The tests were conducted with the help of local experts and practitioner involving local woman to understand real time required to conduct each step and what are major stumbling blocks that affects the efficiency and productivity were identified and accordingly layouts were designed. We also considered logistics issues, integration of storage, resources, recycling, and space for better movement to ensure the production team have better environment to work at both the centers.

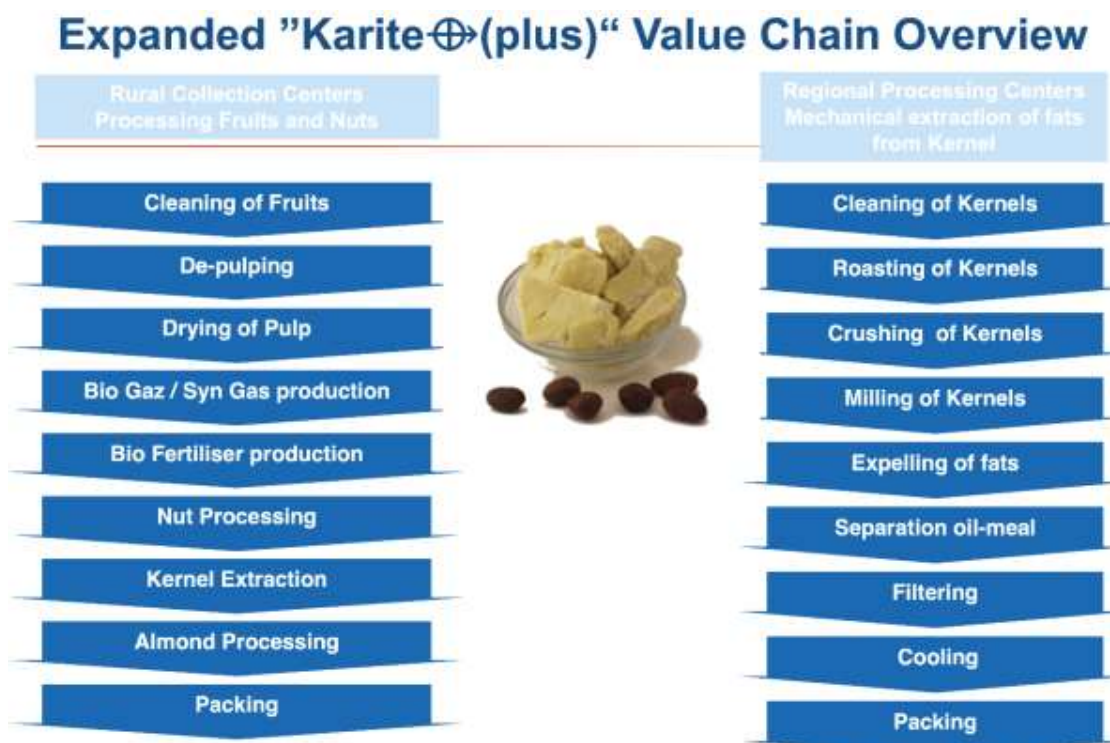
Before finalising the layout and other technological solutions, we had conducted a virtual workshop with all the stakeholders. Before conducting the virtual workshop, detailed report, layout of collection site and processing center, test reports conducted at different stages, and other related documents were shared with them. Virtual workshop was conducted using Skype and WhatsApp and had detailed discussions with the stakeholders. The queries raised by the stakeholders were answered and wherever was possible and agreed were included.

The processing of the whole fruit can provide significant benefits for food and energy security. The fruit can be processed into valuable foodstuff which can expand the value chain and significantly increase the return to the stakeholders. Some Foodstuff which could be obtained from the Shea pulp do not require sophisticate equipment of processes.

Significant quantity of biomass can be collected from the fruit process to contribute to the decarbonisation of the value chain. The pulp wastes which can be estimated to 20% in fruit processing can produce large amount of biogas.

The shells, representing 30% of the dried nut can processed in a Syngas generator to generate steam for the process.

Figure 1: Shea Butter value chain overview



Even though the exploitation of the fruit can contribute to an economic income for women in the Shea sector, the valorisation of the fruit flesh has been very marginal. Through our study we can see that the major reason is the scale at which the collection is made. If overall the collection is

done on a large scale, in the field it is made by small groups and in small quantities. The amount collected by the small group is so small that it is not economically viable to invest in equipment for its transformation. The establishment of collection centres increases the amount to be transform and justifies high investments. Furthermore, since the Shea fruit is seasonal, the equipment installed can, in the off-season, be used for the processing of other local products.

The tests were conducted using fruits collected from the village in the center of Burkina around Ouagadougou. The objective is to evaluate the percentage of macro elements in the Shea fruit to determine the volume available for processing based on the daily collection. The first test was conducted with fruits collected at the end of the Shea season in 2019.

The second tests were conducted with fruit available at the beginning of the season 2020 to assess the variance. A drying test was also conducted to assess the aspect and taste of the resulting foodstuff.

The results where globally consistent with the literature and assumptions, confirming the following respective weight for the elements of interest:

- a) Fruit de-pulping
Pulp: 65%
Nut: 35%
- b) Nut Shelling
Shell: 29 %
Almonds: 71%
- c) Pulp Drying
Dried Pulp: 25%
Water: 75%

The processing of the whole fruit can provide significant benefits for food and energy security. The fruit can be processed into valuable foodstuff which can expand the value chain and significantly increase the return to the stakeholders. Significant quantity of biomass can be collected from the fruit process to contribute to the decarbonisation of the value chain. The pulp wastes which can be estimated to 20% in fruit processing can produce large amount of biogas.

Figure 2: Shea butter making traditional process



- The existing butter-making process in Burkina Faso drives significant deforestation and greenhouse gas emissions, and requires women's groups to conduct heavy physical labour. Furthermore, the lack of access-to-market means that the women's groups do not receive appropriate remuneration for their inputs, and typically fail to improve their socio-economic status despite their hard labour

The shells, representing 30% of the dried nut can processed in a Syngas generator to generate steam for the process.

The Shea wine showed the highest profit margin of over 300% and this is because very few inputs were needed, and its production required less labour. The Shea jam, on the other hand, had the lowest profit margin of 22.8%. The profit margins of the Shea butter-based products ranged from 47.0 to 62.2%.

Unlike the fruit flesh of the Shea fruit, Shea butter has been extracted from the Shea nut since ever. Many technologies have been developed for the extraction. Each technology is adapted to different condition and extraction case. The experts did analysis of two methods and come up with their analysis on seven parameters and given in the following table.

Table 1: Comparative analyse

Criteria	Churning	Press
Steps	The churning process has more processing steps, which means a longer process, and more room for processing mistakes and loss of quality	The press process is more concise. Better for quality control. Low humane errors potential
Cost	The capex is 123 437 000 F Cfa, the costs are 10% lower than the press method	The capex is 136 297 500 F Cfa, the costs are Higer than the press method
Yield	The churning yield is around 22%	The press yield is around 33%
Employment	The churning will create around 20% more employment than the press method	It creates less employment but is more rentable than the churning process
Maintenance	The maintenance issue may have the same rating here: the churning process has more steps and will more maintenance need in term of quantity	The press method has less steps, less machine, but higher demand of maintenance in term of quality (need more qualified technicians)
Energy	The churning process will consume more energy in form of heat than the press process, but will need less energy in form of electricity. We should pay attention to Greenhouse gas emission. the two processes are autonomous in thermal energy. their residues may be enough all the thermal energy they need.	
Water	The churning process is consuming lot of water	The press method consumes very few water. Only the cleaning water
Environment	The wastewater produced in the process must be disposed somewhere! and can lead to environmental problem. Also, the mix of wastewater and energy rich residues is often to be dried in open air, leading to dirty surrounding	The press releases only dried press cake easy to dispose and ready to use. the environment can stay clean
Adaptation	The churning is traditionally implemented and adapted to the women	The press method is new, but simple. It may adapt easily

Detailed quotations were collected from different vendors for supply of equipment for both collection and processing centre from five companies. The detailed quotations are given in the [annex 2](#) and [3](#). Name of the agencies are as follow:

Amount in CFA

Name of the agency	Collection centre	Processing centre	Remark
Burkina Lifestyle Concepts SARL	77,502,163	110,182,700	Lowest quotation for both
ISOMET (Innovation en SOLaire et METallique)	90,564,063	112,430,000	
BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)	86,312,438	110,640,250	
Amisy Food Machine	10,064,600		
Megawatt & Solutions Private Limited, New Delhi	72,266,040		

They will supply the equipment if they receive contract. Lowest quotation provider for both Collection and Processing Centre is given by - Burkina Lifestyle Concepts SARL, Ouagadougou.

To construct collection and processing centre quotation were also collected from the companies and agencies working in Burkina Faso. Following agencies submitted their quotations

Amount in CFA

Name of the agency	Collection centre	Processing centre	Remark
Burkina Lifestyle Concepts SARL	42,300,000	47,300,000	
ISOMET (Innovation en SOLaire et METallique)	42,180,000	47,180,000	
BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)	42,000,000	46,900,000	Lowest quotation for both

Details of machines and instruments proposed for collection and processing centres are given in the following tables.

Table 2: Collection centre - detailing of each section with machine

Sr No	Details of the machines
1.	Washing and sorting
2.	De-pulping
3.	Cooking
4.	Drying
5.	Shelling
6.	Shea almond drying
7.	Packaging - a scale
8.	Packaging - Bag sewing machine
9.	Pulp drying
10.	Dried pulp Packaging
11.	Packaging table
12.	Energy production
Sr No	Details of the machines
13.	De-Sheller- electric
14.	Boiler - electric

Table 3: Details of technologies proposed for processing centre

Sr No	Technology	Model	Production capacity	Power consumption	Dimension
1.	Crusher	Electric	200 kg/hr	2 Kw	(2200X1050X1500)mm
2.	Roaster	Electric	40 kg/hr	0.75 kw	(1200X1200X700)mm
3.	Miller	Electric	1800 kg/hr	90 KW	(1440X630X1420)mm
4.	Boiler	Electric	1000 L/hr	2.2 Kw	(1400X1400X900)mm
5.	Mixer	Electric	200 kg/hr	2 HP	(2200X1200X900)mm
6.	Filter	Electric	40 Lt/hr	2 Kw	(1100X1100X3550)mm
7.	Packaging	Electric	300 bags/min	1.36 Kw	(665X1000X1580)mm

a) Construction costs and timeline (also who would do the construction?)

Details for construction cost and timeline is mentioned in the detailed budgeting attached as separate document ([annex 2](#)). To construct collection centre and processing centre detailed tendering process needs to be adopted. This will be different for each sites which includes distance, availability of material area factor. The detailed time line for construction is as mentioned below:

- 3 months tender
- 3 months delivery of equipment/ Construction of the building
- 3 months installation of equipment and commissioning

Table 4: Construction and land development of collection centre

Designation	Unit	Unit cost	Total cost (including shipment and customs)
Land	1	5 000 000	5 000 000
Landscaping	1	1 000 000	1 000 000
Building	1	17 000 000	17 000 000
Storage	1	7 000 000	7 000 000
Water system	1	7 000 000	7 000 000
Electricity	1	3 000 000	3 000 000
Waste management including biogas	1	6 000 000	6 000 000
Fruit bubble washing machine Contact www.pasenmachinery.com	1	3 850 000	5 696 000
Depulping tables	5	869 745	4 348 725
Steamer	5	900 000	4 500 000
Hangar dryer	4	1 100 000	4 400 000
Shelling machine, Henan My Only Food Machine Co., Ltd.	1	1 050 000	2 196 000
Almond Packaging	2	500 000	1 000 000
Pulp dryer	1	15 000 000	15 000 000
Thermal installation	2	9 300 000	18 600 000
Dried pulp packaging machine Zhejiang Ruibao Packaging Machine Co., Ltd.	2	950 000	2 816 750
Dried pulp packaging table	2	870 000	1 740 000
Production tools	1	4 000 000	4 000 000
Transport, Installation and Commissioning	25% of equipment supply cost		
Vehicles	1	5 800 000	5 800 000
Total			116 097 475

Table 5: Capex of processing centre

Designation		Unit cost	Total cost
Land	1	10,000,000	10,000,000
Landscaping	1	1,000,000	1,000,000
Building	1	25,000,000	25,000,000
Storage	1	7,000,000	7,000,000
Water System	1	7,000,000	7,000,000
Electricity	1	3,000,000	3,000,000
Waste Management	1	3,000,000	3,000,000
Dry cleaner NSD 0	1	18,000,000	18,000,000
Crusher	1	2,946,000	2,946,000
Shea nut drying/roasting unit	1	2,258,500	2,258,500
Press MBU 40	1	20,186,000	20,186,000
Thermal installation	2	9,300,000	18,600,000
Filter Kinetic	1	2,758,500	2,758,500
Packaging Filling and labelling machine	1	4,008,500	4,008,500
Packaging table	2	870,000	1,740,000
Production tools	1	4,000,000	4,000,000
Transport Installation and Commissioning	25%		
Vehicles	1	5,800,000	5,800,000
Total			136,297,500

b) Yearly turnover of collection and processing centers

Table 6: Details of turnover of collection center for next five years

Collection centres	First Year	Second Year	Third Year	Fourth Year	Fifth Year
Yearly processed fruits	700 000	2 100 000	2 100 000	2 100 000	2 100 000
Yearly almond production	283 500	850 500	850 500	850 500	850 500
Total energy cost	15,036,840	37,592,100	37,592,100	37,592,100	37,592,100
Total water cost	42,525	127,575	127,575	127,575	127,575
Total disinfectant cost	283,500	850,500	850,500	850,500	850,500
Total transport cost	2,835,000	8,505,000	8,505,000	8,505,000	8,505,000
Total production cost	3,161,025	9,483,075	9,483,075	9,483,075	9,483,075
Fruit Pulp					
Yearly processed fruit pulp	283 500	850 500	850 500	850 500	850 500
Monthly dried fruit pulp production	70 875	212 625	212 625	212 625	212 625
Total fruit pulp packaging cost	5 670 000	17 010 000	17 010 000	17 010 000	17 010 000
Fruit pulp production cost	5 670 000	17 010 000	17 010 000	17 010 000	17 010 000
Fruit pulp unit price cfa/Kg	1 000	1 000	1 000	1 000	1 000
Total variable cost	8 831 025	26 493 075	26 493 075	26 493 075	26 493 075
Total cost	43 154 325	60 721 223	60 721 223	60 721 223	60 721 223
Total turnover	70 875 000	212 625 000	212 625 000	212 625 000	212 625 000

Table 7: Yearly turnover processing centres Press (CFA)

Processing centres	First Year	Second Year	Third Year	Fourth Year	Fifth Ydear
Yearly processed almond	504 000	504 000	504 000	504 000	504 000
Yearly butter production	176 400	176 400	176 400	176 400	176 400
Total energy cost	13 568 815	13 568 815	13 568 815	13 568 815	13 568 815
Total water cost	42 000	42 000	42 000	42 000	42 000
Total packaging cost	5 292 000	5 292 000	5 292 000	5 292 000	5 292 000
Price / Kg	600	600	600	600	600
Total variable cost	19 364 815	19 364 815	19 364 815	19 364 815	19 364 815
Total turnover	86 475 185	86 475 185	86 475 185	86 475 185	86 475 185

2. Introduction

The Shea tree, (*Vitellaria paradoxa*) belongs to the family of Sapotaceae. Shea trees grow in the wild. Sheanut (*Butyrospermum paradoxum*) is an oil rich tropical tree crop, which is indigenous to the West African savannah zone. As reported by Lovett and Haq (2000), almost every part of the tree has its uses. Its fruit contains one or two nuts which are brown and shiny. The fruit pulp is eaten, but the tree is mainly important for its nut which contains a kernel with an oil content ranging from 45% to 60%. The oil, known as Shea butter, is used in the manufacture of soap, candles, cosmetics, pharmaceutical products and butter substitutes (Aviara et al., 2005).

The Sheanut, fruits and seeds obtained from the Shea tree plays an important role in the nutritional and economy of the people; it is the most important source of fatty acids, glycerol in the diet. It has antimicrobial properties, which gives it a place in herbal medicine. It is also used in the pharmaceutical and cosmetic industries as an important raw materials and/ or a precursor for the manufacture of soaps, candles and cosmetics. Shea butter is used as a sedative or anodyne for the treatment of sprains, dislocation and the relief of minor aches and pains. Shea nut oil obtained from the Shea kernels is the main traditional medicine in many rural areas (Leaky, 1999; Maranz et al., 2004). The seed kernels produces oil which is nutritious with illustrated fatty acids like linoleic and oleic fatty acids and fat soluble vitamins (Karin, 2004). The hard protective shell, are used as a water proofing material on the walls of mid-buildings to protect them from eroding by forces of the wind and rain.¹

To obtain Shea nut oil from the kernel, series of processes must be carried out. The processes include; shelling which is the first process followed by crushing, roasting, milling and kneading, (Koya and Faborode, 2006).

Shea processing is a major industry in Burkina Faso, and conducted almost exclusively by women's groups; however, it still involves outdated methods: potentially degrading habitats including Shea trees, emitting significant greenhouse gas emissions, causing a number of health and hygiene risks, and not realising its potential to support social and -economic development.

A consortium including an international project developer, a renowned local Shea processing training company, and an engineering firm have established **Serious Shea**, which seeks to improve the Shea butter-making process in Burkina Faso.

The project has received the firm support of the Ministry of Environment in Burkina Faso, and grant funding from the African Development Bank and World Bank to conduct the preparation activities, and will ensure women's groups can produce the Shea butter more efficiently and sustainably, and also achieve significantly better access-to-market (receiving 2x - 5x more profit per tonne). The project will use clean technology at each processing centre and drive reforestation / afforestation at the collection sites, generating certified carbon credits by decarbonising the process and REDD+.

3. Burkina Faso context

3.1. Country Background

Burkina Faso is a low-income Sahelian country with limited natural resources. Its economy is largely based on agriculture, which employs close to 80% of the working population, although gold exports have increased in recent years. Despite the progress made in the past two decades, Burkina Faso faces many development challenges, particularly in terms of health and education.

¹ Design, fabrication and testing of Shea nut shelling machine, 1 Ibrahim, M. G., 1 Shehu, A. A., 2 Dauda, S.M. and 3*Ahmad, D, International Food Research Journal 23(Suppl): S71 - S79 (December 2016)

4.1.1. Map of Burkina Faso



Figure 3: Burkina Faso country map

4.1.2. Political context

Growing insecurity, particularly in the north, close to the border with Mali and Niger, has displaced many residents of Burkina Faso. In January 2019, fewer than 50,000 people were internally displaced, while in March 2020, this number stood at roughly 765,000. At that time also, more than 2,000 schools were closed, a situation that deprived some 300,000 students of education and impacted more than 11,000 teachers.

Presidential and parliamentary elections are scheduled for November 2020. For the first time in the country's history, citizens of Burkina Faso in the diaspora will be able to vote from abroad.

4.1.3. Economic Overview

Despite the security and humanitarian crisis, the macroeconomic framework remained broadly stable in 2019. Although real GDP growth fell slightly, it remained fairly strong at 6%, compared to 6.8 % in 2018. Economic activity was led mainly by the services sector, the growth of which offset the decline in activity in the agricultural, mining, and construction sectors.

Without considering the impact of the COVID-19 global pandemic, growth is projected to fall slightly to 5.8% in 2020, with this downward trend continuing owing to the security and humanitarian crisis. These projections (may/will) be impacted by the spread of the coronavirus, climate events, the worsening of the security and humanitarian crisis and, in the event of a global recession, the possible decline in cotton prices.

The current account deficit is projected to increase but to remain fairly well controlled and stable at roughly 4.5% of GDP during the 2020-2022 period. A trade surplus of approximately 1.5% of GDP on average is expected during the same period. Three main factors will contribute to this expected surplus: the increase in gold exports, the decline in oil imports, and the consolidation of the government budget, which is expected to contribute to lower demand for imports.

Burkina Faso 2019 highest dollar value export product groups were the following

1. Gems, precious metals: US\$2.3 billion (69.6% of total exports)
2. Cotton: \$357.8 million (11%)
3. Zinc: \$155.3 million (4.8%)
4. Oilseeds: \$141.8 million (4.3%)
5. Fruits, nuts: \$119.7 million (3.7%)

Shea nuts and butter number 4, are playing an important role in the country economy, and not only. The tree itself is of great environmental importance and is protected under the national forestry code. Shea butter is, of course, well known on international markets, but in Burkina

Faso, all parts of the tree are used: the leaves, the bark, and the roots are used for traditional medicine. The flowers and pulp are eaten, and the nut is used to make Shea butter. No wonder, then, that the Shea tree is often described as the 'green gold of women'.

3.2.in Burkina Faso and other sub-Saharan African countries, Shea butter is produced for both the domestic export markets. It is extracted from the fruit kernel by women, using a method that is both labour- and energy-intensive.

This is a traditional supply chain that pays little heed to economic criteria (strong social role of the activity, use of non-commercial domestic resources, etc), but it does need to be competitive to generate income.

In the savannah of Burkina Faso, the Shea tree plays an important role in the village subsistence economy. Women extract almonds from its fruits and turn them into butter with multiple virtues, used in cooking and for body care. By selling Shea, in the form of almonds or butter, they obtain a small additional income. Between 2012 and 2013, Shea butter export revenues reached 2.9 billion CFA francs compared to 4.8 billion CFA francs in 2008 in Burkina Faso. For 2015, the country expects revenues of around 7 billion CFA francs.

3.3. The Shea Butter Sector in Burkina Faso

Shea (*Vitellaria paradoxa*), the "butter tree" or "women's green gold", grows in the wild on a 5 000 km strip in the wooded savannahs of West and East Africa – the "Shea belt". The Shea tree only grows in Benin, Burkina Faso, Cameroon, Central African Republic, Chad, Côte d'Ivoire, Ethiopia, Ghana, Guinea Bissau, Guinea, Mali, Niger, Nigeria, Senegal, Sierra Leone, South Sudan, Sudan, Togo, The Democratic Republic of the Congo, and Uganda. Shea butter is part of the cultures of peoples of different areas and provides livelihoods for many. Almost everything from the Shea tree – fruit, nuts, leaves, bark, roots, and dead wood – is used for purposes ranging from food, pharmaceutical products, soaps, and cosmetics.

▪ The socio-economic role of Shea

Shea activities provide incomes for women mainly but few for men: they range from the collection of nuts to making raw Shea butter for consumption and treating Shea butter for creams, pomades and cosmetics. Shea is Burkina Faso's fourth export product after gold, cotton and livestock products, and for this reason, it has an important socio-cultural role and is also a critical element in the national economy. In Burkina Faso, the sector contributed more than CFA 10 billion (USD 17 million) through nut sales in 2012 (Rousseau *et al.*, 2015) and currently employs an estimated 1.5 million in the Shea chain (Palmieri 2012). At the end of 2013, the price of Shea nuts was 250 CFA francs per kilogram (Rousseau *et al.*, 2015).

The processing of Shea butter is a social and cultural event that brings women together, usually for the common purpose of generating income, there are often meetings, songs and dances to celebrate. One of the main findings was that women felt significant social empowerment from belonging to Shea groups. A large proportion of butter is used for local consumption, making it an important source of food and hence of food security. Of the estimated 600 000 tonnes of Shea nuts harvested in West Africa, 350 000 tonnes are exported, mostly as raw nuts; the remaining 250 000 tonnes are processed and consumed locally and are not part of the trade market (USAID, 2010).

▪ The poverty of women in the Shea sector

Shea provides a vital source of income for Burkinabé women and can improve the livelihoods of women and their families. The women involved in the Shea sector are usually poor in terms of the international poverty line of USD 1.90/day. Of the women, 50 per cent earned less than CFA 200 000 (USD 341) per year, and fewer than 36 per cent are literate. Women's incomes are low

and unstable because they depend on the availability of nuts and the ability to sell the products. Women in the sector are vulnerable to issues such as climate change, increasing prices of nuts, lack of opportunity for personal and entrepreneurial growth, and poverty. The Shea sector can give poor and vulnerable women opportunities to improve their livelihoods and the living standards of their households.

3.4. Overview of Shea Butter Industry

Shea butter is a type of fat which is used in various skins and hair related products and possess various healing properties. It is derived from the seeds of *Vitellaria paradoxa* (Shea) tree which is mainly found in Africa. Shea butter is free of any artificial chemicals and is composed of stearic acid and oleic acid. Shea butter is used extensively in cosmetic products such as skin moisturizers, hair conditioners and lip gloss. Apart from this, Shea butter is used in various industries such as food and healthcare. The process of extracting Shea butter from nut involves various processes such as cracking, crushing, roasting, grinding, separating the oils and collecting and shaping. The market share of Shea butter has constantly been rising as compared to cocoa butter or palm oils. The market share of the Shea butter is growing due to the increased adoption of Shea butter products over cocoa butter or palm oils. The advantages that it offers over the cocoa and palm oil is that Shea butter has low-fat content and better healing properties than the former.

Shea butter is also much more environmentally viable than cocoa butter and palm oil. The Shea tree is a slow-growing tree and starts producing fruit at around 15 years. The Shea tree is very important in the ecosystem of the so call Shea belt.

On the chemical level, the Shea butter products are powerful antioxidants and restore the damaged skin and hair to health. However, the cocoa butter contains the only polyphenol which helps relieve stress. Shea butter contains vitamin A and vitamin E which are vital for skin and eyes. In addition to this, Shea butter contains cinnamic acid which prevents the skin from ultraviolet rays. Moreover, cocoa butter is a good moisturizer, but Shea butter is more flexible and can be used in different treatment processes such as skin healing, hair healing and lip healing.

The global Shea butter market is anticipated to witness robust growth over the forecast period owing to the rising disposable income and consumer awareness. Europe and North America are holding the largest market share of global Shea butter followed by Asia-Pacific. Further, government policies are bolstering the European Shea butter market. Recently, European Union instructed that a minimum 5% Shea must be present in eatable products like chocolate which is expected to bolster the Shea butter market.

4.4.1. Stakeholders

The stakeholder in the value chain of Shea butter are the following:

- **Farmers:**

Farmers are mostly women. They are collecting the Shea fruits or nuts extract the almond. this important part of the Shea butter value chain had little to say. Today most of the little farmers have found a way to organise themself into CBO.

- **CBO (Community based organization).**

The CBO is organisations that represent the bottom of the line of the Shea butter sector. The CBO is now an important stakeholder of the Shea butter value chain as they are a strong voice of the farmers. The CBO is an important interface between the farmers and the other stakeholders. They input the work of policymaker and researchers, talk with one voice to the

traders and processors. In one word, they are an excellent tool to defend the interests of the farmers.

- **Traders**

There are two kinds of traders. The first type is scouts like and is going to remote villages to buy the Shea almond. This type of traders sells their products to the international traders or collection service of big processors (like AKK). The traders sell their products as well to local processors such as Olvea.

- **Processors**

The processors extract the butter from the Shea almond and conditions for the sale.

The processor maybe is any of the first three stakeholders:

Farmers are mainly extracting the butter for local consumption. the butter extracted by the farmers are basic quality and are not energy effective.

Farmers may also join any a CBO to better defend their interests. sometimes they may event join forces to implement a butter processing facility. together, they may be able to purchase basic materials. The CBO can reach a higher level of quality of butter since they may benefit from training programs and get some funding for basic machinery. The CBOs are selling their butter mostly to the fair-trade market or to other processors for the further refinery.

- **Consumer**

If fact, we have two main types of consumers classified by awareness.

Awareness is especially important for Shea butter since it will define the marketing of the product. Chocolate, ice cream or margarine the consumer is not aware of the Shea butter in the product he is buying, and will not care where the butter is coming from, how it has been produced, or how far the benefits have been shared. often the main criteria lie in the food production industry and it is the cost.

On the other hand, products sold as “Shea butter” mainly in the cosmetic industry, refers in the marketing to properties of the Shea butter, but also to story of the butter: Where it comes from, how environment friendly it has been produced, by whom, and how fair the benefit have been shared. This market is vital for women in the Shea butter business. To secure this market, it is important to develop efficient and adapted production processes, in other to achieve financial, social, and environmental viability.

This study aims to analyse the Shea butter processing sector and processes and develop adapted technologies and methods to valorise the value chain of the Shea butter.

Develop the use of the fruit flesh, study the right mix of existing machines to increase the production yield, reduce the waste by transforming it into energy, reduce the CO2 production, increase the Shea butter quality.

- **Policymaker**

Policymakers include governmental bodies and corporate organisations of the Shea sector. They are in charges of making the rules and the laws of the Shea butter sector. The policymakers consult all the stakeholders to define new rules.

- **Researcher**

The researchers are in charges of detailed study into the Shea butter, concern, or issue using the scientific method. the results of the research are used by all Stakeholders to improve the Shea butter impact on the economic, social and environmental point of view.

4.4.2. Shea Butter production locations

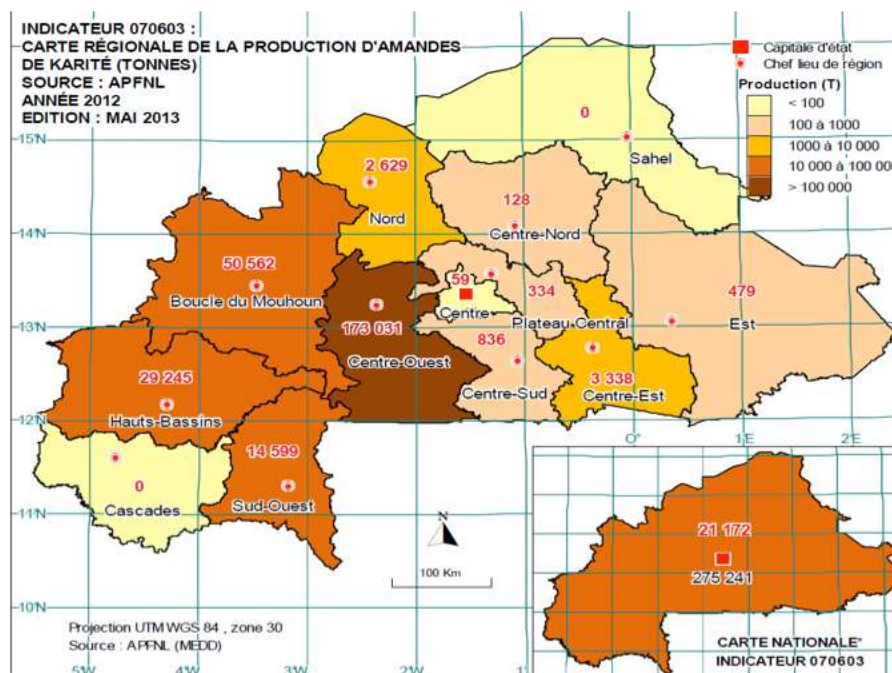


Figure 4: Shea Butter Production locations

3.5. Sector Analysis

The Shea tree is a tree that is only found in Africa. It grows naturally throughout the Sahel-Sudanese strip in about 18 countries: Nigeria, Burkina Faso, Mali, Ghana, Côte d'Ivoire, Benin, Togo, Senegal, Niger, Cameroon, Chad, Zaire, Guinea Bissau, Uganda, Ethiopia, Sierra Leone, Sudan, Central African Republic. There are two species of Shea trees: the *vitellaria paradoxa* variety and the *vitellaria nilotica* variety. In terms of Shea tree stand, Burkina Faso ranks second only to Nigeria. However, in terms of the quality of Shea products, the *vitellaria paradoxa* species is the one that provides the most sought-after ingredients by the food, pharmaceutical and cosmetic industry, and the Shea tree stand in Burkina Faso is exclusive of the *vitellaria paradoxa* species. It is this major asset in particular that justifies the establishment, in Burkina Faso, of all multinationals operating in Shea products. Shea products from the species *vitellaria paradoxa* is in high demand by the food, pharmaceutical and cosmetic industries because Shea butter from this species is richer in stearine. Shea stearine is the most necessary and sought-after ingredient in the global food industry, particularly for the manufacture of chocolate and other food products. Chemical studies have confirmed that the *nilotica* variety is genetically different and that its almonds are less rich in stearic acid. In addition to the confectionery industry, other Shea markets, such as the body care sector, the cosmetics industry in general and the pharmaceutical industry, have formulated their products to use Shea butter from the *vitellaria paradoxa* variety, which is more price competitive, more available and also has a higher melting point.

In terms of the potential for Shea almond production, according to the report of the study on the socio-economic aspects of the priority sectors of non-wood forest products (January 2017), initiated by the Ministry in charge of the environment (Burkina Faso), the Shea tree stand has increased from 190 million trees to 280, 802,860 million trees. As for the production potential of Shea almonds, it has increased from 850,000 tonnes of almonds per year to 1,247,295 tonnes per year with only a real exploitation rate of about 32% or 404,412 tonnes. The distribution of Shea stands by region is given in Table 1 below. About 68% of this potential, or

842,883 tonnes, remains untapped, anything that currently contrasts with the global Shea product environment, which has some interesting prospects.

Estimates are for both conventional and organic production. In terms of the production of almonds and organic butter, there are sufficient classified forests and forest reserves that contain Shea trees and are parks that will increase the supply of almonds and organic butter sufficiently to meet the ever-increasing demand. However, as Shea is growing very slowly, it is appropriate to take appropriate measures to repopulates it.

In view of its potential, the geographical coverage of about 70% of the territory, the market prospects quite interesting for the domestic, sub-regional, regional and international market, the Shea sector can, if it is better value, contribute significantly to the increase of our exports and to the improvement of the situation of Burkina Faso's trade balance, which is structurally in deficit.

In addition to the butter, it could be meaningful to investigate the valorisation of the fruit flesh which represents half the fruit weight. Some tests have been made and the result is a greenish, sweet, and light astringent matter, that may be used for local candies, jam or alcohol production. The potential in term of tons and income is considerable.

Table 8: Information about Shea tree population and potential

REGION	Shea tree population		Potential		Potential actually exploited
	Number of trees	%	Tons of Shea kernel	%	Tons of Shea kernel
Boucle du Mouhoun	29 586 960	10,54	166 119	13,32	80 060
Cascades	43 935 223	15,65	252 119	20,21	75 636
Centre	2 541 139	0,90	3 099	0,25	930
Centre-Est	8 475 318	3,02	10 335	0,83	3 100
Centre-Nord	2 007 043	0,71	2 447	0,20	734
Centre-Ouest	43 455 260	15,48	211 978	17,00	63 593
Centre-Sud	19 819 308	7,06	56 628	4,54	16 988
Est	34 892 868	12,43	42 548	3,41	12 765
Hauts Bassins	45 466 080	16,19	255 274	20,47	76 582
Nord	4 610 532	1,64	5 830	0,47	1 749
Plateau-Central	3 965 164	1,41	4 835	0,39	1 451
Sahel	0	0,00	0	0	0
Sud-Ouest	42 047 965	14,97	236 083	18,93	70 825
National	280 802 860	100,00	1 247 295	100,00	404 412

Source: report of the study on the socio-economic aspects of the priority sectors of non-wood forest products (January 2017).
monographie karité

■ Production capacity

Apart from the industrial Shea butter production units that know their (potential) production capacity, it is difficult to give an estimate of the production capacities of women's organizations and women's individual production. Indeed, the production chain of women's organizations is half mechanized and half-manual, so that production capacity depends on the dynamism of the

manual production teams that make up the production line. The table below shows the production capacity of industrial units in Burkina Faso.

Table 9: Industrial Shea butter processing units in Burkina Faso.

Place	Capacities (tons)	Processing methods
Bobo-Dioulasso	3 000	Mechanical Extraction
Bobo-Dioulasso	75 000	Mechanical and solvent extraction
Ouagadougou	1 500	Mechanical Extraction
Total	79 500	

Source: the national sustainable development strategy document for Burkina Faso's Shea sector

This table shows that the total capacity of the industrial units to process almonds into butter is 79,500 tonnes, but recent studies give a potential production of almonds of 1,247,295 tons or a potential total equivalent of 415,765 tons of butter with a processing coefficient according to traditional methods of 3 kg of almonds per 1 kg of butter. The confrontation between the total processing capacity of the industrial units currently installed in Burkina Faso (79500 tonnes) and the existing processing potential (415,765 tonnes) reveals a positive gap of 336,265 tonnes of Shea butter that can still be produced if additional processing units are established.

In the case of women's Shea butter-producing organizations and individual producers, production capacity is very limited, a few hundred tonnes.

The production of Shea almonds until then is done individually by women. However, there are currently project initiatives to produce Shea almonds in production centres. If this were to happen, women's tasks at the rural level would be limited to picking fruit and nuts and passing them on to the various production centres.

▪ Many challenges ahead

Despite its growth, African Shea faces many challenges that could determine its evolution in the coming years. The sustainability of the resource is thus a major aspect of the development of the Shea sector. The Shea tree is an endangered species. At issue is the exploitation of the tree for its wood used in the construction of houses and palisades or the production of coal and the clearing of land for the benefit of crops such as cotton.

This situation remains complicated, due to the difficulties of regeneration of the Shea arboreal parks, linked in particular to the intrinsic characteristics of the tree itself. It takes the tree nearly 15 to 20 years to produce nuts, with a peak in yield after 40 to 50 years. To make matters worse, fruit production varies from year to year, from place to place, or even from tree to tree, depending on climatic factors and the impact of human activities.

Apart from this major challenge, several reports have also highlighted the need for the industry to improve its practices in order to deliver a high-quality product, in commercially acceptable quantities on the international market. This means addressing the many bottlenecks that directly affect the performance of the sector. These include the hardship and workload of women that limit the significant increase in production, look for sustainable energy solutions for butter production, determine or develop simple and adequate machines for the extraction of good quality butter, find a solution to valorise the fruit flesh in order to increase the income of actors in the Shea butter sector.

In addition, the lack of storage structures also remains a problem for processors and groups wishing to quickly sell large volumes of butter in response to the needs of buyers. Added to this is the lack of information on quality standards and the still insufficient organisation of the sector in some producing countries. Of course, the various projects deployed by some development organisations have provided solutions to the sector, with regard to the

management of cooperatives or professional organisations. But there are still huge projects, both in terms of supporting groups to meet the criteria of fair trade and at the level of processing or export operators who do not always have up-to-date information on the trends of the different Shea markets.

With regard to all the challenges, this study aims to analyse the Shea butter processing sector and processes and develop adapted technologies and methods to valorise the value chain of the Shea butter, Develop the use of the fruit flesh, Study the right mix of existing machines to increase the production yield, reduce the waste by transforming it into energy, reduce the CO2 production, and increase the Shea butter quality.

Figure 5: Schematic presentation of the 12 processing steps from stored seeds to sale²

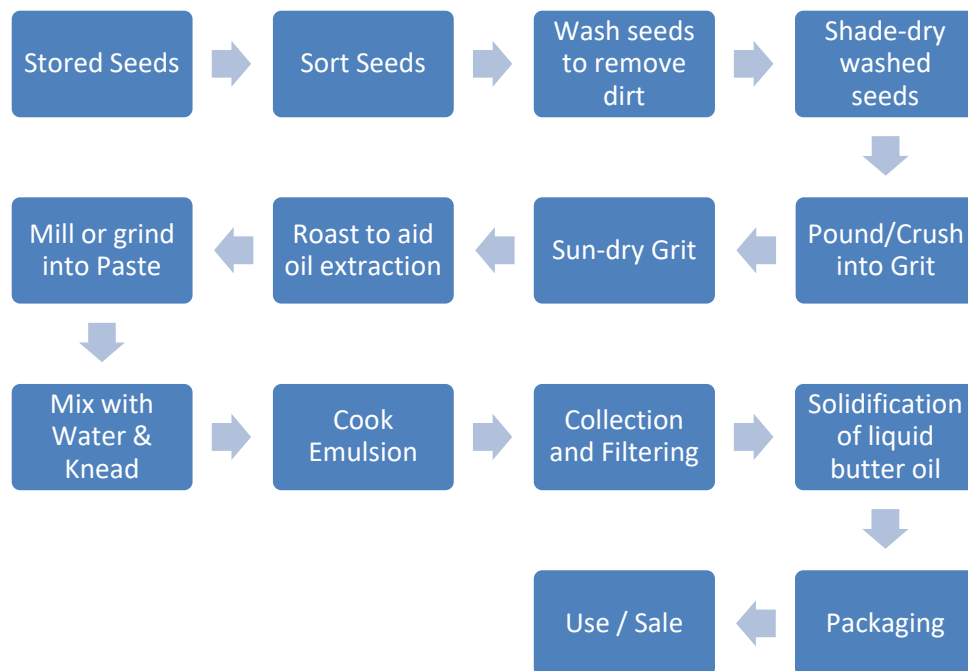


Figure 6: Process of Shea butter making



- The existing butter-making process in Burkina Faso drives significant deforestation and greenhouse gas emissions, and requires women's groups to conduct heavy physical labour. Furthermore, the lack of access-to-market means that the women's groups do not receive appropriate remuneration for their inputs, and typically fail to improve their socio-economic status despite their hard labour.

² Source: SNV Ghana Renewable Energy Team Field Study, 2012

4. Detailed Designs & Specifications – Deliverables

4.1. Prepare detailed schematics, 3D representations of the designs for collection sites and processing centres

Understanding the production process from collection and processing was steep learning because of the challenges at both level. To design technological solutions, is fundamental to understand every step of the value chain. Listing correct processes, its timings, gaps and causes, appropriate solutions, resources required at each stage, expertise to operate machines, seasonality and sourcing of the crops, processing done at village level, quality requirement, procedure for transport, human resources for quality monitoring and ensuring the quality these were questions which were discussed at length before providing any technological solutions.

To maintain the Shea butter quality, it was complex to produce the Shea butter in rural areas. Therefore, we have designed a model with two levels of processing, item one – collection centre - location rural area close to village will process the fruits into high quality dry pulp and high quality shea almonds. Second item – processing centre – it will be located in the bigger city. Five collection centres plus one processing centre will work as a cluster. In each processing centre is designed to process 3000 tons of kernels into 1000 tons of Shea butter.

The production is often conducted by women's groups and helps to improve the living conditions of rural households. In traditional methods, non-renewable biomass is inefficiently consumed in boiling nuts denatures & de-husking, roasting, heating water for kneading, and boiling fat to dry and clean, and accounts for ~25kgCO₂e/kg butter at farm. Both the deforestation and carbon emissions could be reversed and negated respectively through the implementation of the Eco-Processing Centres, as they will minimise the need for fuelwood, and enable the protection and reforestation of forest populations.

The concept of collection centre and processing centre evolved after understanding that to make it more environment and energy conservative, it is important to segregate the pre processing of fruits at collection centre and Shea butter production using environment friendly technologies at processing centre. Each centre is expected to save >32,000tCO₂e/year of GHG emissions and drive widespread reforestation.

The Serious Shea project aims to decarbonise the Shea butter value chain in Burkina Faso through the development of 50 x Collection Sites and 10 x Eco-Processing Centres:

- Each set of 5 collection sites will be fully equipped to process 7,500mt of fruit into 3,750mt of almonds
- Each processing centre will be fully equipped to process 3,750mt of almonds into 1,000 mt of Shea butter per year, with further fractionization and refining potential

Why we need collection sites and processing centres?

- To organize Shea Butter production system
- To ensure better quality production
- To enhance efficiency and effectiveness in entire Shea butter production process
- To build the capacity of women and strengthen their engagement in new structure of the production process
- To increase profit share of women
- To control the production quality
- To ensure sustainability in production system across the units

Key features of collection site

- Designates space to store kernels
- Professionally designed space for pre processing of fruits and kernels

- Separate space to process pulp
- Provides better working environment to the women
- Women groups can use it as ware house for preprocessed kernels
- To maintain the quality of kernels

The women groups can use the services of the collection site on the following conditions

- They will sign an official contract with the group and Serious Shea jointly for the supply of fruits and kernels to the Serious Shea
- Serious Shea team will provide orientation to all the women on the procedure of the collection site and how to use them for shared understanding
- They will provide 50 % of their total fruits/kernels to Serious Shea as fees to use the services of the collection site.
- If the women want to sell the 50% of the kernel to Serious Shea for butter making then they Serious Shea will purchase those seeds on the market rates of that day and make payment to the women

Key features of processing centre

- One stop solution approach
- Well equipped with high quality equipment
- Well trained human resources to operate the equipment and produce quality butter
- Eco friendly
- Production of renewable energy by [bio gas](#) using the fruit waste, [syngas](#) using the nut shell and cake and solar energy to reduce carbon foot prints
- Scientifically storage facility for processed butter and other products

Composition of the clusters

- **Collection centre**

Serious is driving an effort to transform the lives of rural women Shea workers in Burkina Faso through the development of 50 x Collection Sites:

- d. Each set of collection sites will be fully equipped to process Shea fruit into Shea kernels
- e. Each processing centre will be fully equipped to process Shea kernels into Shea butter, with further fractionization and refining potentially
- f. The facilities will draw on renewable energy sources (e.g., biogas and solar PV) and include all the equipment necessary for clean, efficient, and environmentally-friendly Shea butter production

- **Processing centre**

Serious is driving an effort to transform the lives of rural women Shea workers in Burkina Faso through the development of 50 x Collection Sites and 10 x Eco-Processing Centres:

More specifically, each centre to be equipped with an optimal mix of the following:

- e. Solar cookers and steam boilers to dry and boil the nuts cleanly and efficiently
- f. Solar-powered motors, which enable the women's groups to produce the Shea paste quickly without the strenuous physical labour of traditional manual methods
- g. Large-scale improved cookstoves to boil the mixture and extract the butter
- h. All the necessary apparatuses and equipment to store, transport, package and export the final Shea butter product

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 - a. Each set of collection sites will be fully equipped to process Shea fruit into Shea kernels
 - b. Each processing centre will be fully equipped to process Shea kernels into Shea butter, with further fractionization and refining potentially
 - c. The facilities will draw on renewable energy sources (e.g., biogas and solar PV) and include all the equipment necessary for clean, efficient, and environmentally-friendly Shea butter production

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More specifically, each centre to be equipped with an optimal mix of the following:

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- Large-scale improved cook stoves to boil the mixture and extract the butter
- All the necessary apparatuses and equipment to store, transport, package and export the final Shea butter product

Collection centre:

- Semi-mechanised sites used by women's groups to process fruit into Shea kernels, supervised by a Serious Shea employee(s)
- Sites will be located in (i) PIF areas, (ii) areas with high levels of Shea fruit yields and (iii) areas with sufficient women's groups eager to partner with Serious Shea
- Utilise clean technology and renewable energy, but retain the traditional process for women's groups to operate easily
- 20kg per day x 90-day season = 1,800kg of fruit per woman each season = 900kg kernels per women each season (assume 250kg to 500kg of kernels per women each season to be conservative)
- Number and size of collection sites depends on distance women can transport fruit, and the requirements of the processing plant
- 50% of Shea kernels purchased by Serious Shea (3,750mt), and 50% (3,750mt) kept by women's groups for a fee to use themselves
- During the off-season for Shea production, the collection sites will be used to process dried mango (12t/site/year)
- Each collection site will also produce 300t/site/year of compost from the unused fruit meat and 300t/site/year of fuel for sale

Expanded "Karite⊕(plus)" Value Chain Overview



Figure 7: Shea buter making processes

Table 10: Detailing steps at collection centre

Sr No	Functions s	Total Quantity	If the pulp is not used	If the pulp is used for Jam or Squash making	Total days required		Remarks
					5 women	10 women	
1	Processing of 50t of fruit into kernels	50 tons	1 tons /day/women	1ton/day/two women	10 days	5 day	1 ton/day/women if the pulp is not used for jam making 1ton/day/two women
2	In terms of capacity, if the production of fruit is highly seasonal, we are assuming we will need 10 x collection centres, each producing 750t dried kernels/annum (375t kept by women's groups, and 375t sent to the Serious Shea processing centre)	3,750t	300-500 ton/year/center	1000 t butter produced	at 10 collection centres		300-500 ton/year/centre so we need 10 collection centres. Each collection site will need additional storage capacity based on the quantum of production received in that region. The storage facility is calculated on the base of 50tn/site (10%+/-). It will not require any additional space to process the additional quantity of fruits/kernels.
3	Power capacity and electricity usage?		Diesel/ petrol engine of 8 hp and 50 KW connection for collection site or 50 KW capacity solar plant will be installed at each collection site				
4	Water requirement for the process (litres)		To process 100tns fresh fruits total water requirement is 15 kiloliters. The water will be recycled for re-usage as well into the process. To store this quantity water there is a requirement to have storage tank of 15 m3 capacity of a tank - dimension 2.5X1.5X5				
5	Energy generated and capacity of renewable energy installation		50 KW solar plant to power the 10 dryers and ancillary equipment for the collection centre				
6	Human power required to manage the centre (both skilled; semi-skilled; unskilled)		40 women required for de-pulping, 40 women require for boiling and drying (semi-skilled) and 20 women require for processing (skilled) – 5 technicians and 1 manager. This is calculated when the collection site will work on its full capacity i.e. 5 tonnes of fruits per day. 1 ton pre-processing the site will require a team of 20 human resources.				

Processing centre:

A total 10 processing centres will be set up across the country. The processing centre is a medium-size centre with expected an annual Shea butter processing capacity of 500 tonnes, therefore the type of equipment that will be recommended for usage in the centre are semi-automated process equipment powered by diesel/petrol engine of 8 hp. The average daily combined Shea butter processing capacity of the centre will be on an average of 3-5 tonnes of butter per day. If the

centre runs for 20 days in a month and operates at an optimum capacity of 75% (that is 3 tonnes per day) this will translate to 60 tonnes every month (20 days x 3 tonnes = 60 tonnes).

- Pilot plant will be built in Ouagadougou, capable of processing 1,080mt of Shea kernels into 378mt / year of unrefined organic Shea butter
- Operated by Serious Shea employees (women) only
- Exported to international cosmetic buyers initially (and then possibly food buyers)
- 10% of profit provided back to participating women's groups

Table 11: Detailing steps at processing centre

Sr No	Functions	Description
1	Production capacity	4 tons per day production is needed at one processing centre to produce 1000t/annum. So at a 50tns capacity processing centre, it will be completed in 13 days and it will operate on 15 days procurement cycle from all the collection sites. A supply route will be prepared to supply kernels from all the collection sites to run the processing centre on its full capacity.
2	Power requirement and electricity usage	Energy use is 500 KW solar plant for the Shea milling and processing facility, Total Area 2000 SQM
3	Water requirement for the process (litres) -	100,000 ltr capacity tank would be required to meet the water requirement of the 50 t production throughout the year
4	Human resource requirement to manage the centre (both skilled; semi-skilled; unskilled)	10 non-skilled, 5 skilled, one manager is required to manage a processing centre

a) Collection centre:

Detailed literature review helped in defining the key requirements of a collection centre. The design of each collection centre will be depend on - distance, size, number of women's groups, etc. This includes equipment (including capacity, suppliers, lifespan, specifications, etc.), space, energy plan etc. Based on secondary literature review following are essential requirements for a collection centre:

- Loading and offloading space
- Safe storage for nuts
- Weighing space
- Wastage and thrash area
- Record room cum office
- Drying space
- Tools and equipment to collect Shea fruit
- De-pulping
- Solar cookers and steam boilers to dry and boil the nuts cleanly and efficiently
- Warehousing/storage facilities for dried nuts
- Energy or electricity space.

b) Processing centre:

The processing centres, ensuring the options utilize clean technology and renewable energy, are modular and scalable, and cover the full value chain (incl. necessary transport methods and storage facilities). A processing centre must have following:

- Shea nut shelling machine. Shelling machine efficiency is tested on following key parameters
 - a. It's shelling efficiency
 - b. Cleaning efficiency of the machine and
 - c. Recovery efficiency

- Fruit meat processing (drying/cooking)
- Nut boiling, cracking and drying
- Fat extraction from almonds
 - Solar-powered motors, which enable the women's groups to produce the Shea paste quickly without the strenuous physical labour of traditional manual methods
- Large-scale improved cook stoves to boil the mixture and extract the butter
- Fat refining and cooling
- Ecological packing for long-haul road transportation
- Renewable power sources
- Water pumping and recycling
- Waste processing and management
- Refine initial draft designs through interviews with international experts and
- Equipment to assess and monitor quality of final and intermediary products
- All the necessary apparatuses and equipment to transport, package and store the final Shea butter product.

c) Layout of the collection site and processing centre

The team had detailed discussions and also visited factories where Shea Butter production is going on and based on their inputs and discussion with experts finalized the detailed layout for collection centre, processing centre. High definition layout designs are shown in the following sections

d) The layout of the collection centre

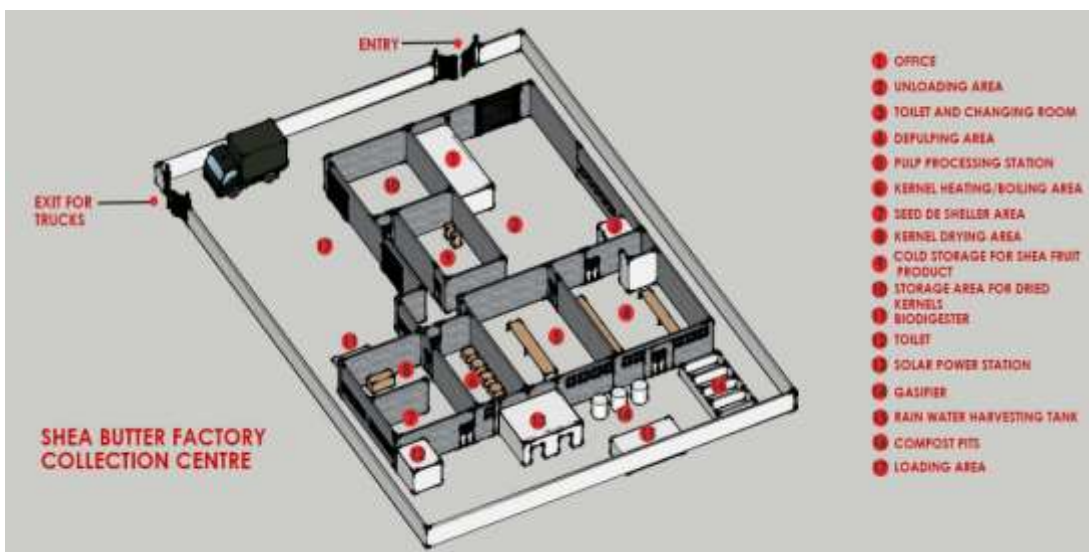


Table 12: High definition layout of the collection centre with the 3D view and detailing of each section of the centre

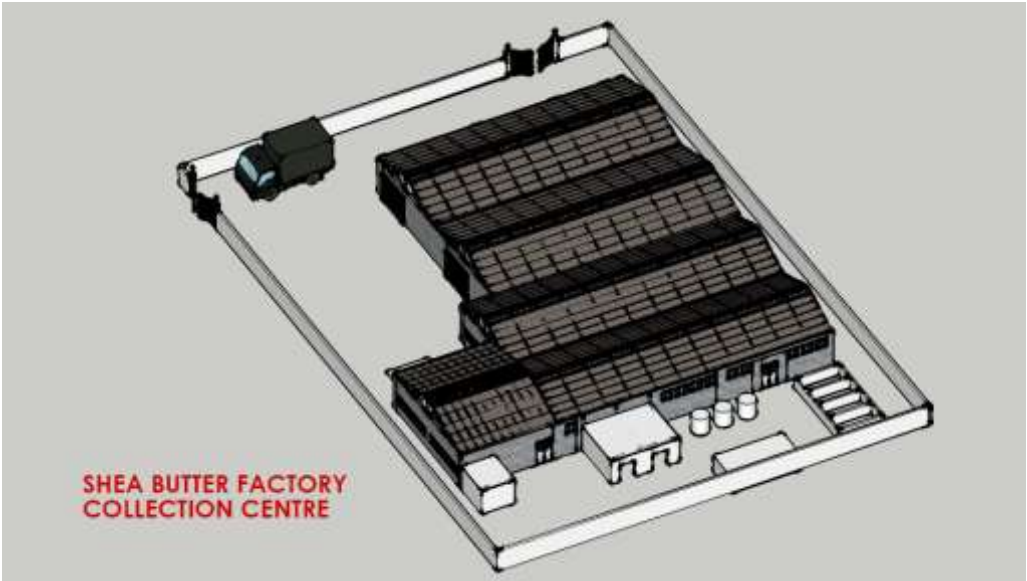


Table 13: 3D high definition top view of the collection centre

e) The layout of the processing centre

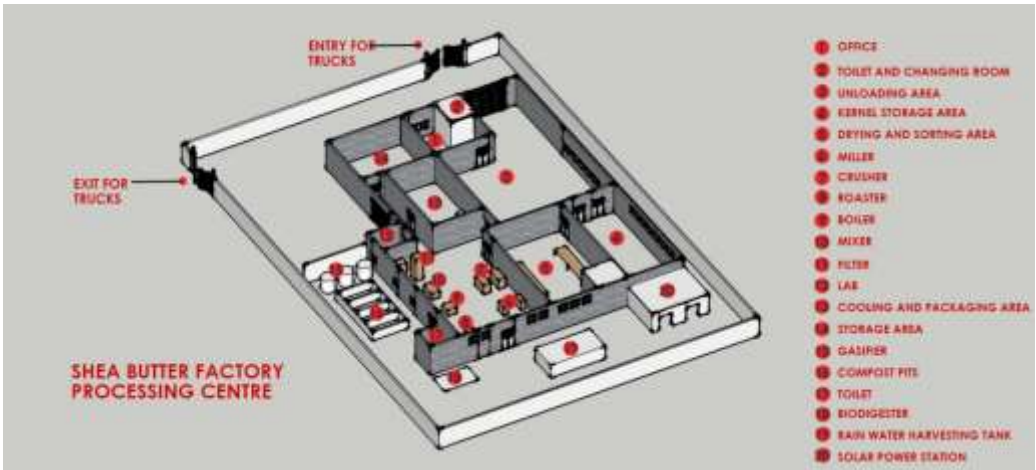


Table 14: Processing centre high definition layout with detailing of each section

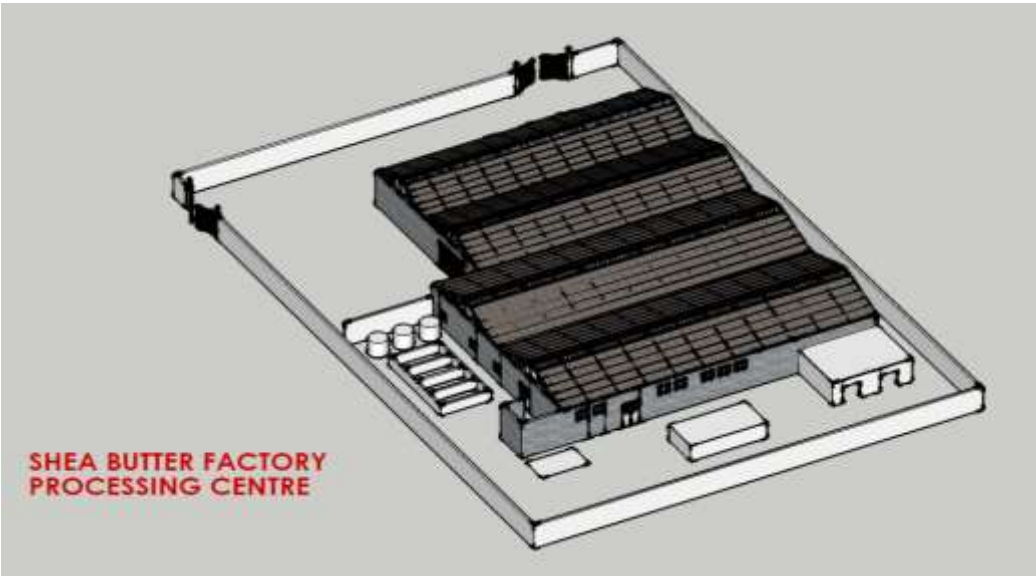


Table 15: 3D high definition top view of the processing centre

f) Energy requirements of both the centre

The gasifier will be used to produce Syn Gas and seed shell and Shea cake will be used as source to generate the renewable energy at collection and processing centres. Bio digester will be used to produce good quality cake which can be used in gasifier.

In each centre solar energy plan will be set to meet the energy requirements. Details are as follows:

- Collection centre: 50 KW solar plant to power the 10 dryers and ancillary equipment for the collection centre
- Processing centre: Energy use is 500 KW solar plant for the Shea milling and processing facility,

4.2. Prepare spreadsheets to calculate implementation costs, and reference original quotations from selected equipment suppliers and contractors

A. Collection Centre

a) Collection centre - detailing of each section with machine and its processes

15. Washing and sorting

After receiving the fruits from the area, they are cleaned and sorted

Material used

Fruit bubble washing machine Contact www.pasenmachinery.com

Production capacity 0.5-2t/h(depends on raw material); Electrical consumption : 2 Kwel, 220V,50HZ, single phase; 304 stainless steel; 4000*850*1200mm

The cost is 5,696,000 from Fcfa and defined as:

FOB: 3,850,000 F cfa
Shipment: 883,500 F cfa
Customs Cost: 962,500 F cfa
The cost is 5,696,000 Fcfa and defined as follows:
Costs of the equipment ex works: 3,850,000 F
Cost of transport to Ouaga: 883,500 F
Customs cost: 962,500 F



16. De-pulping

The pulp of good quality fruit is gently removed in a clean environment. Poor quality pulp is removed as well and used for biogas production.

Material used: The de-pulping is done on stainless steel tables to ensure the cleanliness of this step. The de-pulping tables are produced locally. Their cost is estimated at 869,745 Fcfa per piece or 4,348,725 Fcfa for the 5 units. The de-pulping tables will be delivered by a local company

<https://www.ipadev.net/>

17. Cooking

The nuts harvest from the two batches are cooked to contain microbiological and bacterial infections, but also to avoid any germinations.

Material used

The nuts will be cooked using stainless steel steamers. The energy in the form of steam will be produced by a gasifier and a boiler.

<https://www.ipadev.net/>

Capacity 200 kg/h

Unit Price 900 000 F cfa



18. Drying

After the cooking, the nuts are dried to make it easier to shell.

Material used

<https://www.ipadev.net/>

Solar hangar dryer

Produced by IPADDEV

Capacity 300 kg/day

Unit price: 1 100 000 Fcfa



19. Shelling

The shelling will be done with electrical shellers.

Material use

Shelling machine: Henan My Only Food Machine Co., Ltd.

Power 2,2 Kw; Capacity 500Kg/h

The unit cost is 4,571,000 Fcfa and defined as:

FOB: 1 050 000 F cfa

Shipment: 441 750 F cfa

Customs Cost: 262 500 F cfa



20. Shea almond drying

After the shelling, the almond is dried before packaging.

Material used

<https://www.ipadev.net/>

Solar hangar dryer

Produced by IPADEV

Capacity 300 kg/day

Unit price: 1,100,000 Fcfa



21. Packaging

After de drying, the almonds are pack and shipped to the butter extraction unit.

Material used

The packing equipment consists of:

- a scale

Purchase on local market

Electronic Scale

Capacity 300 kg

Unit price: 300 000 Fcfa



- Bag sewing machine.

Purchase on local market

Bag sewing machine

Unit price: 200 000 Fcfa



22. Pulp drying

Pulp drying tests and a small market study strengthen the idea, that Shea fruit dried pulp could be a valuable product on the local and international market.

Material used

<https://www.ipadev.net/>

Production capacity (Dried pulp)
300Kg/day, Thermal Energy
consumption 100 Kwth

The cost is 15,000,000 Fcfa when locally
produced. The same dryer would cost
25 000 000 Fcfa if imported.



23. Dried pulp Packaging

Dried pulp packaging machine Zhejiang Ruibao Packaging Machine Co., Ltd.

Capacity 2 cycle of 1Kg/ min, input 1Kw

The total unit cost is 1 408 000 de Fcfa and defined as:

FOB: 950 000 F cfa

Shipment: 220 500F cfa

Customs Cost: 237 500 F cfa



24. Packaging table

Material used

The packaging tables are stainless steel and produced locally. The unit cost is 869,745 Fcfa per piece or 1 739 490 Fcfa for the 2 units. The packaging tables will be delivered by a local company <https://www.ipadev.net/>

25. Energy production

While the electrical energy will be provided by a solar energy producer, the thermal energy will be produced from the production waste, such as shells, press and other processing residues. the residues will be transformed into heat by using 2 gasifiers, of 140 Kw each. The gasifiers will produce steam by heating a boiler. The steam will be used for the thermal processes.



Gasifier, boiler and burner IPADEV <https://www.ipadev.net/>, Capacity 140 Khth , Unit price 9 300 000 CFA

Table 16: Details of electriict machines proposed at collection center

Sr. No	Particular	Source of energy	Link	Model	Production capacity	Power consumption	Dimension
	De-Sheller	Electric	https://www.amisyfoodmachine.com/nut-processing-equipment/nut-shell-kernel-separator.html	FL-01	300 kg/hour	2.2 KW	(1970X920X1350) mm
	Boiler	Electric	https://www.amisyfoodmachine.com/food-machinery/jacketed-kettle.html	JGC1000	1000L/hr	2.2 Kw	(1400X1400X900) mm

b) Processing centre - detailing of technologies proposed, their requirements and costing for processing and collecting centres

Table 17: Details of technologies proposed for processing centre

Sr No	Technology	Model	Link	Model	Production capacity	Power consumption	Dimension
8.	Crusher	Electric	https://www.amisyfoodmachine.com/nut-processing-equipment/peanut-kernel-chopping-machine.html	QS-01	200kg/hr	2Kw	(2200X1050X1500)mm
9.	Roaster	Electric	https://www.amisyfoodmachine.com/nut-processing-equipment/small-peanut-roasting-machine.html	AMS-RM50	40kg/hr	0.75kw	(1200X1200X700)mm
10.	Miller	Electric	https://www.amisyfoodmachine.com/food-machinery/paste-grinding-machine.html	AMS-GM-300	1800kg/hr	90 KW	(1440X630X1420)mm
11.	Boiler	Electric	https://www.amisyfoodmachine.com/food-machinery/jacketed-kettle.html	JGC1000	1000L/hr	2.2 Kw	(1400X1400X900)mm
12.	Mixer	Electric	https://www.asengworks.com/search.html?ss=mixer		200 kg/hr	2 HP	(2200X1200X900)mm
13.	Filter	Electric	https://www.indiamart.com/proddetail/liquid-filter-machine-16131924212.html	sacIf01	40Lt/hr	2Kw	(1100X1100X3550)mm
14.	Packaging	Electric	https://www.amisyfoodmachine.com/packing-machine/automatic-liquid-packaging-machine.html	JTBZ-60C	300bags/min	1.36 Kw	(665X1000X1580)mm

4.3. Prepare a detailed presentation of resource flows (energy, water, waste, etc.), scalability, and product flows (fruit, dried nuts, butter, etc.).

a) Finalization of testing protocols – fresh fruits and kernels

The team had the discussion at one production unit and had a discussion on the production processes and developed detailed protocols for testing. A total of 14 different types of tests were identified and detailed procedure for testing was developed ([annex 1](#)). What is the purpose of the test, need for the test and test procedure – these are three-parameter used to define the test procedures. Test results for collection and processing centres as mention in the following section.

This document sets out the tests to be done in the collection centres. They are intended to give us the necessary information for the precise sizing of collection centres and ecological processing centers of Shea butter.

a. Testing procedures at collection centers

Test 1: Fruits sorting

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to assess the quantity of spoiled and good quality fruit in the batch of fruit brought back to the collection unit. This evaluation will allow us to know the quantity of good quality pulp that can be harvested and thus size the pulp recovery line. This operation will also allow us to determine the amount of pulp damaged and thus size the bio digester, which will be implanted for the processing of the damaged pulp.	• 200 kg Shea fruits minimum • Containers (4) large sizes (about 50 l) • Accurate scale • Stopwatch	• Fill a container with a defined volume, unsorted Shea fruit. • Weigh the container. • Pour the fruit into another container • Weigh the container that contained the fruit. The difference in the double weight will give us the exact weight of the fruit • The ratio of weight to volume will give us the density, which we will need for sizing the storage of fruit. • Sort the fruit into two batches. A first batch of good quality fruit whose pulp is valuable, and a second batch of fruit whose pulp is spoiled or not valuable. Determine the weight of both batches.
Collected data		
Fruits total weight	311	kg
Good quality fruit weight	262	kg
Bad quality fruit weight	48.5	kg
Sorting time	81 min	

Test 2: Fruits washing water quantity

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to assess the amount of water that is needed for washing good quality fruit.	• Fruits of a verified weight equal or more than 150Kg • Containers (4) large sizes (about 50 l) • Accurate scale • Clean water • Fruit disinfectant • Accurate scale	1. Weigh the fruit to wash. 2. Set the right amount of washing water (take tips from Felicite) 3. Document the weight of water needed to wash the specified amount of fruit.
Collected data		
Fruit weight	262	kg
Water quantity	90	Litre

Test 3: Pulp extraction

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to assess the weight of the recoverable pulp of good quality fruit. This information will inform us about the pulp ratio and	• Fruit with a determined weight equal or more than 150Kg • Kitchen knives. • Containers (4) large sizes (about 50 l)	• Weigh the fruit to be de-pulped. • De-pulp the fruit by storing the nut pulp separately. • Note the number of people who pulp fruit • Note the nut's depulping time

allow us to size the pulp processing line.	• Stopwatch • Accurate scale	• Weigh the nuts and pulp separately and document
Collected data		
Total fruit weight	311	
Weight of good quality pulp	193	
De/pulping time	6h49min	
Number of persons	4	
Weight of the nuts	82	

Test 4: Pulp drying

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to evaluate the energy needed to dry the pulp.	• Pulp at a certain weight and greater or equal to 25Kg • adapted dryer • Stopwatch • Accurate scale	• Weigh the pulp to dry. • Dry the pulp in a suitable dryer. • Note the drying time • Weigh the pulp after drying.
Collected data		
Drying time	12 h	
Pulp weight	7.5	Kg
Dried pulp weight	4.4	Kg

Test 5: Shea nuts cooking

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to assess the amount of energy needed to cook the nuts.	• Shea nuts at defined weight, but greater or equal to 75Kg • Gas cooker • Containers (4) large sizes (about 50 l) • Chronometer • Accurate scale	• Weigh the Shea nuts to be cooked • Weigh de quantity of water used • Weigh the gas cylinder before cooking • Cook Shea nuts (take advice with Felicity on cooking settings) • Weigh the gas cylinder after cooking. • Note the cooking time
Collected data		
Poids des noix à cuire	32Kg	
Water used	10 Litre	
Water steamed	4 Litre	

Test 6: Shea nuts drying

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to assess the drying speed of the Shea nuts and, if possible, the amount of energy needed to dry the nuts.	• A minimum of 75 kg of fresh Shea nuts • A dryer • Containers (4) large sizes (about 50 l) • Accurate scale	1. Weigh the nuts to dry. 2. Dry the nuts until they reach the drying level requires for crushing. Take advice with Félicite. 3. Weigh the nuts after drying 4. The difference in weight will inform us about the amount of water extracted as well as the amount of energy used

Collected data		
Nuts weight	32.6	Kg
Dried nuts weight	31.6	Kg
Drying time	26	hours

Test 7: Shelling

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to determine the shelling speed, the mass of the recovered hulls, and the amount of energy used for shelling.	- A minimum of 75 kg of dried Shea nuts - A de sheller (talk to IRSAT) - Containers (4) large sizes (about 50 l) - Accurate scale - Stopwatch	1. Weigh the nuts to be shelled. 2. Document engine power. 3. Time the actual shelling time 4. shell the nuts. 5. Weigh the almonds after shelling. 6. Weigh the hulls, and document.
Collected data		
Weight of nuts to be shelled	31.6	Kg
Weight of shelled nuts	24	Kg
Engine power	manual	
Shelling time	2h 30 min	

Test 8: Almond drying test

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to assess the drying speed of the Shea almond and, if possible, the amount of energy needed to dry the almond.	- A minimum of 50 kg of fresh Shea nuts - A dryer - Containers (4) large sizes (about 50 l) - Accurate scale	- Weigh the nuts to dry. - Dry the nuts until they reach the drying level requires for crushing. Take advice with Félicité. - Weigh the nuts after drying - The difference in weight will inform us about the amount of water extracted as well as the amount of energy used
Collected data		
Almond weight	24	Kg
Dried almond weight	23.8	Kg
Drying time	18	high air humidity
Weight difference	0.2	Kg

b. Testing procedures at processing centers

Test 1: Shea Almonds Weighing

Purpose of the test	Needs for the test	Test procedures
Determine the weight of the almonds to be treated for the test.	- A minimum of 150 kg of dried Shea almonds - Containers (4) (about 50 l) - Accurate scale	- Weigh the dried almonds in their containers. - Weigh the empty container of the Shea almonds.

Collected Data		
Total weight	200Kg	
Container weight	1,5	
Container volume	50l	

Test 2: Almond sorting

Purpose of the test	Needs for the test	Test procedures
Determine the amount of bad quality almonds that could be used as energy.	- A minimum of 150 kg of Shea almond - Containers (4) large sizes (about 50 l) - Accurate scale	- Separate poor-quality almonds from good quality almonds by manual sorting. - Weigh both batches accurately and document.

Collected Data

Good quality almonds weight	198.5	Kg
Poor quality almonds weight	1.5	Kg

Test 3 : Shea almond washing

Purpose of the test	Needs for the test	Test procedures
- Eliminate the impurities of the almonds so that they do not affect the final quality of the Shea butter. - Determine how much water is needed at this stage of the transformation.	- A minimum of 100 kg of dried and sorted Shea almonds - Containers (4) large sizes (about 50 l) - Accurate scale	- Weigh the almonds to be washed. - Determine the volume of all water used to wash the almonds. - Wash the almonds according to the usual procedure a serious Shea butter. (Take advice with Félicite).

Collected Data

Water used	900 litre clean water	658 litre wastewater collected after washing
Almond weight	200Kg	

Test 4: Shea almond drying

Purpose of the test	Needs for the test	Test procedures
- Dry the almonds to facilitate the next step, which is crushing. - Determine how much energy is needed for drying.	- A minimum of 200 kg of fresh Shea nuts - A dryer or direct sun drying if possible - Containers (4) large sizes (about 50 l) - Accurate scale	- Weigh the almonds to dry. - Dry the almonds until they reach the drying level needed for crushing. (Take advice with Félicite). - Weigh the almonds after drying - The difference in weight will inform us about the amount of water extracted as well as the amount of energy used

Collected Data

Almond weight before drying	220.5	
Almond weight after drying	203.5	

Test 5: Shea almond crushing

Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to determine the amount of energy needed to crush Shea almonds.	- A minimum of 100 kg of dried Shea almonds. - A crusher.	- Weigh the crushed almonds. - Note the power of the crusher's engine.

	• A stopwatch • Containers (4) large sizes (about 50 l) • Accurate scale	• Take the crushing time. Start the stopwatch at the beginning of the crushing and cut as soon as the crushing is over. (Take advice with Félicité). • Documenting both times
Collected Data		
Engine power	11 kw	
Crushing time	2min15	
Crushed almond weight	203.5kg	
Test 6: Shea almond roasting		
Purpose of the test	Needs for the test	Test procedures
Determine how much energy the roasting requires.	• A minimum of 75 kg of crushed Shea almonds • A roaster • Wood • Containers (4) large sizes (about 50 l) • Accurate scale	• Weigh the almonds to be roasted. • Weigh the wood before roasting • Time the roasting. • Turn off the fire immediately after roasting. • Dry wood and residual charcoal • Weigh the residual wood. • Weigh the residual coal.
Collected Data		
Almond Weight	203.5Kg à 150 Kg	
Wood weight before roasting	43.5	
Crushed almond weight	203.5kg	
Dry wood weight after roasting	36.5	

Test 7 : Shea almond milling		
Purpose of the test	Needs for the test	Test procedures
Determine how much energy to use for grinding.	• A minimum of 75 kg of roasted and crushed almonds. • A mill • Containers (4) large sizes (about 50 l) • Accurate scale	1. Weigh the crushed almonds to grind. 2. Note the power of the mill engine. 3. Time the actual grinding time. Start the timer at the beginning of the grind and cut as soon as the actual grinding time is over. (Take advice with Félicité). 4. Documenting both times
Collected Data		
Milling engine power	11kw	
Milling time	45 min	
Crushed almond weight	150kg	After milling 145 Kg
Test 8 : Churning		
Purpose of the test	Needs for the test	Test procedures
Determine how much energy to use for churning. Determine how much water to use for churning.	• A minimum of 50 kg of Shea almond paste • A churn	• Weigh the churning dough. • Note the power of the churn's engine. • Time the actual time of churning. Start the time at the beginning of the

Determine the weight of usable churning residues such as Energy	- Containers (4) large sizes (about 50 l) - Accurate scale	churning and cut the time as soon as the actual churning is over. (Take advice with Félicité). - Documenting both times - Determine exactly how much water is used for churning. - Decant the churning residue. - Collect churning residue. - Dry the churning residue. - Weighing the churning residue
Collected Data		
Paste weight	144,5	
Churn engine power	11 kw	
Churning time	35min	
Test 9 : Butter thermal dewatering		
Purpose of the test	Needs for the test	Test procedures
Determine how much energy is needed to cook butter	- Emulsion recovered from the previous process - Containers (4) large sizes (about 50 l) - Precise balance - Accurate scale	- Weigh the collected emulsion. - weight the raw butter after cooking
Collected data		
Emulsion weight	167 Kg	Raw butter after cooking 72.5
Amount of water evaporated	94.5	
Test 10 : Butter decantation		
Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to determine how much fat residue can be used as an energy source	- A minimum of 25 kg of raw butter - Containers (2) large sizes (about 10l) - Accurate scale	- Weigh the raw butter (already done in the previous step). - Proceed with decanting. (Take advice with Félicité). - Weigh the decanted butter - Weigh the fatty residues
Collected data		
Raw butter weight	72.5 kg	
Collected butter weight	50 kg	
Fatty residues collected	22.5 kg	
Test 11 : Butter filtration		
Purpose of the test	Needs for the test	Test procedures
Determine the amount of fatty residue after filtration	- A minimum of 20 kg of decanted butter - A filter - Containers (3) of medium sizes (about 10 l) - Accurate scale	- Weigh the decanted butter. - Strain the decanted butter. - Weigh the refined butter - Weigh the fatty residues

Collected data		
Decanted Butter weight	50 Kg	
Refined butter weight	49 Kg	
Fatty residues weight	1 Kg	

b) Resource flow diagram at collection and processing centres

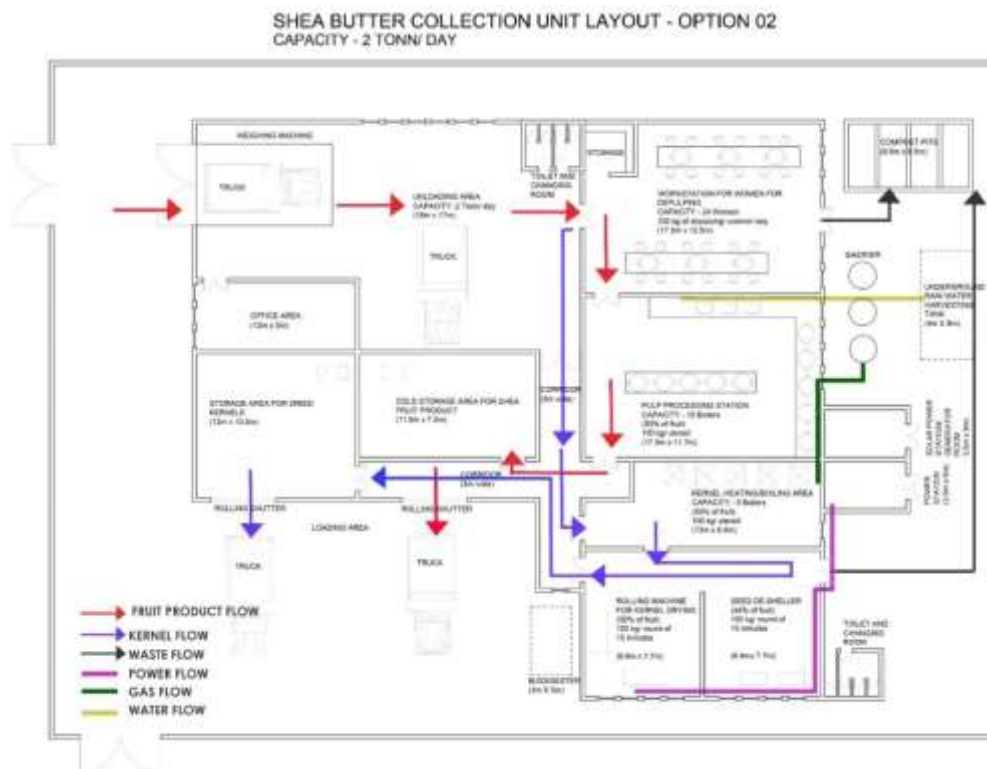


Figure 8: Resource flow of collection center

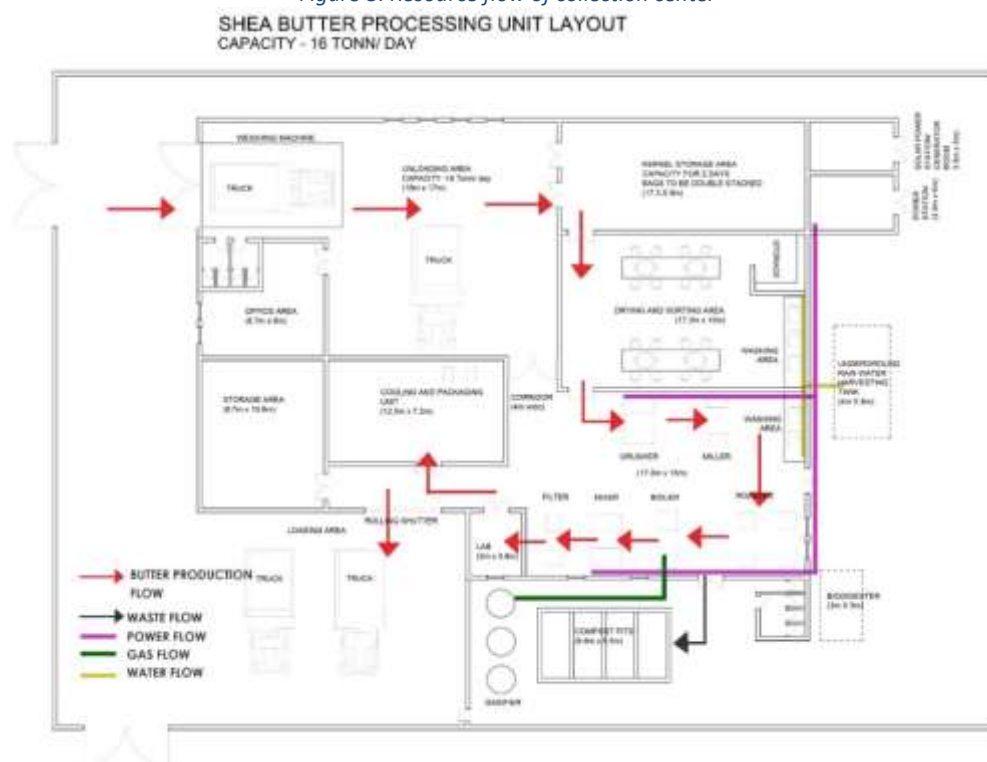


Figure 9: Resource flow of processing center

c) Process flow at collection and processing centres

Figure 10: Collection center - Material flow

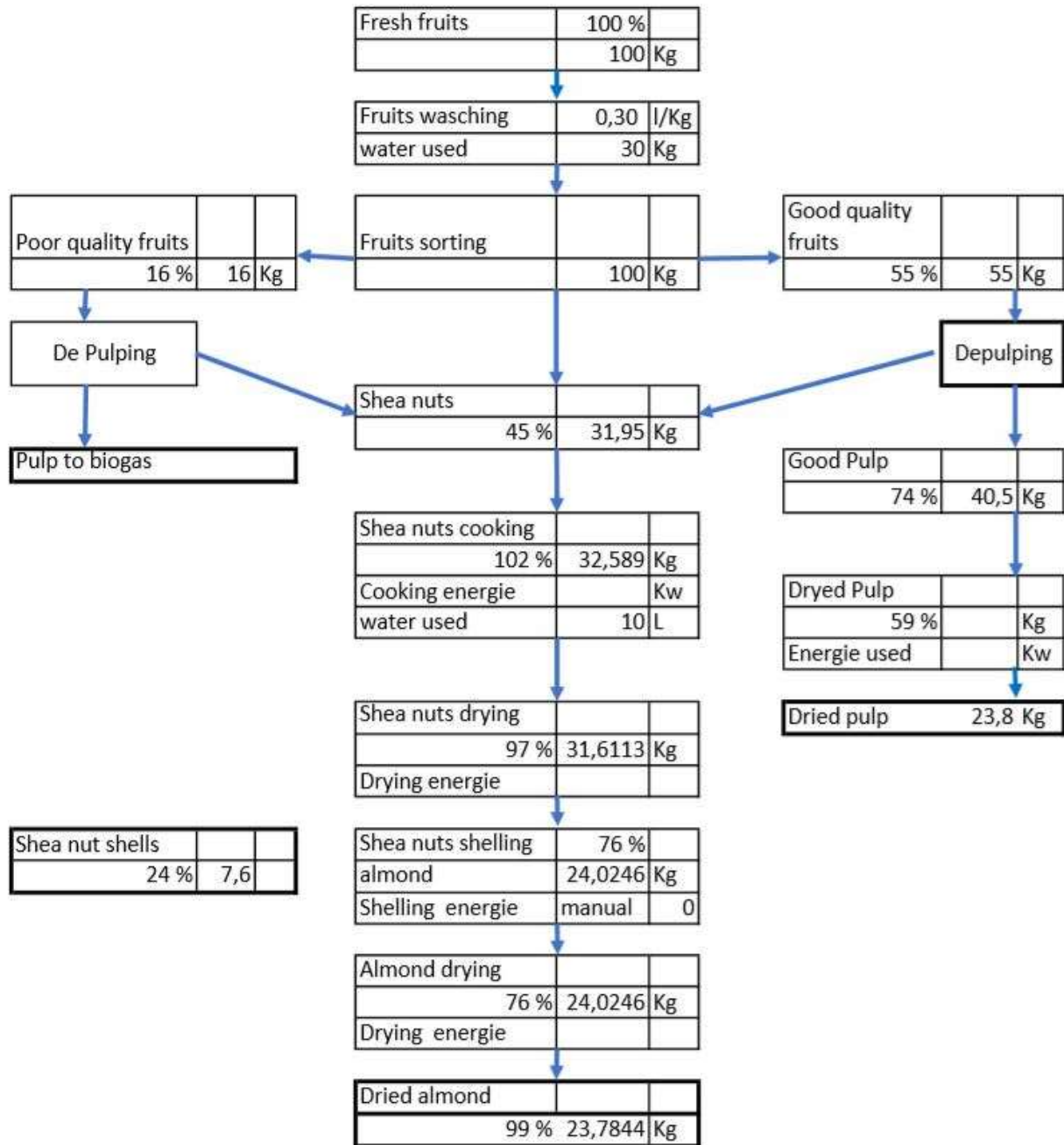


Figure 11: Processing Centre- material flow

Processing Method: Press					
				Dried almond	100%
					200 Kg
1	Poor quality almond	8 Kg		Sorting	
	Energy content	33.6 Kwh		Good almond	166.4 Kg
2				Dry Cleaning	0.00095 Kwh/Kg
				Almond	192 Kg
				Engine power	1.1 Kw
				Cleaning time	10 min
3				Crushing	0.0232 Kwh/Kg
				Dried Almond	189 Kg
				Crushed almond	188 Kg
				Engine power	5.5 Kw
				Crushing time	47.8 min
4				Roasting	0.18 Kwh/kg
				Crushed almond	188 Kg
				Roasted almond	185 Kg
				Engine power	42 Kw
				Roasting time	47.6 min
5	Press Cake	111 Kg		Pressing	0.089 Kwh/kg
	Energy content	466.2 Kwh		Roasted almond	185 Kg
				Raw butter	74 Kg
				Engine Power	15 Kw
				Pressing time	66 min
6	Fat residues	1 Kg		Filtering	0.012 Kwh/Kg
	Energy content	4.2 Kwh		Raw butter	74 Kg
				Refined Butter	73 Kg
				Engine power	2.2 Kw
				Filtering time	24.8 min
				Yield	40%
7				Packaging	0.0069 Kwh/kg
				Refined butter	73 Kg
				Engine power	2.2 Kw
				Packaging time	13.65 min
	Potential residu energy	504 Kwh			
				Electric energy/Kg almond	0.13 Kwh
				Electrical energy cost/Kg	26.49 Cfa
				Total electrical energy for	26.49 Kwh
				Thermal energie per Kg	0.18 Kwh
				Thermal energie for 200Kg	35.45 Kwh
				No water is used in the	
				Water cost per Kg	1 Cfa

with 1 kwh cost 200 cfa

From processing waste

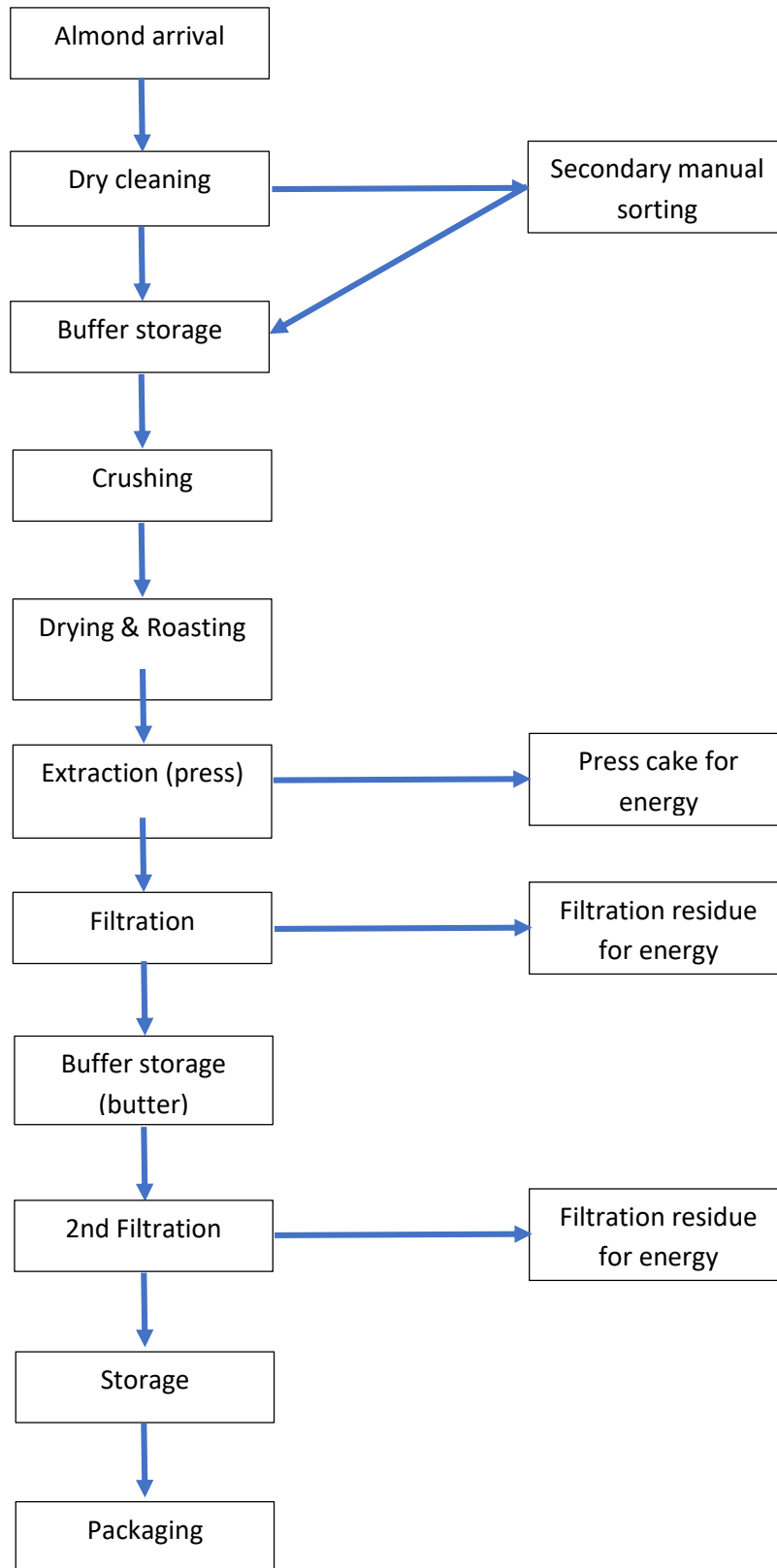
From processing waste

Figure 12: Processing method: Churning

Processing Method: Churning									
					Dried almond	100%			
						200 Kg			
1	Poor quality almond	8 Kg			Sorting				
	Energy content	33.6 Kwh			Good almond	166.4 Kg			
2	Waste water	748			Washing	0.0050 Kwh/kg		Water/h	
		4.50 L/Kg			Water ratio	4.4071 l/Kg			2000 l/h
					engine power	2.75 Kw			
					Washing energy	1.01 Kwh			
					washing time	22 min			
					Water	733.33 l			
3					Drying	208.0 Kg			
					Dried almond	189.0 Kg			
					Dried away water	19.0 Kg			
					Drying energy/kg	0.127 Kwh			
4					Crushing	0.0232 Kwh/Kg			
					Dried Almond	189.0 Kg			
					Crushed Almond	188 Kg			
					Engine power	5.5 Kw			
					Crushing time	47.8 min			
5					Roasting	0.16 Kwh/kg			
					Crushed almond	188 Kg			
					Roasted almond	185 Kg			
					engine power	42 Kwh			
					Roasting time	47.6 min			
6					Milling	0.02418 Kwh/kg			
					Roasted almond	185 Kg			
					Braun paste	183.5 Kg			
					Engine power	2.2 Kw			
					Milling time	121 min			
7	Waste after drying	51 Kg			Schurning	0.018 Kwh/Kg			
	energy content	212 Kwh			Braun paste	183.500 Kg			
					Emulsion	133 Kg			
					Engine power	3 Kw			
					Churning energy	3.5 Kwh			
					Churning time	70 min			
8	Fat Waste	61 Kg			Emulsion cooking	0.945 Kwh/kg			
	energy content	254 Kwh			Cooking energy/Kg emulsion	1.421 Kwh/kg			
					Emulsion	133 Kg			
					Raw butter	72.5 Kg			
					Energie used	188.993 Kwh			
9	Fat Waste	19.5 Kg			Decantation				
	Energy content	82 Kwh			Raw Butter	72.5 Kg			
					Butter after decantation	53 Kg			
10	Fat Waste	9 Kg			Filtration	0.004 Kwh/Kg			
	Energy content	37.8 Kwh			Butter after decantation	53 Kg			
					Refined Butter	44 Kg			
					Engine power	2.2 Kw			
					Filtration time	20 min			
					Filtration energy	0.73333 Kwh			
	Potential residue energy	620 Kwh							
					Electric energy/Kg almond	0.0736 Kwh			
					Electrical energy cost	14.7140 Cfa			with 1 kwh cost 200 cfa
					Total electrical energy for 200Kg	14.7140 Kwh			
					Thermal energie per Kg almond	1.24 Kwh			
					Thermal energy for 200Kg	247.36 Kwh			From processing waste
					Water used/Kg	4.41 l			
					Water cost per m2	663 Cfa			with 1 m3 cost 663 cfa
					Water cost per Kg almond	2.92188 Cfa			

Figure 13: Processing diagram for a large commercial Shea butter production center

(The yield at olvea is 42%)



4.4. Feasibility study of production processes at processing centres

▪ Implementation of a CO2 free Shea butter processing company

The aim of this study is to develop and implement an ecological way of Shea fruits processing based on mature operations and methods.

If until now, the main interest of the players in the Shea sector was limited to the collection and processing of Shea almonds, our process will include the collection and processing of the Shea fruit as well. enhancement of all Shea fruit. To do this, the processing concept will be based on two centers.

▪ Collection Centers

The first center, called the collection center will specialize in collecting and processing the fruit into dried fruit pulp and dried almonds. In our collection centers, the fruit will be sorted. The nuts of the poor-quality fruits will be removed and processed to dried almonds, where the pulp will be transformed into biogas. the biogas slurry will be used as fertilizer.

The pulp of the good quality fruits will be dried to a snack. The nut will be turned into dried almond for further processing in the Butter extraction unit.

The Shea nut will be boiled, dried, shelled, and dried again. The good quality almonds harvested will be packaged and transferred to an extraction center, dedicated to the grinding of Shea butter.

After deshelling, the shells will be used as energy source and produce the processing heat through a gasifier.

The Shea nut will be boiled, dried, shelled, and dried again. The good quality almonds will be packaged and transferred to the processing centers to extract the butter.

4.5. Analytical report of Shea fruit tests

a) Scope

The objective of tests is to determine in vivo the percentage of the macro elements of the Shea fruits which could be processed to induce economic value as products for consumption or sources of energy. The initial assumption classified the macro elements in two categories:

A) Tradable parts:

- Pulp
- Almonds

B) Energy sources

- Skin
- Shells

The value chain of the Shea fruits should be extended to the potential of processing the Shea fruit pulp as foodstuff, we therefore included basic tests relative to processing the fruits

b) Background and References on Processing and utilization for the Shea Fruits

Figure 14: Shea Butter products and biproducts



Normally, after harvesting of Shea fruits, they are de-pulped by hand or the whole fruit is fed to animals before processing the nuts into butter. The animals eat the pulp and leave the nut which is not edible to them. De-pulping is preceded by initial fermentation or boiling of the fruits. However, to process the Shea pulp, fresh and wholesome fruits are required.

Utilization of Shea fruit pulp, Collection, handling and processing of Shea fruits Methodology³

Fresh and wholesome Shea fruits were collected from CRIG experimental plots at Bole and were washed and frozen to make removal of the pulp easier. The pulp of the frozen fruits was removed by scraping with a knife. It was then blended into slurry. Proximate analysis was conducted on the fresh pulp by determining the pH using a Mettler Toledo GmbH pH meter, total sugar concentration using the Dubois et al. (1956) method and protein content using the Kjeldahl's method (AOAC 1990). Total phenolics were assessed by the Folin-Ciocalteu assay (Asami et al. 2003). Minerals such as potassium, sodium, magnesium, iron and zinc were determined by Atomic Absorption spectroscopy (Unicam 929 AAS model, UK). Phosphorous was determined by the colorimetric method using the Cecil CE 3041 spectrophotometer.

Preparation of Shea jam⁴

The blended pulp was sieved through plastic mesh to get smooth slurry. The sieved pulp was placed on a heater to boil. Sugar was added to the pulp and allowed to boil until about two-thirds of the water in the mixture had been evaporated. The proportion of Shea pulp to sugar was 7:3 respectively. Pectin prepared locally from lemons was added at 0.5% concentration to allow the mixture to set. Sodium metabisulphite, ascorbic and citric acids were also added at 0.01% concentration each. The set jam was then allowed to cool to about 85° C and filled into glass jars, which had been pre-sterilized in hot water, and capped immediately.

Production of Shea wine⁵

Water was added to scraped Shea pulp in a ratio of 3:1 and boiled to prevent the growth of undesirable microorganisms in the mixture. The must was allowed to cool and then poured into plastic barrels for fermentation. Sufficient sugar was added to the must to bring it to a specific gravity (SG) of 1.090. Wine yeast (Bourgovin RC 212) was activated according to the manufacturer's instruction and added to the must. The barrels were then covered for fermentation of the must. During the fermentation period, the specific gravity, temperature and pH of the fermenting must be monitored. Potential alcohol content of the must was determined using the formula below:

$\% \text{alcohol} = (\text{SGi} - \text{SGf}) / 0.00736$, where SGi is the initial SG and SGf is the final SG.

At the end of the fermentation, the wine produced was decanted from the macerated fruit tissues,

³ Ref 1: Utilization of Shea fruit to enhance food security and reduce poverty in Ghana

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¹New Product Development Unit, Cocoa Research Institute of Ghana, P.O. Box 8, Akim-Tafo, Ghana ²Social Science and Statistics Unit, Cocoa Research Institute of Ghana, P.O. Box 8, Akim-Tafo, Ghana

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dead yeast cells and other products into new plastic barrels. The barrels were covered with lids which had air-traps fitted in them. The young wine was then allowed to mature for two years

c) Shea pulp and derive foodstuff specifications

The oven dried Shea (*vitellaria paradoxa*) fruit pulp were analysed for its nutritional composition and some elemental constituents. The result showed mean value as moisture 4.58%, ash 8.95%, crude fat 1.35%, crude protein 3.5%, crude fibre 9.6% and carbohydrate 72.02%. Results of mineral analysis showed that calcium had the highest concentration of all the elements analysed and were found to be of the order: Ca > Ni > Co > Mg > Pb > Mn > Fe > Cd. These values suggest that the fruit can be used as nutrient supplement and as a source of micro and macro elements to the body.⁶

Composition of Shea pulp, kernels, and butter⁷

- **Macronutrients**
- **Minerals**
- **Vitamins**
- **Amino acids**

	Pulp			Reference
	Min.	Average	Max.	
Macronutrients				
Moisture (%)	67.0	74.2	80.3	(Maranz et al., 2004a; Mbaiguinam et al., 2007)
Energy (kcal/100 g dw)		179.5		(Ugese et al., 2008a)
Carbohydrates (g/100 g dw)	8.1	22.6	37.2	(Mbaiguinam et al., 2007; Ugese et al., 2008a)
Crude protein (g/100 g dw)	4.2	5.2	5.6	(Maranz et al., 2004a; Mbaiguinam et al., 2007; Ugese et al., 2008a)
Crude lipid (g/100 g dw)		1.3		(Ugese et al., 2008a)
Crude fiber (g/100 g dw)		42.2		(Ugese et al., 2008a)
Ash (g/100 g dw)	4.7	5.1	5.4	(Mbaiguinam et al., 2007; Ugese et al., 2008a)

	Pulp			Reference
	Min.	Average	Max.	
Mineral (mg/100 g dw)				
Ca	2.5	117.3	426.0	(Eromosele et al., 1991; Maranz et al., 2004a; Mbaiguinam et al., 2007; Ugese et al., 2008b)
Cu	0	0.1	1.1	(Eromosele et al., 1991; Maranz et al., 2004a)
Fe	0.4	8.5	16.0	(Eromosele et al., 1991; Maranz et al., 2004a; Mbaiguinam et al., 2007; Ugese et al., 2008b)
K	21.7	830.3	1686.0	(Maranz et al., 2004a; Mbaiguinam et al., 2007; Ugese et al., 2008b)
Mg	11.1	57.2	129.0	(Eromosele et al., 1991; Maranz et al., 2004a; Mbaiguinam et al., 2007; Ugese et al., 2008b)
Mn	0.3	0.6	0.9	(Eromosele et al., 1991; Maranz et al., 2004a)
P	1.0	39.8	71.4	(Eromosele et al., 1991; Maranz et al., 2004a; Mbaiguinam et al., 2007; Ugese et al., 2008b)
Na		19.3		(Ugese et al., 2008b)
Zn	0.5	2.1	4.0	(Eromosele et al., 1991; Maranz et al., 2004a; Ugese et al., 2008b)
Vitamin (mg/100 g)				
B		7.0		(Maranz et al., 2004a)
C		196.1		(Eromosele et al., 1991)

⁶ Ref 2: Nutritional and some elemental composition of Shea (*vitellaria paradoxa*) fruit pulp

O. C. Aguzue*, Akanji F. T., Tafida M. A. and Kamal M. J.

Chemistry Advanced Laboratory, Sheda Science and Technology Complex, Garki, Abuja

⁷ Ref 2: Critical Reviews in Food Science and Nutrition

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Nutritional Composition of Shea Products and Chemical Properties of Shea Butter: A Review

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Table 18: Chemical analysis of fresh Shea pulp and jam

Parameter	Pulp	Jam
Phenolics (mg/100 g)	226.25 ± 1.56	4.53 ± 0.28
Fat (%)	0.96 ± 0.06	–
Protein (%)	1.09 ± 0.01	3.80 ± 0.05
Sugar (mg/g)	74.60 ± 4.22	300.20 ± 8.61
K (mg/kg)	4.16	1.46
Mg (mg/kg)	3.43	0.11
Ca (mg/kg)	3.79	0.29
Mn (µg/kg)	12.50	12.02
Cu (µg/kg)	85.70	0.19
Fe (µg/kg)	27.65	2.82

Table 19: Chemical analysis of young wine and lees after racking

Parameter	Young wine	Mature wine
Phenolics (mg/100 g)	5097.06 ± 0.01	40.40 ± 0.31
Protein (%)	1.26 ± 0.01	0.92 ± 0.02
Total acidity (%)	0.35 ± 0.01	0.71 ± 0.01
Alcohol (%)	12.65 ± 0.93	13.69 ± 0.15
pH	3.92 ± 0.08	3.79 ± 0.02
Specific gravity	0.99 ± 0.00	0.99 ± 0.00
K (mg/Kg)	24.12	1,431.40
Mg (mg/Kg)	3.43	114.30
Ca (mg/Kg)	2.12	135.14
Mn (µg/kg)	26.60	110.02
Cu (µg/kg)	62.50	–
Fe (µg/kg)	9.90	10.00

d) Tests of Shea Fruits

The tests were conducted using fruits collected from the village in the center of Burkina around Ouagadougou. The objective is to evaluate the percentage of macro elements in the Shea fruit to determine the volume available for processing based on the daily collection. The first test was conducted with fruits collected at the end of the Shea season in 2019.

We conducted a second test recently with fruit available at the beginning of the season 2020 to assess the variance.

A drying test was also conducted to assess the aspect and taste of the resulting foodstuff.



Figure 15: Shea fruits testing



Figure 16: Shea fruit testing procedure

e) Results

The results were globally consistent with the literature and assumptions, confirming the following respective weight for the elements of interest:

- d) Fruit de-pulping
 - Pulp: 65%
 - Nut: 35%
- e) Nut Shelling
 - Shell: 29 %
 - Almonds: 71%
- f) Pulp Drying
 - Dried Pulp: 25%
 - Water: 75%

f) Analysis

The processing of the whole fruit can provide significant benefits for food and energy security. The fruit can be processed into valuable foodstuff which can expand the value chain and significantly increase the return to the stakeholders. Some Foodstuff which could be obtained from the Shea pulp do not require sophisticated equipment or processes.

Significant quantity of biomass can be collected from the fruit process to contribute to the decarbonisation of the value chain. The pulp wastes which can be estimated to 20% in fruit processing can produce large amount of biogas.

The shells, representing 30% of the dried nut can be processed in a Syngas generator to generate steam for the process.

g) Economic Benefits: Economic feasibility studies

The estimated production costs of all the products developed is shown in Table 9. The Shea wine showed the highest profit margin of over 300% and this is because very few inputs were needed, and its production required less labour. The Shea jam, on the other hand, had the lowest profit margin of 22.8%. The profit margins of the Shea butter-based products ranged from 47.0 to 62.2%.

Table 9: Estimated production cost analysis for producing 100 kg of each product.

Product	Production cost (\$ ^a)	Labour cost (\$)	Revenue (\$)	%Profit
Shea jam	277.78	236.61	666.67	22.8
Shea wine	112.50	121.63	1,000.00	327.1
Shea butter lotion	249.63	134.16	625.00	62.8
Shea butter pomade	336.95	83.85	659.21	56.7
Shea butter alata samina	307.70	103.33	666.67	62.2
Shea butter shower gel	161.19	105.51	400.00	50.0
Shea butter toilet soap	353.84	101.67	669.64	47.0



Figure 17: Shea fruit and seed weighing during testing procedure

4.6. Detail operational and maintenance requirements and associated processes

▪ Feasibility study: Implementation of a CO₂ free Shea butter processing company

The aim of this study is to develop and implement an ecological way of Shea fruits processing based on mature operations and methods.

If until now, the main interest of the players in the Shea sector was limited to the collection and processing of Shea almonds, our process will include the collection and processing of the Shea fruit as well. Enhancement of all Shea fruit. To do this, the processing concept will be based on two centers.

A. Collection Centers

The first center, called the collection center will specialize in collecting and processing the fruit into dried fruit pulp and dried almonds. In our collection centers, the fruit will be sorted. The nuts of the poor-quality fruits will be removed and processed to dried almonds, where the pulp will be transformed into biogas. the biogas slurry will be used as fertilizer.

The pulp of the good quality fruits will be dried to a snack. The nut will be turned into dried almond for further processing in the Butter extraction unit.

The Shea nut will be boiled, dried, shelled, and dried again. The good quality almonds harvested will be packaged and transferred to an extraction center, dedicated to the grinding of Shea butter.

After deshelling, the shells will be used as energy source and produce the processing heat through a gasifier.

The Shea nut will be boiled, dried, shelled, and dried again. The good quality almonds will be packaged and transferred to the processing centers to extract the butter.

a) Capex of collection centre

Table 20:Details of Capex of collection centre

Designation		Unit cost	Total cost (including shipment and customs)
Construction and land development			
Land	1	5 000 000	5 000 000
Landscaping	1	1 000 000	1 000 000
Building	1	17 000 000	17 000 000
Storage	1	7 000 000	7 000 000
Water system	1	7 000 000	7 000 000
Electricity	1	3 000 000	3 000 000
Waste management including biogas	1	6 000 000	6 000 000
Fruit bubble washing machine Contact www.pasenmachinery.com	1	3 850 000	5 696 000
Depulping tables	5	869 745	4 348 725
Steamer	5	900 000	4 500 000
Hangar dryer	4	1 100 000	4 400 000
Shelling machine, Henan My Only Food Machine Co., Ltd.	1	1 050 000	2 196 000
Almond Packaging	2	500 000	1 000 000
Pulp dryer	1	15 000 000	15 000 000
Thermal installation	2	9 300 000	18 600 000
Dried pulp packaging machine Zhejiang Ruibao Packaging Machine Co., Ltd.	2	950 000	2 816 750
Dried pulp packaging table	2	870 000	1 740 000
Production tools	1	4 000 000	4 000 000
Transport Installation and Commissioning	25% of equipment supply cost		
Vehicules	1	5 800 000	5 800 000
Total			116 097 475

c) Yearly turnover of collection centers

Table 21: Details of turnover of collection center for next five years

Collection centres	First Year	Second Year	Third Year	Fourth Year	Fifth Ydear
Yearly processed fruits	700 000	2 100 000	2 100 000	2 100 000	2 100 000
Yearly almond production	283 500	850 500	850 500	850 500	850 500
Total energy cost	15,036,840	37,592,100	37,592,100	37,592,100	37,592,100
Total water cost	42,525	127,575	127,575	127,575	127,575
Total disinfectant cost	283,500	850,500	850,500	850,500	850,500
Total transport cost	2,835,000	8,505,000	8,505,000	8,505,000	8,505,000
Total production cost	3,161,025	9,483,075	9,483,075	9,483,075	9,483,075
Fruit Pulp					
Yearly processed fruit pulp	283 500	850 500	850 500	850 500	850 500
Monthly dried fruit pulp production	70 875	212 625	212 625	212 625	212 625
Total fruit pulp packaging cost	5 670 000	17 010 000	17 010 000	17 010 000	17 010 000
Fruit pulp production cost	5 670 000	17 010 000	17 010 000	17 010 000	17 010 000
Fruit pulp unit price cfa/Kg	1 000	1 000	1 000	1 000	1 000
Total variable cost	8 831 025	26 493 075	26 493 075	26 493 075	26 493 075
Total cost	43 154 325	60 721 223	60 721 223	60 721 223	60 721 223
Total turnover	70 875 000	212 625 000	212 625 000	212 625 000	212 625 000

d) Conclusion

Even though the exploitation of the fruit can contribute to an economic income for women in the Shea sector, the valorisation of the fruit flesh has been very marginal. Through our study we can see that the major reason is the scale at which the collection is made. If overall the collection is done on a large scale, in the field it is made by small groups and in small quantities. The amount collected by the small group is so small that it is not economically viable to invest in equipment for its transformation. The establishment of collection centres increases the amount to be transform and justifies high investments. Furthermore, since the Shea fruit is seasonal, the equipment installed can, in the off-season, be used for the processing of other local products.

B. Processing centres

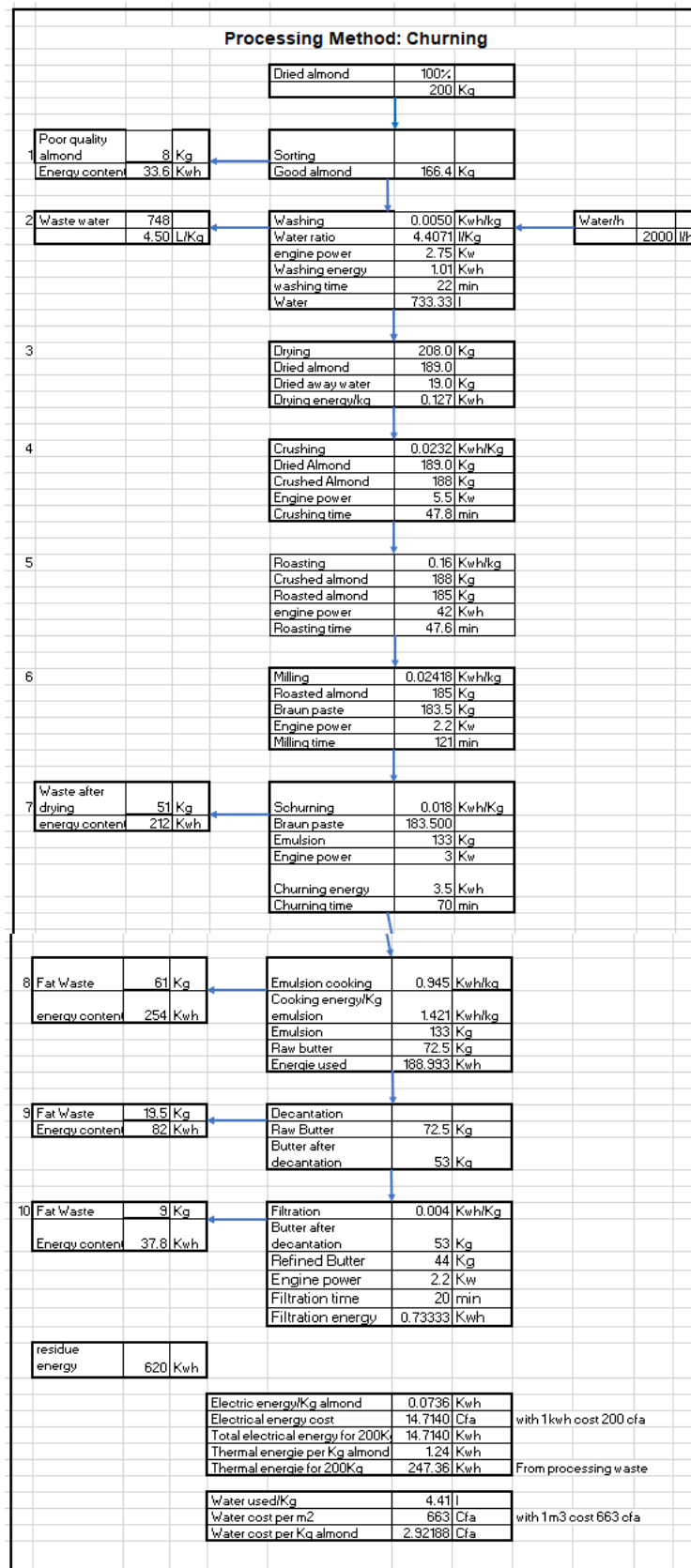
Unlike the fruit flesh of the Shea fruit, Shea butter has been extracted from the Shea nut since ever. Many technologies have been developed for the extraction. Each technology is adapted to different condition and extraction case.

In this study we will analyse the two most common technologies and choose the best adapted one to the project we are developing.

a) The churning

Shea butter extraction by churning is the oldest extraction method. It is for sure the most common method used by the women groups. It has today been improved and mechanized. the churning method is today less time and muscular effort demanding, but it is still room for improvement, particularly in the field of energy consumption, water consumption, and waste reduction.

Figure 18: Details of Churning processing method



b) The press Method

Shea butter extraction using the press method is mostly used by big companies. Despite their fair yield the press technology did not find its way to the “groupement de productrices”. High investment cost, high technical skill for operation and maintenance, availability of energy. The reasons are as follows:

Figure 19: Details of press method of Shea butter processing

Processing Method: Press				
			Dried almond	100% 200 Kg
Poor quality almond	8 Kg		Sorting	
Energy content	33.6 Kwh		Good almond	166.4 Kg
			Dry Cleaning	0.00095 Kwh/Kg
			Almond	192 Kg
			Engine power	1.1 Kw
			Cleaning time	10 min
			Crushing	0.0232 Kwh/Kg
			Dried Almond	189 Kg
			Crushed almond	188 Kg
			Engine power	5.5 Kw
			Crushing time	47.8 min
			Roasting	0.18 Kwh/kg
			Crushed almond	188 Kg
			Roasted almond	185 Kg
			Engine power	42 Kw
			Roasting time	47.6 min
Press Cake	111 Kg		Pressing	0.089 Kwh/kg
Energy content	466.2 Kwh		Roasted almond	185 Kg
			Raw butter	74 Kg
			Engine Power	15 Kw
			Pressing time	66 min
Fat residues	1 Kg		Filtering	0.012 Kwh/Kg
Energy content	4.2 Kwh		Raw butter	74 Kg
			Refined Butter	73 Kg
			Engine power	2.2 Kw
			Filtering time	24.8 min
			Yield	40%
			Packaging	0.0069 Kwh/kg
			Refined butter	73 Kg
			Engine power	2.2 Kw
			Packaging time	13.65 min
Potential residu energy	504 Kwh			
			Electric energy/Kg almond	0.13 Kwh
			Electrical energy cost/Kg	26.49 Cfa
			Total electrical energy for 200Kg	26.49 Kwh
			Thermal energie per Kg almond	0.18 Kwh
			Thermal energie for 200Kg	35.45 Kwh
			No water is used in the	
			Water cost per Kg	1 Cfa
				with 1 kwh cost 200 cfa
				From processing waste
				From processing waste

c) Comparative analysis of two methods

Table 22: Comparative analyse

Criteria	Churning	Press
Steps	The churning process has more processing steps, which means a longer process, and more room for processing mistakes and loss of quality	The press process is more concise. Better for quality control. Low humane errors potential
Cost	The capex is 123 437 000 F Cfa, the costs are 10% lower than the press method	The capex is 136 297 500 F Cfa, the costs are Higer than the press method
Yield	The churning yield is around 22%	The press yield is around 33%
Employment	The churning will create around 20% more employment than the press method	it creates less employment but is more rentable than the churning process
Maintenance	The maintenance issue may have the same rating here: the churning process has more steps and will more maintenance need in term of quantity	the press method has less steps, less machine, but higher demand of maintenance in term of quality (need more qualified technicians)
Energy	The churning process will consume more energy in form of heat than the press process, but will need less energy in form of electricity. We should pay attention to Greenhouse gas emission. the two processes are autonomous in thermal energy. their residues may be enough all the thermal energy they need.	
Water	The churning process is consuming lot of water	The press method consumes very few water. Only the cleaning water
Environment	The wastewater produced in the process must be disposed somewhere! and can lead to environmental problem. Also, the mix of wastewater and energy rich residues is often to be dried in open air, leading to dirty surrounding	the press releases only dried press cake easy to dispose and ready to use. the environment can stay clean
Adaptation	the churning is traditionally implemented and adapted to the women	the press method is new, but simple. It may adapt easily

d) Capex of processing centre

Table 23: Details of processing centre

Designation		Unit cost	Total cost
Land	1	10,000,000	10,000,000
Landscaping	1	1,000,000	1,000,000
Building	1	25,000,000	25,000,000
Storage	1	7,000,000	7,000,000
Water System	1	7,000,000	7,000,000
Electricity	1	3,000,000	3,000,000
Waste Management	1	3,000,000	3,000,000
Dry cleaner NSD 0	1	18,000,000	18,000,000
Crusher	1	2,946,000	2,946,000
Shea nut drying/roasting unit	1	2,258,500	2,258,500
Press MBU 40	1	20,186,000	20,186,000
Thermal installation	2	9,300,000	18,600,000
Filter Kinetic	1	2,758,500	2,758,500
Packaging Filling and labelling machine	1	4,008,500	4,008,500
Packaging table	2	870,000	1,740,000
Production tools	1	4,000,000	4,000,000
Transport Installation and Commissioning	25% of equipment supply cost		
Vehicles	1	5,800,000	5,800,000
Total			136,297,500

Table 24: Yearly turnover processing centres Press (CFA)

Processing centres	First Year	Second Year	Third Year	Fourth Year	Fifth Ydear
Yearly processed almond	504 000	504 000	504 000	504 000	504 000
Yearly butter production	176 400	176 400	176 400	176 400	176 400
Total energy cost	13 568 815	13 568 815	13 568 815	13 568 815	13 568 815
Total water cost	42 000	42 000	42 000	42 000	42 000
Total packaging cost	5 292 000	5 292 000	5 292 000	5 292 000	5 292 000
Price / Kg	600	600	600	600	600
Total variable cost	19 364 815	19 364 815	19 364 815	19 364 815	19 364 815
Total turnover	86 475 185	86 475 185	86 475 185	86 475 185	86 475 185

Table 25: Costing of processing centre- part 2

Designation	Number	Capacity	Unit cost	Total cost	Photo
Fruit bubble washing machine	1	1t/h, 2 Kwel	4,800,000	4,800,000	
Band dryer and roaster	1	400Kg/h, 70 Kwth	4,500,000	4,500,000	
Crusher	1	250 Kg/h, Engine 5,5 Kw	1,900,000	1,900,000	

Technical report - Site Technology Design and Development

Mill	1	1000kg/h, Engine 2,2kw	2,800,000	2,800,000	
Churn	1	500 L, Engine 3Kw	3,000,000	3,000,000	
Thermal installation	3	140 Kwth	9,300,000	27,900,000	
Filter	1	300Kg/h, engine 2,2Kw	1,500,000	1,500,000	
Packaging Filling and labelling	1	20-50 bags/min, bag maxi size 100mmx 180mm	2,500,000	2,500,000	
Packaging table	2		870,000	1,740,000	
Production tools	1		4,000,000	4,000,000	
Vehicles	1		5,800,000	5,800,000	
Total			116,440,000		

e) Quotations

The study team received quotations for supply of equipment for both collection and processing centre from five companies. The detailed quotations are given in the annex 2 and 3. Name of the agencies are as follow:

Amount in CFA

Name of the agency	Collection centre	Processing centre	Remark
Burkina Lifestyle Concepts SARL	77,502,163	110,182,700	Lowest quotation for both
ISOMET (Innovation en SOLaire et METallique)	90,564,063	112,430,000	
BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)	86,312,438	110,640,250	
Amisy Food Machine	10,064,600		
Megawatt & Solutions Private Limited, New Delhi	72,266,040		

They will supply the equipment if they receive contract. Lowest quotation provider for both Collection and Processing Centre is given by - Burkina Lifestyle Concepts SARL, Ouagadougou.

To construct collection and processing centre quotation were also collected from the companies and agencies working in Burkina Faso. Following agencies submitted their quotations.

Amount in CFA

Name of the agency	Collection centre	Processing centre	Remark
Burkina Lifestyle Concepts SARL	42,300,000	47,300,000	
ISOMET (Innovation en SOLaire et METallique)	42,180,000	47,180,000	
BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)	42,000,000	46,900,000	Lowest quotation for both

4.7. Prepare detailed quality management processes (including assessing Shea nuts harvested, intermediary products, final products). Summarise results from initial tests with existing producers and compare them with potential quality from Serious Shea.

a) Quality of Shea butter generated

The general quality of Shea butter, according to most international standards, is pinned on four Shea nut quality parameters: peroxide value, moisture content, free fatty acid levels, and the insoluble impurities. Peroxide value (PV) is an indicator for stability and level of deterioration of Shea butter and it measures the milli equivalents of oxygen or hydroperoxides in 1 g of fat or oil (Ikya et al., 2013). It is a valuable measure of oil quality as it serves as an indicator of degradation of the long fatty acid chains through auto-oxidation into peroxides that can later break down into other chemicals including foul smelling ketones and aldehydes. The parameter deprives the butter of its stability and thus butter with high peroxide value ($PV > 10 \text{ meq kg}^{-1}$) is associated with the development of rancidity, which eventually limits their use in the food industry (Shahidi, 2005). Peroxide value is the most common determinant of lipid oxidation (Shahidi, 2005). Hydro peroxides under normal condition are remarked to have no flavour or odour of their own, they are however unstable and usually break down rapidly to

other products such as aldehydes and thereby developing strong, disagreeable flavour and scent.

Free Fatty Acids (FFA) by definition is the fatty acids present in oil or fat which has not been neutralized (Guy, 2009) or just unattached fatty acids present in a fat (Sapna and Nirmali, 2009). FFAs are related to their acid values. The acid value is a parameter expressed as the number of milligrams of potassium hydroxide required to neutralize the free fatty acids contained in one gram of fat or oil (Kardash and Turyan, 2004) and it is twice the FFA of fat and therefore acid value is directly proportional to free fatty acids (Roger et al., 2010) and thus the lower the acid value of oil, the fewer FFA it contains.

Moisture is a chemical contaminant usually mixed with oil. Presence of moisture in oil affects the quality of the oil and significant amount of moisture in oil support microbial growth (Alirezalu et al., 2011; Hee, 2011) and lipid oxidation leading to rancidity (Hee, 2011) thereby reducing the shelf life of the fat and its corresponding products. Conversely, the low moisture content of Shea butter is indicative of good quality (Olaniyan and Oje, 2007b). The difference in the moisture content of Shea butter can be attributed to Shea vegetation (Quainoo et al., 2013), although the minimum

moisture content of Shea butter is 0.05% but can go as high as 2.0% (West African Regional Standards, 2006). Insoluble impurities refer to dirt and other foreign materials in Shea butter (Hamilton et al., 1986; Hee, 2011). It has been reported that some of these materials are bonded to the butter via the machinery employed in the extraction of the butter. Insoluble impurities may also make their way into the butter through physical contact of the butter with the soil, water, ground as well as packaging materials. The amount of insoluble impurities is an important quality parameter that determines Shea butter deterioration since metals can catalyze the oxidation of Shea butter and thus decreases its market value (Hee, 2011).

Shea butter quality degradation factor

There are several factors which may lead to the degradation of Shea butter such as microbiological contamination, chemical degradation, refining processes, and enzymatic degradation, and most occur post the collection stage during handling, processing and storage.

Microbiological Contamination

One of the major challenges facing Shea butter imports and trade (even the importation of raw or dried Shea nuts) into the US is microbiological contamination and quality degradation. Sources of contamination are many but can include the following:

- Fruit diseases and surface mold cause fruit decay. Microorganism contaminations reduce fruit quality and negatively impact Shea butter color.
- Leaving the ground and crushed material for too long (even just over night) before hot water extraction. This allows lipase enzymes to breakdown fatty acids into free fatty acids and promotes oxidative degradation.
- Using contaminated water in the extraction process introduces contaminants to the extracted Shea butter.
- Open containers in wet or damp areas allow contaminants to enter Shea butter.
- Non-sterile packaging by wholesalers and retailers also introduces contaminants.
- Time before boiling, roasting and drying increases fungal infections. Improved efficiency is required at these steps.
- Chemical Degradation (due to oxidation and hydrolysis)
- Agents or factors that result in reduction or destruction of cinnamic acid in the non-saponifiable fraction reduce the anti-histaminic and anti-inflammatory activity of Shea butter.
- Exposure of boiled Shea nuts to air for too long results in higher fungal attacks and peroxide levels.

- U V damage due to prolonged and inefficient sun drying promotes peroxide damage.
- Oxidation of unsaturated fatty acids causes the rancidity of Shea butter and reduces quality.
- Exposure of Shea butter to factors such as heat, light, and traces of metals particularly copper and iron promote rancidity.
- Microorganisms producing oxidizing enzymes also promote oxidative rancidity
- Refining

Refining is the normative process of Shea in Europe, but only because the European market has historically incorporated Shea butter as a cocoa butter substitute. Refining of Shea butter results in the loss of a significant portion, or even the entire unsaponifiable fraction. The unsaponifiable fractions are undesirable when the product is to be used as a cocoa butter substitute and when the product needs to be bleached and uniform in color and consistency. However, those same products which are removed in that refining process are also the groups of compounds considered to contribute to quality when the Shea butter is used for alternative markets, such as those which focus on the healing properties critical to pharmaceutical, cosmeceutical and personal care uses.

Enzymatic Degradation

When extraction is done by cold pressing, enzymes released from the tissue of the Shea nut can advance decay and degradation. Heating the Shea butter is necessary to denature these enzymes. Additionally, the longer the time before boiling, roasting or drying begins, the greater the likelihood of enzymatic degradation.

Butter Color

Shea butter color can be a determining quality factor for some industries. Final Shea color is related to the quality of the kernels processed. The presence of fungal infections (black nuts) and 'charring' increases the yellowness of the butter and can be prevented or reduced by more efficient drying and roasting techniques. Generally, the best color range for un-refined Shea butter is light yellow to pale beige or off white. Apart from dyes that may traditionally be added, the sources of Shea butter yellowness are: fungal filaments or spores, tannins from kernels and kernel-coating, and contaminants formed during heating or roasting.

Quality Control Considerations

Contamination enters Shea butter at the processing and preparation stages. To avoid microbiological contamination of Shea butter, extraction should use clean well (or sterilized) water. Storage should be in airtight, cool, and dry containers; and packaging should be in sterilized containers. Post-processing chemical contaminants come from contaminated water and/or unclean containers used during extraction or for storage. Rancidity is another quality reducing factor in Shea butter and results from oxidation of the unsaturated fatty acids. Polyunsaturated fatty acids are particularly susceptible to oxidative rancidity because they contain one or more double bonds which are, by their nature, highly reactive. The level of polyunsaturated fatty acids is at least one of the determinants of shelf life. Thus, factors that accelerate the reaction (such as heat, light, traces of 'pro-oxidant' metals such as copper and iron) contribute directly to degradation of quality and reduction of shelf life. Microorganisms produce oxidizing enzymes (insensitive to antioxidants), oxygen, high temperatures, and oxidative products such as lipid peroxides, oxidative aldehydes, and alkenals, and free fatty acids from the breakdown of unsaturated fatty acids all promote the onset of rancidity of (unrefined) Shea butter. Recent analyses of Ghanaian Shea butter in our laboratory concluded that several interventions in the Shea butter primary processing are critical to lowering the chances of degradation and attaining higher quality Shea butter. These interventions include

- a) First, the practice of using heaped versus fresh fruit shows that there was no significant improvement due to heaping and that using fresh fruit will yield acceptable results. However,

logistics and practical aspects of the extraction process may preclude this from becoming the accepted norm.

- b) Second, raw nuts yield much better quality butter than boiled nuts. This would also be a time and resource saving modification in the process. Boiling damages cell walls and makes nuts more prone to heat caused quality degradation. In addition, temperatures in the nuts may not reach high enough to denature all lipases and this may allow these degrading enzymes to become active, yet the temperatures may sufficiently damage membranes as to cause greater lipase release than would normally occur.
- c) Third, drying the boiled nuts on an open courtyard gave the worst quality Shea butter within boiled nuts. This is probably due to the boiled nuts being open microbiological degradation. Nuts dried in closed containers gave better quality nuts probably due to minimal exposure to micro-organisms. However, the best quality shea butter at this stage came from raw nuts dried on the open courtyard. This can be easily explained by the protection the flesh of the fruit gives the nuts, plus the preservation of nut integrity due to not boiling it (which allows microorganisms to enter). Intact fresh, raw nuts dry well and keep the kernel safe from microbiological degradation. Fried nuts gave lower quality Shea within raw nuts than from non fried nuts. This can also be explained by the degrading effects of the heat (speeds up oxidation reactions), and that many nuts will have cracked open due to the heat and become accessible to micro-organism degradation.
- d) Fourth, flying the nuts within the boiling treatment gives better quality Shea butter (probably due to the heat preventing any microbial degradation of the boiled nuts). However, if the nuts are used raw and not boiled, not flying the nuts gives the lower degradation and better Shea butter quality.
- e) The overall best quality Shea butter comes from fresh fruit, raw nuts, courtyard dried kernels and no frying. Low quality water and containers with metals that catalyze chemical degradation reactions could be additional factors in lowering quality Shea of butter. Though our analysis shows higher quality for certain treatments, realizing successful adoption by the processors (typically women) may be difficult due to the lower yield of butter that is achievable. Higher quality comes at a cost of greater effort and lesser return in form of Shea butter extracted. Our investigation of organoleptic properties found that forty percent of individuals who were given unrefined Shea butter to assess reported unfavorable reactions to the odor. This negative response to the odor is a major problem in US (and European) public acceptance of unrefined Shea butter. Refining Shea butter removes all phenolic compounds (degraded and undegraded) and removes the 'unpleasant' aroma. Yet, unrefined Shea butter also contains the unsaponifiable fraction that is associated with its medicinal and healing properties such as the antioxidant activity of several phenolic compounds. Our results also showed that the percent unsaponifiables is inversely correlated to the peroxide value, while FFA is directly and positively correlated to peroxide value. This is expected since both peroxide value and FFA are indicators of degradation.

Quality requirement

- Fatty Acid composition: requirement for low Free Fatty Acid content (FFA preferably <3%)
- Impurities: Requirements for low levels of impurities <0.1%
- Moisture content: Requirements for low moisture content (<0.1%)
- ISO requirements for natural and organic cosmetics: ISO 16128 standard as an alternative minimum standard, which covers definitions and criteria for natural and organic cosmetic ingredients and products
- Fair Production: Fairtrade and FairWild (for wild-collected ingredients)
- Manufacturing processes: Good Manufacturing Practices of the European Federation for Cosmetic Ingredients

- EU Legal requirements: Relevant European cosmetics legislation (Regulation (EC) 1223/2009); Registration Evaluation and Authorisation of Chemicals (REACH), although non-chemically modified
- Classification, Labelling and Packaging of chemicals (CLP): Specific packaging requirements. (e.g. Re-use or recycle packaging materials, storage of containers, Separation of organic certified Shea butter) as specified by freight forwarder or transport company

4.8. Prepare summary of technical project description to be used in Serious Shea business plan document (Component 1)

c) Construction costs and timeline (also who would do the construction?)

Details for construction cost and timeline is mentioned in the detailed budgeting attached as separate document. To construct collection centre and processing centre detailed tendering process needs to be adopted. This will be different for each sites which includes distance, availability of material area factor. The detailed time line for construction is as mentioned below:

- 3 months tender
- 3 months delivery of equipment/ Construction of the building
- 3 months installation of equipment and commissioning

d) Operational and maintenance requirements

To run a centre for a shift of 8 hours following team is essential

- One site supervisor/Admin
- Two technicians
- Two laborers
- Two guards

e) Detail resource flows (amounts and costs of energy, water, waste, etc.), scalability, and product flows (volumes of fruit, dried nuts, butter, etc.)

- Volume of raw material depends on the supply to equipment maximum capacity and number of shifts
- Scalability is achieving by adding additional line of production with the same number of equipment
- Material flow is as follow:

f) Monitoring requirements

- All processes will be monitored by the team of two technicians,
- They will be responsible to start and complete stop all batches following the standard protocols.
- These team of technicians will be trained on the standard protocols.

g) The technical team considered the following criteria in the following order for priority:

- Certified for food production
- Certified for organic food production
- Engine efficiency to reduce power usage
- Adaptability to solar, biogas, syngas and steam generators
- Reliability, maintenance requirements
- Availability of spare parts

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Annex 1

Procédures de test pour les centres de transformation

TEST 1: Fruits de Karité

But	Besoins	Procédure de test
Poids moyen Volume moyen Poids et volume pulpe Poids et volumes coque Poids et volume amande	- Un minimum de 3 Kg d'amandes séchées de karité - Balance de Précision	- Peser et mesurer les fruits - Séparer la pulpe, les amandes et la coque - Peser les éléments séparés
Purpose of the test	Needs for the test	Test procedures
Données à collecter / Temps		
Poids total		
Poids du récipient		
Volume total du récipient		

Échantillonnage pour Analyse : Pulpe, Coques, Amandes / Sucres, Pouvoir Calorifique, FFA/humidité/ Contenu en huile

Test 1: Pesée des amandes de Karité

But	Besoins	Procédure de test
Déterminer le poids des amandes à traiter pour le test.	- Un minimum de 150 Kg d'amandes séchées de karité - Récipients (4) (environ 50 l) - Balance précise	- Peser les amandes séchées dans leurs récipients. - Peser le récipient des amandes de karité vide.
Purpose of the test	Needs for the test	Test procedures
Determine the weight of the almonds to be treated for the test.	- A minimum of 150 kg of dried Shea almonds - Containers (4) (about 50 l) - Accurate scale	- Weigh the dried almonds in their containers. - Weigh the empty container of the Shea almonds.
Données à collecter / Temps		
Poids total		
Poids du récipient		
Volume total du récipient		

Échantillonnage pour Analyse : Coques, Amandes / Pouvoir Calorifique, FFA/humidité/ Contenu en huile

TEST 2: Tri des amandes de Karité

But	Besoins	Procédure de test
Déterminer la quantité d'amandes de mauvaises qualités qui pourraient être utilisées comme énergie.	- Un minimum de 150 Kg d'amandes de karités séchées. - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	- Séparer par triage manuel les amandes de mauvaise qualité de celles de bonne qualité. - Peser les deux lots de façon précise et documenter.
Purpose of the test	Needs for the test	Test procedures
Determine the amount of bad quality almonds that could be used as energy.	- A minimum of 150 kg of Shea almond - Containers (4) large sizes (about 50 l) - Accurate scale	- Separate poor-quality almonds from good quality almonds by manual sorting. - Weigh both batches accurately and document.

Donnés à collecter / Temps		
Poids amandes de bonnes qualité		
Poids amandes de mauvaises qualité		

Échantillonnage pour Analyse : Amandes / FFA/humidité/ Contenu en huile

TEST 3: Masse volumique des amandes de Karité

But	Besoins	Procédure de test
Déterminer la masse volumique des amandes. La masse volumique nous permet de calculer le stockage	- Un minimum de 30 Kg d'amandes de karité séchées. - Récipients (2) de grandes tailles (environ 50 l) - Une Balance précise	- Remplir un récipient a volume défini, d'amande de karité a ras le bord. - Déterminer le poids du récipient remplis, peser ensuite le récipient rempli et documenter. - Le poids total moins le poids du récipient, nous donnera les poids des amandes.
Purpose of the test	Needs for the test	Test procedures
Determine the density of almonds. The density allows us to calculate storage	- A minimum of 30 kg of dried Shea almonds. - Containers (2) large sizes (about 50 l) - Acurate scale	- Fill a container with a defined volume, with Shea almond at the edge. - Determine the weight of the filled container, then weigh the filled container and document. - The total weight minus the weight of the container, will give us the weights of the almonds.

Donnés à collecter / Temps		
Volume du recipient (1)	en litre	
Poids total pesé (2)		
Poids recipient (3)		
Poids des amandes (4)	Difference (2-3)	
Masse volumique des amandes	(4/1)	

TEST 4: Lavage des amandes de Karité

But	Besoins	Procédure de test
- Eliminer les impuretés des amandes afin qu'elles n'affectent pas la qualité finale du beurre de karité. - Déterminer la quantité d'eau nécessaire a cette étape de la transformation.	- Un minimum de 100 Kg d'amandes séchées et triées de karité - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	- Peser les amandes a laver. - Déterminer le volume de toutes l'eau utilisée pour le lavage des amandes. - Laver les amandes selon la procédure usuelle a serious Shea butter.
Purpose of the test	Needs for the test	Test procedures
- Eliminate the impurities of the almonds so that they do not affect the final quality of the Shea butter. - Determine how much water is needed at this stage of the transformation.	- A minimum of 100 kg of dried and sorted Shea almonds - Containers (4) large sizes (about 50 l) - Acurate scale	- Weigh the almonds to be washed. - Determine the volume of all water used to wash the almonds.

		Wash the almonds according to the usual procedure a serious Shea butter.
Donnés à collecter / Temps		
Volume de l'eau utilisée		
Poids des amandes		

Échantillonnage pour Analyse : Amandes / FFA/humidité/ Contenu en huile

TEST 5: Séchage des amandes de Karité

But	Besoins	Procédure de test
- Sécher les amandes afin de faciliter l'étape suivante qui est le concassage. - Déterminer la quantité d'énergie nécessaire pour le séchage.	- Un minimum de 200 Kg d'amandes triées et lavées de karité - Un séchoir ou solaire par jours ensoleillé. - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	- Peser les amandes à sécher. - Sécher les amandes jusqu'à atteindre niveau de séchage requis pour le concassage. - Peser les amandes après séchage - La différence de poids nous informera sur la quantité d'eau extraite ainsi que la quantité d'énergie utilisée
Purpose of the test	Needs for the test	Test procedures
- Dry the almonds to facilitate the next step, which is crushing. - Determine how much energy is needed for drying.	- A minimum of 200 kg of fresh Shea nuts - A dryer or direct sun drying if possible - Containers (4) large sizes (about 50 l) - Accurate scale	- Weigh the almonds to dry. - Dry the almonds until they reach the drying level needed for crushing. - Weigh the almonds after drying - The difference in weight will inform us about the amount of water extracted as well as the amount of energy used
Donnés à collecter / Temps		
Poids des amandes avant séchage		
Poids des amandes après séchage		

Échantillonnage pour Analyse : Amandes / FFA/humidité/ Contenu en huile

TEST 6: Concassage des amandes de Karité

But	Besoins	Procédure de test
Le but de ce test est de déterminer la quantité d'énergie nécessaire au concassage des amandes de karité.	- Un minimum de 100 Kg d'amandes de karité séchées. - Un concasseur. - Un chronomètre - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	- Peser les amandes à concasser. - Noter la puissance du moteur du concasseur. - Chronométrer le temps de concassage. Démarrer le chrono au début du concassage effectif et couper dès que le concassage effectif est terminé. - Documenter les deux temps
Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to determine the amount of energy needed to crush Shea almonds.	- A minimum of 100 kg of dried Shea almonds. - A crusher. - A stopwatch	- Weigh the crushed almonds. - Note the power of the crusher's engine.

	- Containers (4) large sizes (about 50 l) - Accurate scale	- Take the crushing time. Start the stopwatch at the beginning of the crushing and cut as soon as the crushing is over. - Documenting both times
Donnés à collecter /temps		
Puissance du moteur		
Temps effectif de concassage		
Poids des amandes concassées		

TEST 7: Torréfaction des amandes de Karité

But	Besoins	Procédure de test
Déterminer la quantité d'énergie utiliser pour la torréfaction.	- Un minimum de 75 Kg d'amandes concassées de karité - Un torréfacteur - Du bois - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	1. Peser les amandes à torréfier. 2. Peser le bois avant torréfaction 3. Chronométrer le temps de torréfaction. 4. Eteindre immédiatement le feux apres torréfaction. 5. Sécher le bois et le charbon résiduel 6. Peser le bois résiduel. 7. Peser le charbon résiduel.
Purpose of the test	Needs for the test	Test procedures
Determine how much energy the roasting requires.	- A minimum of 75 kg of crushed Shea almonds - A roaster - Wood - Containers (4) large sizes (about 50 l) - Accurate scale	• Weigh the almonds to be roasted. • Weigh the wood before roasting • Time the roasting. • Turn off the fire immediately after roasting. • Dry wood and residual charcoal • Weigh the residual wood. • Weigh the residual coal.
Donnés à collecter /temps		
Poids des amandes		
Températures / Temps		
Poids du bois avant torréfaction		
Poids du bois séché apres torréfaction		
Poids du charbon séché		

Échantillonnage pour Analyse : Amandes torréfiées / FFA/humidité/ Contenu en huile

TEST 7 bis: Torréfaction des amandes de Karité

But	Besoins	Procédure de test
Déterminer la quantité d'énergie utiliser pour la torréfaction.	- Un minimum de 75 Kg d'amandes concassées de karité - Un torréfacteur - Une bouteille a gaz plus bruleur adapté au torréfacteur. - Récipients (4) de grandes tailles (environ 50 l)	• Peser les amandes à torréfier. • Peser la bouteille de gaz avant torréfaction • Chronométrer le temps de torréfaction. • Eteindre immédiatement le feux apres torréfaction. • Peser la bouteille de gaz apres torréfaction.

	Balance précise	
Purpose of the test	Need for the test	Test procedures
Determine how much energy the roasting requires.	- A minimum of 75 kg of crushed Shea almonds - A roaster - A gas cylinder with a burner suitable for roaster. - Containers (4) large sizes (about 50 l) Accurate scale	- Weigh the almonds to be roasted. - Weigh the gas cylinder before roasting - Time the roasting time. - Immediately turn off the fire after roasting. - Weigh the gas cylinder after roasting.

Donnés à collecter /temps

Poids des amandes

Températures / Temps

Poids du gaz avant torréfaction

Poids du gaz apres torréfaction

Échantillonnage pour Analyse : Amandes torréfiées / FFA/humidité/ Contenu en huile

TEST 8: Mouture des amandes de Karité

But	Besoins	Procédure de test
Déterminer la quantité d'énergie utiliser pour la mouture.	- Un minimum de 75 Kg d'amandes torréfiées et concassées. - Un moulin - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	- Peser les amandes concassées à moudre. - Noter la puissance du moteur du moulin. - Chronométrer le temps effectif de mouture. Démarrer le chrono au debut effectif de la mouture et couper dès que le concassage effectif est terminé. - Documenter les deux temps
Purpose of the test	Needs for the test	Test procedures
Determine how much energy to use for grinding.	- A minimum of 75 kg of roasted and crushed almonds. - A mill - Containers (4) large sizes (about 50 l) - Accurate scale	- Weigh the crushed almonds to grind. - Note the power of the mill engine. - Time the actual grinding time. Start the timer at the beginning of the grind and cut as soon as the actual grinding time is over. - Documenting both times

Donnés à collecter /temps

Puissance moteur

Températures / Temps

Temps effectif mouture

Poids des amandes

Échantillonnage pour Analyse : Amandes torréfiées pétrifiées / FFA/humidité/ Contenu en huile

TEST 9: Baratt

But	Besoins	Procédure de test
Déterminer la quantité d'énergie utiliser	Un minimum de 50 Kg de pâte d'amandes de karité	- Peser la pâte à baratter. - Noter la puissance du moteur de la baratte. - Chronométrer le temps effectif du barattage. Démarrer le chrono au debut effectif du

pour le barattage. - Déterminer la quantité d'eau utiliser pour le barattage. - Déterminer le poids des résidus de barattage utilisable comme Energie	- Une Baratte - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	barattage et couper le chrono dès que le barattage effectif est terminé. - Documenter le temps - Déterminer exactement la quantité d'eau utilisée pour le barattage. - Décanner le résidu de barattage. - Récupérer les résidus de barattage. - Sécher le résidu de barattage. - Peser le résidu de barattage
Purpose of the test	Needs for the test	Test procedures
Determine how much energy to use for churning. Determine how much water to use for churning. Determine the weight of usable churning residues such as Energy	- A minimum of 50 kg of Shea almond paste - A churn - Containers (4) large sizes (about 50 l) - Accurate scale	- Weigh the churning dough. - Note the power of the churn's engine. - Time the actual time of churning. Start the time at the beginning of the churning and cut the time as soon as the actual churning is over. - Documenting both times - Determine exactly how much water is used for churning. - Decant the churning residue. - Collect churning residue. - Dry the churning residue. - Weighing the churning residue

Donnés à collecter /temps

Poids de la pâte a baratté		
Températures / Temps		
Poids ou volume de l'eau		
Poids du résidu séché.		
Puissance du moteur de la baratte		
Temps effectif de barattage		

Échantillonnage pour Analyse : Emulsion : FFA/humidité/ Contenu en huile , Résidus : FFA/humidité/ Contenu en huile

TEST 10: Collection et lavage de l'émulsion

But	Besoins	Procédure de test
Déterminer la quantité d'eau utilisée pour le lavage du beurre brut.	- Emulsion récupérée dans le processus antérieur - Un séchoir - Récipients (4) de grandes tailles (environ 50 l) - Balance précise	- Peser l'émulsion récupérée. - Déterminer la quantité d'eau utilisée pour laver l'émulsion.
Purpose of the test	Needs for the test	Test procedures
Determine the amount of water used to wash the raw butter.	- Emulsion recovered from the previous process - Containers (4) large sizes (about 50 l) - Precise balance - Accurate scale	- Weigh the collected emulsion. - Determine the amount of water used to wash the emulsion
Donnés à collecter /temps		

Poids de l'émulsion		
Températures / Temps		
Quantité d'eau utilisée pour le lavage		

Échantillonnage pour Analyse: Emulsion: FFA/humidité/ Contenu en huile , Résidus : FFA/humidité/ Contenu en huile

TEST 11: Cuisson du beurre

But	Besoins	Procédure de test
Déterminer la quantité d'énergie nécessaire a la cuisson du beurre	- Un minimum de 25 Kg de beurre brut - Un cuiseur - Récipients (2) de grandes tailles (environ 10l) - Balance précise	- Peser le beurre à cuire. - Peser le combustible utilisé pour la cuisson du beurre. (Bouteille de gaz). - Peser le beurre a la fin de la cuisson (attention chaud). - Peser le combustible résiduel (Bouteille de gaz).
Purpose of the test	Needs for the test	Test procedures
Determine how much energy is needed to cook butter	- A minimum of 25 kg of raw butter - A cooker - Containers (2) large sizes (about 10l) - Accurate scale	- Weigh the butter to cook. - Weigh the fuel used to cook butter. (Gas bottle). - Weigh the butter at the end of cooking (warm attention). - Weigh the residual fuel (Gas bottle).
Donnés à collecter /temps		
Poids du beurre à cuire		
Températures / Temps		
Poids du beurre après cuisson		
Poids du combustible		
Poids du combustible résiduel		

Échantillonnage pour Analyse : Émulsion : FFA/humidité/ Contenu en huile , Résidus : FFA/humidité/ Contenu en huile

TEST 12: Décantation du beurre

But	Besoins	Procédure de test
Le but de ce test est de déterminer la quantité de résidus gras qui peut etre utilisé comme source d'énergie	- Un minimum de 25 Kg de beurre brut - Récipients (2) de grandes tailles (environ 10l) - Balance précise	- Peser le beurre brute (déjà extrait dans l'étape précédente). - Procéder a la décantation. - Peser le beurre décanté - Peser les résidus gras
Purpose of the test	Needs for the test	Test procedures
The purpose of this test is to determine how much fat residue can be used as an energy source	- A minimum of 25 kg of raw butter - Containers (2) large sizes (about 10l) - Accurate scale	- Weigh the raw butter (already done in the previous step). - Proceed with decanting. - Weigh the decanted butter - Weigh the fatty residues
Donnés à collecter /temps		
Poids du beurre brut		
Températures / Temps		

Poids du beurre décanté		
Poids des résidus gras		

TEST 13: Filtration du beurre

But	Besoins	Procédure de test
Déterminer la quantité de résidu gras après filtration	• Un minimum de 20 Kg de beurre décanté • Un filtre • Récipients (3) de moyennes tailles (environ 10 l) • Balance précise	• Peser le beurre décanté. • Filtrer le beurre décanté. • Peser le beurre raffiné • Peser les résidus gras
Purpose of the test	Needs for the test	Test procedures
Determine the amount of fatty residue after filtration	• A minimum of 20 kg of decanted butter • A filter • Containers (3) of medium sizes (about 10 l) Accurate scale	• Weigh the decanted butter. • Strain the decanted butter. • Weigh the refined butter • Weigh the fatty residues
Donnés à collecter /temps		
Poids du beurre décanté		
Poids du beurre raffiné		
Poids des résidus gras		

Échantillonnage pour Analyse : Émulsion : FFA/humidité/ Contenu en huile , Résidus : FFA/humidité/ Contenu en huile

Déterminer les quantités d'eau utilisées pour le nettoyage, a chaque étapes de production.

Quotations 1 for Supply of Equipment for Collection Centre



BURKINA LIFESTYLE CONCEPTS SARL

Siège social : Ouagadougou 04, Burkina Faso
Email: ble.infos@gmail.com
RCCM : BF OUA 2012 B3297
IFU : 00043302R
Tel : +226-50411719/70604936

To

Serious Shea SARL
Registered Office: Ouagadougou,
Secteur 29, 01 BP 6433
Ouagadougou 01, Burkina Faso

Date: September 18, 2020

Subject: Quotation for construction of collection centre

Details of the quotation

Sr No	Details of line items	No of units	Unit cost	Total cost (including shipment and customs)	Remark
1.	Washing and sorting	1	CFA 3,700,000	CFA 5,550,000	
2.	De-pulping Tables	5	CFA 845,900	CFA 4,652,450	
3.	Cooking	5	CFA 940,000	CFA 4,794,000	
4.	Drying	4	CFA 900,000	CFA 3,690,000	
5.	Shelling	1	CFA 950,000	CFA 1,710,000	
6.	Shea almond drying	4	CFA 900,000	CFA 3,690,000	
7.	Packaging - a scale	2	CFA 1,027,000	CFA 2,156,700	
8.	Packaging - Bag Hand sewing machine	10	CFA 150,300	CFA 1,518,030	
9.	Pulp drying Unit	1	CFA 12,350,000	CFA 14,820,000	
10.	Dried pulp Packaging	2	CFA 1,000,250	CFA 2,200,550	
11.	Packaging table	2	CFA 900,000	CFA 1,980,000	
12.	Production tools	1	CFA 4,200,000	CFA 5,040,000	
13.	Energy production	1	CFA 8,500,000	CFA 10,200,000	
	Sub Total			CFA 62,001,730	
	Transport Installation and Commissioning	25%		CFA 15,500,432.50	
	Total			CFA 77,502,163	

Amount in words: CFA seventy-seven million five hundred two thousand one hundred sixty three

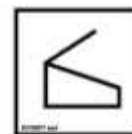
Offer Valid for 3 Months

We certify that in case we are awarded the contract , we are able to build the facilities in the areas of Koudougou , Didyr and Djigologo .

Quotations 2 for Supply of Equipment for Collection Centre

ISOMET (Innovation en Solaire et METallique)

Adresse : 09 BP 122, Ouagadougou 09
Tél. : +226 50 36 64 94
Email : info@isomet-bf.com



SERIOUS SHEA SARL
REGISTERED OFFICE: OUAGADOUGOU,
SECTEUR 29, 01 BP 6433
OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBER 19, 2020

SUBJECT: QUOTATION FOR THE EQUIPMENT OF THE COLLECTION CENTRE, INSTALLATION AND COMMISSIONING

DETAILS OF THE QUOTATION

Sr No	Details of line items	No of units	Unit cost	Total cost (including shipment and customs)	Remark
1.	Washing and sorting	1	CFA 3,900,000	CFA 5,850,000	
2.	De-pulping Tables	5	CFA 810,000	CFA 4,455,000	
3.	Cooking	5	CFA 925,000	CFA 4,717,500	
4.	Drying	4	CFA 950,000	CFA 3,895,000	
6.	Shelling	1	CFA 950,000	CFA 1,710,000	
6.	Shea almond drying	4	CFA 885,000	CFA 3,628,500	
7.	Packaging - a scale	2	CFA 1,050,000	CFA 2,205,000	
8.	Packaging - Bag Hand sewing machine	10	CFA 152,500	CFA 1,540,250	
9.	Pulp drying Unit	1	CFA 13,500,000	CFA 16,200,000	
10.	Dried pulp Packaging	2	CFA 1,150,000	CFA 2,530,000	
11.	Packaging table	2	CFA 900,000	CFA 1,980,000	
12.	Production tools	1	CFA 4,200,000	CFA 5,040,000	
13.	Thermal energy unit	2	CFA 8,500,000	CFA 18,700,000	
	Sub Total			CFA 72,451,250	
	Transport Installation and Commissioning	25%		CFA 18,112,812.50	
	Total			CFA 90,564,063	

Amount in words: cfa ninety million five hundred sixty-four thousand sixty-three

Offer validity: 3 months

R .C.C.M: BF OUA 2002 B 541:

Coris Bank 002091324101-54 Régime d'imposition normal Division Fiscale BOGODOGO N°IFU 00004975Y

Quotations 3 for Supply of Equipment for Collection Centre

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)

Adresse : 09 BP122, Ouagadougou 09

Tél : +226 69366666

Email : info@ilboudo.com

SERIOUS SHEA SARL

REGISTERED OFFICE: OUAGADOUGOU,

SECTEUR 29, 01 BP 6433

OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBRE 19, 2020

SUBJECT: QUOTATION FOR THE EQUIPMENT OF THE COLLECTION CENTRE, INSTALLATION AND COMMISSIONING

DETAILS OF THE QUOTATION

Sr No	Details of line items	No of units	Unit cost	Total cost (including shipment and customs)	Remark
1.	Washing and sorting	1	CFA 3,800,000	CFA 5,700,000	
2.	De-pulping Tables	5	CFA 800,000	CFA 4,400,000	
3.	Cooking	5	CFA 920,000	CFA 4,794,000	
4.	Drying	4	CFA 910,000	CFA 3,690,000	
6.	Shelling	1	CFA 930,000	CFA 1,710,000	
6.	Shea almond drying	4	CFA 880,000	CFA 3,690,000	
7.	Packaging - a scale	2	CFA 1,027,000	CFA 2,156,700	
8.	Packaging - Bag Hand sewing machine	10	CFA 142,500	CFA 1,518,030	
9.	Pulp drying Unit	1	CFA 12,000,000	CFA 14,820,000	
10.	Dried pulp Packaging	2	CFA 975,000	CFA 2,200,550	
11.	Packaging table	2	CFA 920,000	CFA 1,980,000	
12.	Production tools	1	CFA 4,200,000	CFA 5,040,000	
13.	Thermal energy unit	2	CFA 8,200,000	CFA 18,040,000	
	Sub Total			CFA 69,049,950	
	Transport Installation and Commissioning	25%		CFA 17,262,487.50	
	Total			CFA 86,312,438	

AMOUNT IN WORDS: CFA EIGHTY-SIX MILLION THREE HUNDRED TWELVE THOUSAND FOUR HUNDRED THIRTY EIGHT

OFFER VALIDITY: 3 MONTHS

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique) – Société A Responsabilité Limitée au Capital social de 1 000 000 FCFA. Siège Social : Ouagadougou, Secteur 44 – 01 BP 5659 Ouagadougou 01 – Burkina Faso
Tel : (226) 70 29 60 81 - N°RCCM : BF OUA2020B1490 - N° IFU : 00132861M- Régime fiscal : RSI - Compte Coris Bank
N° : BF148 01005 42862724101 37

Quotations 1 for Supply of Equipment for Processing Centre



BURKINA LIFESTYLE CONCEPTS SARL

Siège social : Ouagadougou 04, Burkina Faso
Email: ble.infos@gmail.com
RCCM : BF OUA 2012 B3297
IFU : 00043302R
Tel : +226-50411719/70604936

To

Serious Shea SARL

Registered Office: Ouagadougou,

Secteur 29, 01 BP 6433

Ouagadougou 01, Burkina Faso

Date: September 18, 2020

Subject: Quotation for the equipment of the processing center, installation and commissioning

Details of the quotation

Sr No	Details of line items	No of units	Unit cost	Total cost (including shipment and customs)	Remark
1.	Crusher/Hammer	1	CFA 2,126,800	CFA 2,552,160	
2.	Roaster	1	CFA 1,679,000	CFA 2,014,800	
3.	Miller/Press	1	CFA 16,789,000	CFA 20,146,800	
4.	Boiler/Thermal set	2	CFA 7,500,000	CFA 18,000,000	
5.	Mixer	1	CFA 3,460,000	CFA 4,152,000	
6.	Filter	1	CFA 1,950,000	CFA 2,340,000	
7.	Cleaning line/dry bed	1	CFA 15,900,000	CFA 19,080,000	
	Packaging Table	2	CFA 700,000	CFA 1,540,000	
	Production Tools	1	CFA 15,267,000	CFA 18,320,400	
	Sub Total			CFA 88,146,160	
	Transport Installation and Commissioning	25%		CFA 22,036,540	
	Total			CFA110,182,700	

Amount in words: CFA One hundred ten million one hundred eighty-two thousand seven hundred.

Valid for 3 months

We certify that in case we are awarded the contract , we are able to build the facilities in the areas of Koudougou , Didyr and Djigologo .

Quotations 2 for Supply of Equipment for Processing Centre

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)

Adresse : 09 BP122, Ouagadougou 09

Tél : +226 69366666

Email : info@ilboudo.com

SERIOUS SHEA SARL

REGISTERED OFFICE: OUAGADOUGOU,

SECTEUR 29, 01 BP 6433

OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBRE 19, 2020

SUBJECT: QUOTATION FOR THE EQUIPMENT OF THE PROCESSING CENTRE, INSTALLATION AND COMMISSIONING

DETAILS OF THE QUOTATION

Sr No	Details of line items	No of units	Unit cost	Total cost (including shipment and customs)	Remark
1.	Crusher/Hammer	1	CFA 2,100,000	CFA 2,520,000	
2.	Roaster	1	CFA 1,600,000	CFA 1,920,000	
3.	Miller/Press	1	CFA16,900,000	CFA 20,280,000	
4.	Boiler/Thermal set	2	CFA 7,600,000	CFA 18,240,000	
5.	Mixer	1	CFA 3,400,000	CFA 4,080,000	
6.	Filter	1	CFA 2,000,000	CFA 2,400,000	
7.	Cleaning line/dry bed	1	CFA15,975,000	CFA 19,170,000	
8.	Packaging Table	2	CFA 701,000	CFA 1,542,200	
9.	Production Tools	1	CFA15,300,000	CFA 18,360,000	
	Sub Total			CFA 88,512,200	
	Transport Installation and Commissioning	25%		CFA 22,128,050	
	Total			CFA 110,640,250	

AMOUNT IN WORDS: CFA HUNDRED AND TEN MILLION SIX HUNDRED FORTY THOUSAND AND TWO HUNDRED AND FIFTY

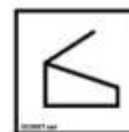
OFFER VALIDITY: 3 MONTHS

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique) – Société A Responsabilité Limitée au Capital social de 1 000 000 FCFA. Siège Social : Ouagadougou, Secteur 44 – 01 BP 5659 Ouagadougou 01 – Burkina Faso
Tel : (226) 70 29 60 81 - N°RCCM : BF OUA2020B1490 - N° IFU : 00132861M- Régime fiscal : RSI - Compte Coris Bank
N° : BF148 01005 42862724101 37

Quotations 3 for Supply of Equipment for Processing Centre

ISOMET (Innovation en Solaire et METallique)

Adresse : 09 BP 122, Ouagadougou 09
Tél : +226 50 36 64 94
Email : info@isomet-bf.com



SERIOUS SHEA SARL
REGISTERED OFFICE: OUAGADOUGOU,
SECTEUR 29, 01 BP 6433
OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBER 19, 2020

SUBJECT: QUOTATION FOR THE EQUIPMENT OF THE PROCESSING CENTRE, INSTALLATION AND COMMISSIONING

DETAILS OF THE QUOTATION

Sr No	Details of line items	No of units	Unit cost	Total cost (including shipment and customs)	Remark
1.	Crusher/Hammer	1	CFA 2,200,000	CFA 2,640,000	
2.	Roaster	1	CFA 1,750,000	CFA 2,100,000	
3.	Miller/Press	1	CFA17,100,000	CFA 20,520,000	
4.	Boiler/Thermal set	2	CFA 7,900,000	CFA 18,960,000	
5.	Mixer	1	CFA 3,800,000	CFA 4,560,000	
6.	Filter	1	CFA 1,900,000	CFA 2,280,000	
7.	Cleaning line/dry bed	1	CFA16,065,000	CFA 19,278,000	
8.	Packaging Table	2	CFA 730,000	CFA 1,606,000	
9.	Production Tools	1	CFA15,000,000	CFA 18,000,000	
	Sub Total			CFA 89,944,000	
	Transport Installation and Commissioning	25%		CFA 22,486,000	
	Total			CFA 112,430,000	

Amount in words: cfa hundred and twelve million four hundred thirty thousand

Offer validity: 3 months

R .C.C.M: BF OUA 2002 B 541:

Coris Bank 002091324101-54 Régime d'imposition normal Division Fiscale BOGODOGO N°IFU 00004975Y

Quotation 4 for collection and processing centre from Amisy Food Machine.

Website: www.amisyfoodmachine.com

Sr No	Technology	Model	Link	Model	Production capacity	Power consumption	Dimension	Cost (USD)
15.	Crusher	Electric	https://www.amisyfoodmachine.com/nut-processing-equipment/peanut-kernel-chopping-machine.html	QS-01	200kg/hr	2Kw	(2200X1050X1500)mm	3450
16.	Roaster	Electric	https://www.amisyfoodmachine.com/nut-processing-equipment/small-peanut-roasting-machine.html	AMS-RM50	40kg/hr	0.75kw	(1200X1200X700)mm	2300
17.	Miller	Electric	https://www.amisyfoodmachine.com/food-machinery/paste-grinding-machine.html	AMS- GM-300	1800kg/hr	90 KW	(1440X630X1420)mm	1650
18.	Boiler	Electric	https://www.amisyfoodmachine.com/food-machinery/jacketed-kettle.html	JGC1000	1000L/hr	2.2 Kw	(1400X1400X900)mm	5200
19.	Mixer	Electric	https://www.asengworks.com/search.html?ss=mixer		200 kg/hr	2 HP	(2200X1200X900)mm	-
20.	Filter	Electric	https://www.indiamart.com/proddetail/liquid-filter-machine-16131924212.html	sac1f01	40Lt/hr	2Kw	(1100X1100X3550)mm	-
21.	Packaging	Electric	https://www.amisyfoodmachine.com/packing-machine/automatic-liquid-packaging-machine.html	JTBZ-60C	300bags/min	1.36 Kw	(665X1000X1580)mm	5600

Quotation 5 for collection and processing centre from Megawatt & Solutions Private Limited

▪ **Name of the supplier:**

Megawatt & Solutions Private Limited

MWS Contact: Mr. Siddharth Malik M: +91-9654451401

T-93, 4th Floor, Gautam Nagar, Yousuf Sarai Commercial Complex, New Delhi-110049

Email ID: smalik@megawattsolutions.in Tel: 011-40536349, 41047606

Solution provided for as plants with following details

Capacity : 300 kg per Day of Shea Butter Peak Production

Energy : 100% of Renewable Energy for Thermal and Power Requirements comprising of solar and wood energy sources

a. System offered : 100% Renewable Energy Driven Semi-Automated Manually Driven System

- Integrated Solar PV System along with Battery back-up and off grid inverter for driving motors
- Heat Provided by Solar Thermal System
- Jacketed cooking kettles for boiling and roasting
- In non-solar days, heat can be provided by wood thereby allowing processing during



b. Operator-Friendly Jacketed Vessels for Boiling & Roasting



Salient characteristics

- Jacketed boiling & roasting vessels ensure no heat loss
- Jacketed boiling & roasting vessels ensure no fire hazard, burns and is operator friendly
- Heat is provided indirectly in form of hot water inside the vessel and is not visible
- Both boiling and roasting operations can be performed in these vessels
- Safe and Easy tilting for taking material out
- Electro-mechanical optional stirrer for ease of stirring specifically required for Shea butter
- Complete Heat by Renewable Energy (Solar Thermal + wood)
- Complete Power by Solar Power

c. Commercial

Sr No	Column A Details of 100% Renewable Energy Driven Semi-Automated Manually Driven System set to provide	Unit	Quoted cost of complete set of items mention in the column A
1.	Concentrated Solar Thermal System of peak capacity 40,000 kcal/hr at 120degC at DNI1kW/m2	1	USD 130,680 Per Set of eight items
2.	2 Thermic oil circulation pumps operating temperature of 150 degC (2 nos per set)	1	
3.	3 Pre-fabricated Insulated piping & fittings for solar, back up and vessels (50 meters per Set) MoC MS ERW C Class	1	
4.	4 Wood based back up heater of 40,000 kcal/hr for 150degC operating temperature (1 nos per Set)	1	
5.	5 Vessel for Boiling and Roasting with wheel based manual tilting in SS304 MoC each of 100 Ltrs capacity with detachable stirrer (2 nos per Set)	1	
6.	6 Solar PV System along with off grid inverter and battery back-up (6kWp capacity per Set)	1	
7.	Balance of plant including electrical cables, protections and safeties	1	
8.	8 Turnkey Installation, Erection & Commissioning of Supplies Included	1	

In Words: USD One Hundred Thirty Thousand and Six Hundred Eighty Only Per Set

Note minimum order is 50 sets

d. Terms & conditions:

- Price basis is exclusive of all taxes and statutory levies applicable at time of billing
- Prices are as per MWS standard make & specifications. Deviations will be charged extra.
- Prices are ex-works Vadodara, Gujarat, India; transportation and clearing charges to customer's accounts

Technical report - Site Technology Design and Development

- 40% Advance along with PO, 55% against pro-forma invoice prior to dispatch, balance upon commissioning. All terms on pro-rata basis
- Supplies will be ready at our works within 20 weeks from date of advance
- User shall own system without any design rights. User will have option to opt O&M or AMC Contracts
- Offer validity for a period of 30 days from date of this offer

Quotation 1 for Construction of Processing Centre



BURKINA LIFESTYLE CONCEPTS SARL

Siège social : Ouagadougou 04, Burkina Faso
 Email: ble.infos@gmail.com
 RCCM : BF OUA 2012 B3297
 IFU : 00043302R
 Tel : +226-50411719/70604936

To

Serious Shea SARL

Registered Office: Ouagadougou,

Secteur 29, 01 BP 6433

Ouagadougou 01, Burkina Faso

Date: September 20, 2020

Subject: Quotation for construction of Processing center

Details of the quotation

Specifications :

Sr No	Particulars	Dimensions /Specifications
1.	Loading and offloading space	50 SQM outside open
2.	Office area +lab	20 Sqm Main Building
3.	Toilet and changing space Staff	4 sqm outside building
4.	Toilet and changing space Management	6 sqm outside building
5.	Processing Area	150 Sqm Main Building
6.	Storage area Raw Material	50 Sqm Main Building
7.	Storage area Butter	20 Sqm Main Building
8.	Sorting Cleaning Area	50 Sqm Outside Covered terrace

BURKINA LIFESTYLE CONCEPTS SARL

Siège social : Ouagadougou 04, Burkina Faso
 Email: blc.infos@gmail.com
 RCCM : BF OUA 2012 B3297
 IFU : 00043302R
 Tel : +226-50411719/70604936

Cost can vary according to Soil Specifications :

Details of line items	Unit	Unit cost	Total cost
Perimeter Wall 30 m x 20 m, 100 Linear meter	100	CFA 10,000	CFA 1,000,000
Land preparation on the entire surface and walls 600 Sqm	600	CFA 10,000	CFA 6,000,000
Building construction	250	CFA 100,000	CFA 25,000,000
Water system	1	CFA6,500,000	CFA 6,500,000
Electricity	1	CFA3,200,000	CFA 3,200,000
Construction of waste management Area including biogas/ Syngas	1	CFA3,400,000	CFA 3,400,000
Thermal installation Construction	1	CFA2,200,000	CFA 2,200,000
Total			CFA 47,300,000

Amount in words:

Forty seven Million three Hundred Thousand CFA.

We certify that in case we are awarded the contract , we are able to build the facilities in the areas of Koudougou , Didyr and Djigologo .

Quotation 2 for Construction of Processing Centre

ISOMET (Innovation en SOLaire et METallique)

Adresse : 09 BP 122, Ouagadougou 09
Tél. : +226 50 36 64 94
Email : info@isomet-bf.com



SERIOUS SHEA SARL
REGISTERED OFFICE: OUAGADOUGOU,
SECTEUR 29, 01 BP 6433
OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBER 21, 2020

SUBJECT: QUOTATION FOR CONSTRUCTION OF PROCESSING CENTRE

DETAILS OF THE QUOTATION

SPECIFICATION

Sr No	Particulars	Dimensions /Specifications
1.	Loading and offloading space	50 SQM outside open
2.	Office area +lab	20 Sqm Main Building
3.	Toilet and changing space Staff	4 sqm outside building
4.	Toilet and changing space Management	6 sqm outside building
5.	Processing Area	150 Sqm Main Building
6.	Storage area Raw Material	50 Sqm Main Building
7.	Storage area Butter	20 Sqm Main Building
8.	Sorting Cleaning Area	50 Sqm Outside Covered terrace

R.C.C.M: BF OUA 2002 B 541:

Coris Bank 002091324101-54 Régime d'imposition normal Division Fiscale BOGODOGO N°IFU
00004975Y

ISOMET (Innovation en SOLaire et METallique)

Adresse : 09 BP 122, Ouagadougou 09

Tél. : +226 50 36 64 94

Email : info@isomet-bf.com



COST MAY VARY ACCORDING TO SOIL SPECIFICATIONS:

Details of line items	Unit	Unit cost	Total cost
Perimeter Wall 30 m x 20 m, 100 Linear meter	100	CFA 10,000	CFA 1,000,000
Land preparation on the entire surface and walls 600 Sqm	600	CFA 9,800	CFA 5,880,000
Building construction	250	CFA 100,000	CFA 25,000,000
Water system	1	CFA6,300,000	CFA 6,300,000
Electricity	1	CFA3,200,000	CFA 3,200,000
Construction of waste management Area including biogas/ Syngas	1	CFA3,500,000	CFA 3,500,000
Thermal installation Construction	1	CFA2,300,000	CFA 2,300,000
TOTAL			CFA 47,180,000

Amount in words: cfa forty-seven million one hundred eighty thousand

Offer validity: 3 months

We certify that in case we are awarded the contract, we can build the facilities in the areas of Koudougou, Didyr and Djigologo

R. C. C. M: BF OUA 2002 B 541:

Coris Bank 002091324101-54 Régime d'imposition normal Division Fiscale BOGODOGO N°IFU 00004975Y

Quotation 3 for Construction of Processing Centre

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)

Adresse : 09 BP122, Ouagadougou 09

Tél : +226 69366666

Email : info@ilboudo.com

SERIOUS SHEA SARL

REGISTERED OFFICE: OUAGADOUGOU,

SECTEUR 29, 01 BP 6433

OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBER 21, 2020

SUBJECT: QUOTATION FOR CONSTRUCTION OF PROCESSING CENTRE

DETAILS OF THE QUOTATION

SPECIFICATION

Sr No	Particulars	Dimensions /Specifications
1.	Loading and offloading space	50 SQM outside open
2.	Office area +lab	20 Sqm Main Building
3.	Toilet and changing space Staff	4 sqm outside building
4.	Toilet and changing space Management	6 sqm outside building
5.	Processing Area	100 Sqm Main Building
6.	Storage area Raw Material	30 Sqm Main Building
7.	Storage area Dried Kernel	30 Sqm Main Building
8.	Storage area Dried Pulp	10 Sqm Main Building
9.	Preparation Area	50 Sqm Outside Covered terrace

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)

Adresse : 09 BP122, Ouagadougou 09

Tél : +226 69366666

Email : info@ilboudo.com

COST MAY VARY ACCORDING TO SOIL SPECIFICATIONS:

Details of line items	Unit	Unit cost	Total cost
Perimeter Wall 30 m x 20 m, 100 Linear meter	100	CFA 10,000	CFA 1,000,000
Land preparation on the entire surface and walls 600 Sqm	600	CFA 10,000	CFA 6,000,000
Building construction	250	CFA 98,000	CFA 24,500,000
Water system	1	CFA 6,500,000	CFA 6,500,000
Electricity	1	CFA 3,000,000	CFA 3,000,000
Construction of waste management Area including biogas/ Syngas	1	CFA 3,500,000	CFA 3,500,000
Thermal installation Construction	1	CFA 2,400,000	CFA 2,400,000
TOTAL			CFA 46,900,000

Amount in words: cfa forty-six million nine hundred thousand

Offer validity: 3 months

We certify that in case we are awarded the contract, we can build the facilities in the areas of Koudougou, Didyr and Djigologo

Quotation 1 for Construction of Collection Centre



BURKINA LIFESTYLE CONCEPTS SARL

Siège social : Ouagadougou 04, Burkina Faso
Email: blc.infos@gmail.com
RCCM : BF OUA 2012 B3297
IFU : 00043302R
Tel : +226-50411719/70604936

To

Serious Shea SARL
Registered Office: Ouagadougou,
Secteur 29, 01 BP 6433
Ouagadougou 01, Burkina Faso

Date: September 20, 2020

Subject: Quotation for construction of collection center

Details of the quotation

Specifications :

Sr No	Particulars	Dimensions /Specifications
1.	Loading and offloading space	50 SQM outside open
2.	Office area +lab	20 Sqm Main Building
3.	Toilet and changing space Staff	4 sqm outside building
4.	Toilet and changing space Management	6 sqm outside building
5.	Processing Area	100 Sqm Main Building
6.	Storage area Raw Material	30 Sqm Main Building
7.	Storage area Dried Kernel	30 Sqm Main Building
8.	Storage area Dried Pulp	10 Sqm Main Building
9.	Preparation Area	50 Sqm Outside Covered terrace

Technical report - Site Technology Design and Development

BURKINA LIFESTYLE CONCEPTS SARL

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Cost can vary according to Soil Specifications :

Details of line items	Unit	Unit cost	Total cost
Perimeter Wall 30 m x 20 m, 100 Linear meter	100	CFA 10,000	CFA 1,000,000
Land preparation on the entire surface and walls 600 Sqm	600	CFA 10,000	CFA 6,000,000
Building construction	200	CFA 100,000	CFA 20,000,000
Water system	1	CFA 6,500,000	CFA 6,500,000
Electricity	1	CFA 3,200,000	CFA 3,200,000
Construction of waste management Area including biogas/ Syngas	1	CFA 3,400,000	CFA 3,400,000
Thermal installation Construction	1	CFA 2,200,000	CFA 2,200,000
Total			CFA 42,300,000

Amount in words:

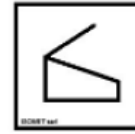
Forty two Million three Hundred Thousand CFA.

We certify that in case we are awarded the contract , we are able to build the facilities in the areas of Koudougou , Didyr and Djigologo .

Quotation 2 for Construction of Collection Centre

ISOMET (Innovation en SOLaire et METallique)

Adresse : 09 BP 122, Ouagadougou 09
Tél. : +226 50 36 64 94
Email : info@isomet-bf.com



SERIOUS SHEA SARL
REGISTERED OFFICE: OUAGADOUGOU,
SECTEUR 29, 01 BP 6433
OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBER 21, 2020

SUBJECT: QUOTATION FOR CONSTRUCTION OF COLLECTION CENTER

DETAILS OF THE QUOTATION

Sr No	Particulars	Dimensions /Specifications
1.	Loading and offloading space	50 SQM outside open
2.	Office area +lab	20 Sqm Main Building
3.	Toilet and changing space Staff	4 sqm outside building
4.	Toilet and changing space Management	6 sqm outside building
5.	Processing Area	100 Sqm Main Building
6.	Storage area Raw Material	30 Sqm Main Building
7.	Storage area Dried Kernel	30 Sqm Main Building
8.	Storage area Dried Pulp	10 Sqm Main Building
9.	Preparation Area	50 Sqm Outside Covered terrace

R.C.C.M: BF OUA 2002 B 541:

Coris Bank 002091324101-54 Régime d'imposition normal Division Fiscale BOGODOGO N°IFU 00004975Y

ISOMET (Innovation en SOlaire et METallique)

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COST MAY VARY ACCORDING TO SOIL SPECIFICATIONS:

Details of line items	Unit	Unit cost	Total cost
Perimeter Wall 30 m x 20 m, 100 Linear meter	100	CFA 10,000	CFA 1,000,000
Land preparation on the entire surface and walls 600 Sqm	600	CFA 9,800	CFA 5,880,000
Building construction	200	CFA 100,000	CFA 20,000,000
Water system	1	CFA 6,300,000	CFA 6,300,000
Electricity	1	CFA 3,200,000	CFA 3,200,000
Construction of waste management Area including biogas/ Syngas	1	CFA 3,500,000	CFA 3,500,000
Thermal installation Construction	1	CFA 2,300,000	CFA 2,300,000
TOTAL			CFA 42,180,000

Amount in words: cfa forty-two million one hundred eighty thousand

Offer validity: 3 months

We certify that in case we are awarded the contract, we can build the facilities in the areas of Koudougou, Didyr and Djigologo

R.C.C.M: BF OUA 2002 B 541:

Coris Bank 002091324101-54 Régime d'imposition normal Division Fiscale BOGODOGO N°IFU 00004975Y

Quotation 3 for Construction of Collection Centre

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)

Adresse : 09 BP122, Ouagadougou 09

Tél : +226 69366666

Email : info@ilboudo.com

SERIOUS SHEA SARL

REGISTERED OFFICE: OUAGADOUGOU,

SECTEUR 29, 01 BP 6433

OUAGADOUGOU 01, BURKINA FASO

DATE: SEPTEMBER 21, 2020

SUBJECT: QUOTATION FOR CONSTRUCTION OF COLLECTION CENTER

DETAILS OF THE QUOTATION

Sr No	Particulars	Dimensions /Specifications
1.	Loading and offloading space	50 SQM outside open
2.	Office area +lab	20 Sqm Main Building
3.	Toilet and changing space Staff	4 sqm outside building
4.	Toilet and changing space Management	6 sqm outside building
5.	Processing Area	100 Sqm Main Building
6.	Storage area Raw Material	30 Sqm Main Building
7.	Storage area Dried Kernel	30 Sqm Main Building
8.	Storage area Dried Pulp	10 Sqm Main Building
9.	Preparation Area	50 Sqm Outside Covered terrace

BEAM (Bureau d'études en Energie, Agroalimentaire et construction Mécanique)

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COST MAY VARY ACCORDING TO SOIL SPECIFICATIONS:

Details of line items	Unit	Unit cost	Total cost
Perimeter Wall 30 m x 20 m, 100 Linear meters	100	CFA 10,000	CFA 1,000,000
Land preparation on the entire surface and walls 600 Sqm	600	CFA 10,000	CFA 6,000,000
Building construction	200	CFA 98,000	CFA 19,600,000
Water system	1	CFA 6,500,000	CFA 6,500,000
Electricity	1	CFA 3,000,000	CFA 3,000,000
Construction of waste management Area including biogas/ Syngas	1	CFA 3,500,000	CFA 3,500,000
Thermal installation Construction	1	CFA 2,400,000	CFA 2,400,000
TOTAL			CFA 42,000,000

Amount in words: cfa forty-two million one hundred eighty thousand

Offer validity: 3 months

We certify that in case we are awarded the contract, we can build the facilities in the areas of Koudougou, Didyr and Djigologo

Annex 6

Approval process of proposed technology and designs

1. Background

The virtual workshop was conducted looking at the availability of all the stakeholder but was not practically possible. This was discussed with the Serious Shea team and requested them to use the technology to meet this deliverable. The Serious Shea team also agreed and suggested conducting this workshop using Skype and WhatsApp.

To ensure that the virtual workshop is effective and fulfill our objectives, detailed reports, documents, designs and test reports were shared with the stakeholders and Serious Shea team.

2. The participants

Name	Agency
▪ Jon White	Serious Shea
▪ David Lyon	Serious Shea
▪ William Kwende	Serious Shea
▪ Yameogo Felicite	Serious Shea
▪ Boffa, Jean-Marc (ICRAF)	Senconomics
▪ Jessica Antista	Techno Serve

3. Schedule of virtual workshop

- Invitation sent November 25, 2019
- Details of the discussion with the stakeholders as

Name	Agency	Date of discussion
○ Jon White	Serious Shea	December 11, 2019
○ David Lyon	Serious Shea	December 02, 2019
○ William Kwende	Serious Shea	December 11, 2019
○ Yameogo Felicite	Serious Shea	January 10, 2020
○ Boffa, Jean-Marc (ICRAF)	Senconomics	December 03, 2019
○ Jessica Antista	Techno Serve	November 27, 2019

4. Details of feedback from Serious Shea and other stakeholders**a. Jon White**

Hi Sandeep,

I approve this stage report, good job.

We expect that there will be more detail and design work in the final report, plus obviously completion of the missing test data.

Answer: Test data for processing center is attached

Hopefully you can also improve the report structure in the final version as 10-20 attachments, even with the good content, is difficult to present and share.

Answer: Yes

b. David Lyon

Feedback

Quality requirements

- The designs and processes should incorporate the key quality and management processes. Those listed by our customers are included in yellow highlight in the attached (with web-links).
- The report should also have a table with: a) overview of requirement; b) list of how the design meets those requirements

Answer: It will be included in the final report

Organic production

- The same approach can be used to demonstrate the design to meet the key organic certification requirements (inclusion in the design and a table to demonstrate how they are met). Also see the attached.
- If the organic production and conventional production are separate, it is important to identify where there are areas of shared personnel or equipment (e.g. vehicles, computers) to ensure efficiency; it should also indicate any differences between a conventional and organic design (with associated costs)

Answer: It will be included in the final report.

Thanks for reviewing the document and sharing your thoughts and putting questions to start the discussion and fine tune the document. We tried to answer most of the questions. A few still we are working to give you clear answers. The response to your question is as given below

The process flow indicates the key parameters. It would be good to discuss the:

1. Variability around the key metrics. This includes including what will make some of the parameters change. (e.g. if fruits have not immediately been taken to a collection center)

Answer: This is more a question for the logistic study, but I can put some ideas on it. the fruit has more pulp than nut at the end of the season. The yield of good quality pulp will go down in case is not collected on time

2. The source of information for the key metrics

Answer: The tests conducted on ground

3. I would like to discuss opportunities to valorize any waste in the process.

Answer: The presented process aims at valorizing all the residues. The residues in the form of shells, bad quality nuts, and press cake will turn it to energy. In case we have more residues than we use, the households around the unit will use it as fuel wood. The poor-quality pulp will be used to feed the biogas unit. As we mentioned, the pulp will be dried and sold on the local or international market as snacks.

4. For the capex items:

- Can you include the currency and the exchange rate at the time of listing the data.

Answer: The currency used is the Cfa or XOF The rate is 01-12 19 is 1 cfa = 0,0016792\$

- Can you specify what is included under 'Vehicles' on p21

Answer: The vehicle will be a second hand pick up, four-wheel drive.

- What are the assumptions under packaging?

Answer: A packaging unit as describe in the capex.

5. For electricity: can you include the expected source? What provisions are included in the event of unreliability (how expensive are they).

Answer: A solar unit can be installed if local electricity is not available

6. Operational and Maintenance requirements: It is not clear what the scope for this staff is for – is this a collection center or processing center? What are the assumptions as to how much work third parties need to do for the processing?

Answer: Light maintenance will be performed by the maintenance personal of the unit, the heavy maintenance is not predictable right now, according to Felicite not high.

7. A quotation for Megawatt solutions is provided. Please can you indicate what you see as the strengths and weaknesses of this approach?

Answer: We are getting inputs from our team and share reply ASAP.

8. When there is not a lot of Shea available, can the equipment be used for drying other fruit?

Answer: The collection center can be used for drying other fruits, just, the trails have to be changed and the dryer well clean, otherwise we will have a mix of flavors.

9. In the XLS file, I notice that the cost/revenue is the same for each month of the year. How will you capture the natural seasonal variation?

Answer: The cost/revenue is the same for the processing centers because we expect to have a certain quantity of nut per year stored, which will be treated all the yearlong. Of course, in reality we won't have the same monthly production. But the real distribution is hardly predictable right now. the collection centers are working 5 months per year.

c. William Kewende

Greetings,

This work was coordinated with me in real time, I have no comments unless specific questions.

d. Yameogo Felicite

An email was written to Ms. Felicite on July 23, 2019 and she gave approval to our designs shared with you on January 10, 2020. Her approval given on mail is as given below:

Hi Sandeep,

All designs are ok for me because all respected the biological units mentioned to us by Ecocert and respect of environment.

I appreciated your job and thing that our project will help rural women collectors and producers.

Be blessed

Mail 23 July, 2019 at 04:39, Sandeep Khanwalkar <skhanwalkar@devalt.org> the email :

Dear Felicite

I hope you are doing well. Please send the document ASAP.

I am attaching you a final layout of the collection site and processing center. Please go through it and give your approval. After receiving your approval, I will convert them into the high-level 3-D design.

e. Boffa, Jean-Marc (ICRAF)

Detailed discussion held with Mr. Jean-Marc on the design and layout of the collection center and processing center. His feed is mentioned below:

A couple of weeks ago we discussed the designs you shared. With apologies for the delay in writing out my observations, as mentioned then they all look very sound and carefully prepared. The concern I shared about the choice of technology was about women's engagement, which I understand is not in your ToR to address.

Answer: The higher women's engagement in the processing of Shea butter is justified when all the processing steps are done manually. In case operations such as crushing, roasting, milling, churning is not done manually, the working load will result in loading and unloading the processing machines and so reducing the engagement of women in the processing process.

It is also useful to mention the pros and cons of various technologies proposed for Serious Shea in your report. In our joint conversation with Rigobert Yaméogo, ex DG of IRSAT (Research Institute in Applied Sciences and Technologies) in Ouagadougou and now independent technology artisan and largest producer of Shea processing technologies in Burkina, it was mentioned that Shea presses/expellers can process 5 to 100T/day and about 30% fat yield but the quality of the resulting butter is low with 6 to 12% of free fatty acids which leads to large losses during refinement and makes it difficult to use as unrefined butter. Industries bypass this issue by doing solvent (hexane) extraction followed by refining and fractioning,

Large quantities of nuts can also be processed with churning technology and it provides good quality butter with a free fatty acid content of 2 to 6% that can be used unrefined in food and cosmetics. Butter yields are around 40%. Refining this butter is profitable due to its low acidity and the virtual absence of impurities.

Answer: We have done two tests at Felicity place and the yield was about 24% which was under my expectation, considering the documentation. Furthermore, olvea (<https://www.olvea-vegetable-oils.fr/producteur-beurre-karite-huile-sesame-biologique-equitable-burkina-faso/>) has a yield of 40% using the press.

f. Jessica Antista, Techno Serve

I reviewed TARA's designs and spoke with Sandeep this morning. There were 2 main concerns that I had, both of which I feel have been well addressed:

1. We feel it is important that the collection centers be equipped to receive both fruit and nut. The layout has been designed for this, which will accommodate the model as it evolves. This is good, as we anticipate it will be much easier to source nut than fruit, certainly at the onset.
2. If Serious Shea wants to pursue organic certification, this will require separate collection and processing facilities, to avoid contamination. So for now, we are assuming that the models' TARA has designed are for the first centers to be installed (which will be used for non-certified) and that if Serious Shea pursues certification in the future, separate centers will need to be built, using the same designs/budget. Sandeep may wish to comment on this further.

We spent quite a bit of time going through my questions, to understand how the model will integrate women's labor at the collection center level, and how this will impact the operational plan. I have a much clearer understanding now. **One additional piece of information that I will need from Sandeep will be the time it takes to process one batch:**

- From fruit to kernel
- From nut to kernel

Once I have that data, I should have everything I need from TARA to move forward with the ops plan. **Please consider this Techno Serve's formal acknowledgement that we have reviewed and validated TARA's designs.**

Answer to the query: Information about the time it takes to process one batch is as given below

Operation	Weight in Kg	Nbre persons	Time	
Sorting	311	5 persons	81 min	Fruit
Depulping	311	4 persons	6h49min	fruit
cooking	82	4	2h15	nut
Nut Drying	80,5	2	18h	Nut
Deshelling	61,2	3	5h20	Kernel-almond
Please, note that those operations are experimental one, Regular processing operation will take 25% to 35% less time.				

a) Layout of the collection site and processing center

The team had detailed discussions and also visited factories where Shea Butter production is going on and based on their inputs and discussion with experts finalized the detailed layout for collection center, processing center. High definition layout designs are shown in the following sections.

b) Finalization of processing center

For processing the Shea butter, there are two options available. Before finalising the process, we need to compare the critical criteria.

Comparative analyse		
Criteria	Churning	Press
Steps	The churning process as more processing which means a longer process, and more room for processing mistakes and loss of quality	The press process is more concise. Better for quality control. Low humane errors potential
Cost	The capex is 123 437 000 F Cfa, the costs are 10% lower than the press method	The capex is 136 297 500 F Cfa, the costs are Higer than the press method
Yield	The churning yield is around 22%	The press yield is around 33%
Employment	The churning will create around 20% more employment than the press method	it creates less employment but is more rentable than the churning process
Maintenance	The maintenance issue may have the same rating here: the churning process has more steps and will more maintenance need in term of quantity	the press method has less steps, less machine, but higher demand of maintenance in term of quality (need more qualified technicians)
Energy	The churning process will consume more energy in form of heat than the pressing process but will need less energy in form of electricity. we should pay attention to Greenhouse gas emission. the processes are autonomous in thermal energy. their residues may all the energy they need.	
Water	The churning process is consuming a lot of water	The press method consumes very few waters. only the cleaning water
Environment	The wastewater produced in the process must be disposed of somewhere! and can lead to an environmental problem. Also, the mix of wastewater and energy-rich residues is often be dried in the open air, leading to dirty surrounding	the press release only dried press cake easy to dispose of and ready to use. the environment can stay clean
Adaptation	the churning is traditionally implemented and adapted to the women	the press method is new but simple. It may adapt easily

From the above-mentioned comparison we can conclude that the press method is the most beneficial for the following reasons:

- Higher yield.
- Much less water in use.

- Clean environment indeed, the churning method leaves a high quantity of wastewater and churning waste that is difficult to dispose of. The press method, on the other hand, leaves dry press cake that can be used as energy in the process.
- Fewer processing steps.
- Higher financial viability due to higher yield, lower energy and water cost.

Tests results of Shea Fruits and Kernel

Sr No	Hauteur (mm)	Diameter (mm)	Circumference (mm)	Pods (mm)
1.	45	46	144.4	51
2.	44	47	143.58	49
3.	43	37	116.18	33
4.	46	45	141.3	51
5.	44	42	131.88	41
6.	41.5	40.5	127.17	34
7.	40	44	138.16	43
8.	48	39	122.46	38
9.	39	41	128.74	33
10.	49.5	42	131.88	44
11.	36	45	141.3	39
12.	41	42	131.88	38
13.	39	42.5	133.45	38
14.	44	38	119.32	36
15.	42	39	122.46	32
16.	40	37	116.18	32
17.	42	37	116.18	30
18.	45	36	113.04	24
19.	47.5	37	116.18	30
20.	41	35	101.9	25
21.	46	37	116.18	30
22.	40	38	119.32	31
23.	48	35	109.9	31
24.	43	33	103.62	23
25.	46.2	44	138.16	48
26.	43.8	39	122.46	36
27.	42.5	39	122.46	33
28.	40	36	113.04	28
29.	53	38	119.32	39
30.	51	36	113.04	31
31.	45	38	119.32	33
32.	42.5	37.5	117.75	30
33.	37	35	109.9	27
34.	53	34	106.76	30
35.	38	35	109.9	24
36.	43.5	34	106.75	23
37.	41	34	106.76	31
38.	35.5	36	113.04	25
39.	42	37	116.18	28
40.	38	35	109.9	22
41.	46	35	109.9	29
42.	44	36.3	113.98	32
43.	45	34	106.75	25
44.	42	35	109.9	27
45.	42	33	103.62	26
46.	46	32	100.48	23
47.	41	31	97.34	21
48.	43	30	94.7	18
49.	41.5	32	100.48	21
50.	39	35	109.9	22

Annex 8

Shea butter Collection Centre - Monthly Turnover														
Particular	%	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	1. Year
Fruits processing capacity in Kg/year		700,000												
Monthly processed fruits		0	0	0	0	140,000	140,000	140,000	140,000	140,000	0	0	0	700,000
Monthly almond production	41%	0	0	0	0	56,700	56,700	56,700	56,700	56,700	0	0	0	283,500
Daily fruits processing		0	0	0	0	5,600	5,600	5,600	5,600	5,600	0	0	0	
Daily almond production		0	0	0	0	2,268	2,268	2,268	2,268	2,268	0	0	0	
Monthly working days		25	25	25	25	25	25	25	25	25	25	25	25	
Energy cost/Kg		44.2	44.2	44.2	44.2	44.2	44.2	44.2	44.2	44.2	44.2	44.2	44.2	
Water cost/Kg		0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	
Transport cost		10	10	10	10	10	10	10	10	10	10	10	10	
Disinfectant cost		1	1	1	1	1	1	1	1	1	1	1	1	
Total energy cost		0	0	0	0	3,007,368	3,007,368	3,007,368	3,007,368	3,007,368	0	0	0	15,036,840
Total water cost		0	0	0	0	8,505	8,505	8,505	8,505	8,505	0	0	0	42,525
Total disinfectant cost		0	0	0	0	56,700	56,700	56,700	56,700	56,700	0	0	0	283,500
Total transport cost		0	0	0	0	567,000	567,000	567,000	567,000	567,000	0	0	0	2,835,000
Total production cost		0	0	0	0	632,205	632,205	632,205	632,205	632,205	0	0	0	3,161,025
Fruit Pulp		Fruit Pulp												
Monthly processed fruit pulp	41%	0	0	0	0	56,700	56,700	56,700	56,700	56,700	0	0	0	283,500
Daily processed fruit pulp		0	0	0	0	2,268	2,268	2,268	2,268	2,268	0	0	0	
Monthly dried fruit pulp production	25%	0	0	0	0	14,175	14,175	14,175	14,175	14,175	0	0	0	70,875
Fruit pulp packaging cost/Kg		20	20	20	20	20	20	20	20	20	20	20	20	
Total fruit pulp packaging cost		0	0	0	0	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	0	0	0	5,670,000
Fruit pulp production cost		0	0	0	0	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	0	0	0	5,670,000
Fruit pulp unit price		1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	
Fruit pulp turnover		0	0	0	0	14,175,000	14,175,000	14,175,000	14,175,000	14,175,000	0	0	0	70,875,000
Fruit pulp specific margin		0	0	0	0	13,041,000	13,041,000	13,041,000	13,041,000	13,041,000	0	0	0	65,205,000
Margin %						92%	92%	92%	92%	92%				
Total production cost		0	0	0	0	1,766,205	1,766,205	1,766,205	1,766,205	1,766,205	0	0	0	
Total turnover		0	0	0	0	14,175,000	14,175,000	14,175,000	14,175,000	14,175,000	0	0	0	
Total specific margin		0	0	0	0	12,408,795	12,408,795	12,408,795	12,408,795	12,408,795	0	0	0	
General margin %						88%	88%	88%	88%	88%				

Technical report - Site Technology Design and Development

Shea butter Collection Centre – Monthly Cost

Particular	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	1. année
Production cost	0	0	0	0	1,766,205	1,766,205	1,766,205	1,766,205	1,766,205	0	0	0	8,831,025
Production utensils	0	0	0	0	7,065	7,065	7,065	7,065	7,065	0	0	0	35,324
Salaries	1,410,000	1,410,000	1,410,000	1,410,000	1,418,831	1,418,831	1,418,831	1,418,831	1,418,831	1,410,000	1,410,000	1,410,000	16,964,155
Maintenance	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	600,000
miscellaneous	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	120,000
Communication cost	10,000	10,000	10,000	10,000	18,831	18,831	18,831	18,831	18,831	10,000	10,000	10,000	164,155
Buildings maintenance	50,000	50,000	50,000	50,000	58,831	58,831	58,831	58,831	58,831	50,000	50,000	50,000	644,155
Annuities	0	0	0	0	0	0	0	0	0	0	0	0	0
Depreciation	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	1,450,350	17,404,205
Tax													0
Total cost	2,980,350	2,980,350	2,980,350	2,980,350	4,780,113	4,780,113	4,780,113	4,780,113	4,780,113	2,980,350	2,980,350	2,980,350	44,763,019

Shea butter Collection Centre – Benefit and loss

Month	January	February	March	April	May	June	July	August	September	October	November	December
Turnover	0	0	0	0	14,175,000	14,175,000	14,175,000	14,175,000	14,175,000	0	0	0
Costs	2,980,350	2,980,350	2,980,350	2,980,350	4,780,113	4,780,113	4,780,113	4,780,113	4,780,113	2,980,350	2,980,350	2,980,350
Initial losses or profits	-2,980,350	-2,980,350	-2,980,350	-2,980,350	9,394,887	9,394,887	9,394,887	9,394,887	9,394,887	-2,980,350	-2,980,350	-2,980,350
Accumulated starting losses or profits	-2,980,350	-5,960,701	-8,941,051	-11,921,402	-2,526,515	6,868,372	16,263,259	25,658,145	35,053,032	32,072,682	29,092,331	26,111,981

Shea butter Collection Centre –Investment and Financial Plan

Necessary	Costs in CFA		Costs in CFA
Capex	115,401,975	Own contribution	20,000,000
Commissioning	8,700,000	Confirmed subsidies	111,593,975
Installation	2,900,000	Loan to contract	0
Local transport and installation	4,592,000		
Start losses			
Working capital			
Total investment	131,593,975	Total financing	131,593,975

Shea butter Collection Centre – Calculation of depreciation costs

Designation Construction and land development	Unit	Unit cost	Total cost	Life time	Yearly rate of depreciation	Yearly depreciation	Monthly depreciation
Land	1	5,000,000	5,000,000	15	7%	333,333	27,778
Landscaping	1	1,000,000	1,000,000	15	7%	66,667	5,556
Building	1	17,000,000	17,000,000	15	7%	1,133,333	94,444
Storage	1	7,000,000	7,000,000	15	7%	466,667	38,889
Water system	1	7,000,000	7,000,000	15	7%	466,667	38,889
Electricity	1	3,000,000	3,000,000	15	7%	200,000	16,667
Waste management including biogas	1	6,000,000	6,000,000	7	14%	857,143	71,429
Fruit bubble washing machine Contact www.pasenmachinery.com	1	3,850,000	5,696,000	5	20%	1,139,200	94,933
Depulping tables	5	869,745	4,348,725	5	20%	869,745	72,479
Shea nut drying/roasting unit Gelgoog (gelgoog.com.cn)	1	2,950,000	4,571,000	5	20%	914,200	76,183
Shelling machine Henan My Only Food Machine Co., Ltd.	1	1,050,000	2,196,000	5	20%	439,200	36,600
Pulp dryer	1	15,000,000	19,633,500	5	20%	3,926,700	327,225
Thermal installation	2	9,300,000	18,600,000	5	20%	3,720,000	310,000
Dried pulp packaging machine Zhejiang Ruibao Packaging Machine Co., Ltd.	2	950,000	2,816,750	5	20%	563,350	46,946
Dried pulp packaging table	2	870,000	1,740,000	5	20%	348,000	29,000
Production tools	1	4,000,000	4,000,000	5	20%	800,000	66,667
Vehicles	1	5,800,000	5,800,000	5	20%	1,160,000	96,667
Total			115,401,975			17,404,205	1,450,350

Shea butter Collection Centre – Salaries

1	Manager processing centre	300,000	cfa
2	Accountant	200,000	cfa
3	Maintenance	200,000	cfa
4	Clerk	100,000	cfa
5	Production leader	150,000	cfa
6	Production worker	60,000	cfa
7	Production worker	60,000	cfa
8	Production worker	60,000	cfa
9	Production worker	60,000	cfa
10	Janitor	60,000	cfa
11	Janitor	60,000	cfa
12	Janitor	100,000	cfa
	Total	1,410,000	cfa

Bio Gas production technical doc⁸

Bio-digester

A bio-digester is a structure that facilitates the decomposition of organic materials such as manure, one example being pig waste to produce methane gas (biogas) that can be used for cooking, lighting and as fuel for the operation of generators. While a bio-digester is normally made of concrete or metal, this manual highlights the low-cost polyethylene plastic bio-digester (Maximiliano, 2009).

Biogas

Biogas is a form of energy produced by anaerobic digestion (the decomposition of constituents of biodegradable matter in an oxygen-free environment). It is a mixture of gases mainly carbon dioxide and methane. Approximate biogas composition:

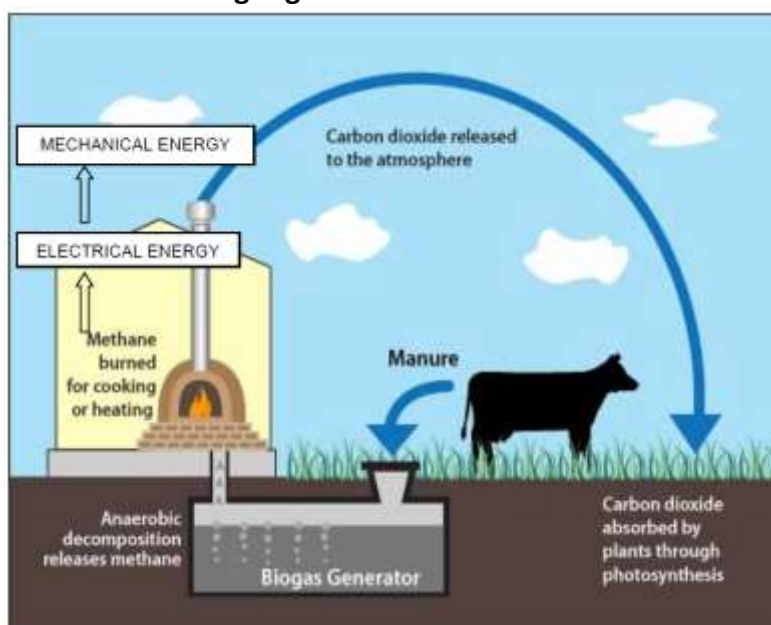
- Methane (CH₄) 55 to 65%
- Carbon dioxide (CO₂) 30 to 35%
- Water vapour 1 to 5%
- Hydrogen sulphide (H₂S) 0 to 3%
- Hydrogen (H₂) 0 to 1%

The decomposition of waste material during anaerobic digestion is caused by bacterial action rather than high temperatures. It takes place in almost any biological environment, but is favoured by warm, wet and low oxygen conditions. Anaerobic digestion also occurs in two major situations created by human activities:

- Sewage (human waste) or animal manure.
- Landfill gas produced by domestic refuse buried at landfill sites.

In nature, there are also many raw materials (organic materials) from which biogas can be extracted: human and animal manure, leaves, twigs, grasses, garbage, agricultural and industrial wastes with organic content greater than 2%. The biogas produced can be used for cooking, heating, lighting (using gas lamps), electricity generation, operation of farm machinery and other energy needs. The effluent produced from the process can be used as a fertilizer for crops.

Biogas generation from wastes



⁸ Guyana Energy Agency, bio-digester information and construction manual for small farmers

Potential feedstock for bio-digesters

The raw materials that can be considered as suitable substances for the production of biogas through the bioconversion process in a digester are:

- Crop Residue - Sugar cane trash, weed, corn and crop stubble, straw, spoiled fodder, etc. Studies have shown that the content of water soluble substances such as sugars, amino acids, proteins and mineral constituents decreases with age of the plant and it is low enough to limit the rate of the digestion process. Thus, the decomposition of crop residues takes a longer time than manures due to their fibrous content and larger particle sizes.
- By-products and waste from Agriculture-based industries and aquatic growth – Oil cakes, bagasse, rice bran, seeds, water from fruit and vegetable processing, filter press mud from sugar factories, marine algae, seaweed, etc., can also be used as feedstock for biodigesters.

Requirements for a successful system

- 1. Acceptance by potential users.
- 2. Ability to use the gas when produced.
- 3. Sufficient demand for gas.
- 4. Availability of sufficient raw materials to meet the production requirements.
- 5. Adequate maintenance and operational control.

Advantages of bio-digesters

- Reduces organic content of waste materials by 30-50% and produces a stabilized liquid effluent.
- Provides a sanitary way for disposal of human and animal wastes.
- Produces large amounts of methane gas which can be stored at ambient temperature.
- Produces free flowing, almost clear liquid odourless effluent. The effluent is a good liquid fertilizer.
- Weed seeds are destroyed and pathogens are either destroyed or greatly reduced in number.
- Rodents and flies are not attracted to the end product of the process. Access of pests and vermin to wastes is limited.
- Helps to conserve on imported energy sources.

Disadvantages of bio-digesters

- Liquid sludge presents a potential water pollution problem if handled incorrectly.
- Proper operating conditions must be maintained in the digester for maximum gas production.

Syngas – alternative source of energy for Shea Butter production⁹

Syngas is an important intermediate product in chemical industry. Annually, a total of about 6 EJ of syngas is produced worldwide, corresponding to almost 2% of the present total worldwide primary energy consumption. The world market for syngas (mainly from fossil energy sources like coal, natural gas and oil/residues) is dominated by the ammonia industry. Other main applications are the production of hydrogen for use in refineries, e.g. hydrogenation steps, and for the production of methanol.

Biomass is heading for a great future as renewable energy source. It not only is available in large quantities, it also is the only renewable energy source that is suitable for the sustainable production of (generally carbon containing) transportation fuels and chemicals. Therefore, the application of biomass as feedstock for the production of fuels and chemicals allows the reduction of fossil fuel consumption and the accompanying CO₂-emissions.

Implementation of biosyngas

Several large biosyngas plants are required to meet the ambitious renewable synthesis gas energy targets [4]. This requires a robust, fuel-flexible, and high-efficient technology for optimum biomass utilisation and good availability.

Syngas production

There are two thermo-chemical ways to produce synthesis gas (H₂ and CO) from biomass: either by applying high temperatures or by using a catalyst at a much lower temperature. The first route generally requires temperatures as high as 1300°C and generally involves an entrained flow gasifier. The second route includes a fluidised bed gasifier and a downstream catalytic reformer, both operating at approximately 900°C. The two options concern two different evolution trajectories. The following sections cover the two options.

Biofuels from syngas

A cleaned and conditioned bio-syngas can be used for the synthesis of many products such as biofuels. In Figure 1.1, the most popular second generation biofuels have been mentioned: Fischer-Tropsch diesel and other oil products, methanol, dimethyl-ether (DME), mixed alcohols (MA), and even pure H₂. The production efficiency depends on the type of biofuel. Theoretically, the LHV efficiency with natural gas as feedstock is 84%, 82% and 78% for methanol, DME and Fischer-Tropsch diesel respectively. The actual LHV efficiencies are lower: 68%, 71% and 63% respectively [46]. In Japan, JFE reached 69% DME production efficiency from natural gas in the 100 ton/day (~20 MWth) slurry phase DME reactor demonstration plant in Kushiro.

In principle, biofuels production is not different from processes where fossil syngas is used. There are however a few biomass-related items that might change the concept. One of those already mentioned is the scale of operation. Whereas fossil based syngas plants typically are huge, some believe that biomass plants may not be larger than several 100 MWth, the size of a typical paper and pulp mill. The limiting factor could be the availability of biomass feedstock at a negotiated price. Downscaling the size might lead to different biofuel synthesis concepts.

Another difference between biofuels synthesis and the conventional processes for the production of products and chemicals from syngas relates to process integration. The option to combine chemical recovery and biofuel synthesis in a paper and pulp mill by the gasification of black liquor. Another option that might change the concept is the possibility to recycle by-products. This is mentioned by

⁹ ECN-C--06-001, SYNTHESIS GAS FROM BIOMASS for fuels and chemicals - A. van der Drift H. Boerrigter

Choren, where naphtha produced in the Fischer-Tropsch synthesis is recycled to the gasifier to increase the efficiency of the diesel production.

A completely different way of increasing the renewable character of transportation fuels is to add H₂ from biomass in a refinery as green substitute for H₂ made by reforming natural gas or residues. Although this option is limited in volume, it is relatively cheap because it does not involve biofuel synthesis and upgrading. Instead, clean syngas will suffice. Legislation and calculation methods however need to be adapted to this alternative “biofuel” option.

Terms of reference

1. Background

Shea processing is a major industry in Burkina Faso, and conducted almost exclusively by women's groups; however, it still involves outdated methods: potentially degrading habitats including Shea trees, emitting significant greenhouse gas emissions, causing a number of health and hygiene risks, and not realising its potential to support social and -economic development.

A consortium including an international project developer, a renowned local Shea processing training company, and an engineering firm have established **Serious Shea**, which seeks to improve the Shea butter-making process in Burkina Faso.

The project has received the firm support of the Ministry of Environment in Burkina Faso, and grant funding from the African Development Bank and World Bank to conduct the preparation activities, and will ensure women's groups can produce the Shea butter more efficiently and sustainably, and also achieve significantly better access-to-market (receiving 2x - 5x more profit per tonne). The project will use clean technology at each processing centre and drive reforestation / afforestation at the collection sites, generating certified carbon credits by decarbonising the process and REDD+.

1.1. Design

Serious Shea has been established to enable the sustainable sourcing, processing and export of Shea butter, displacing traditional methods, which drive significant deforestation and greenhouse gas emissions, and offer no socio-economic improvement to the women's groups.

The foundation of the agribusiness is the implementation of clean and efficient processing centres, known as *Eco-Processing Centres* where the training and technology is provided to the women's groups to collect the Shea nuts sustainably, and process Shea butter cleanly and efficiently.

The project will transform the Shea sector in Burkina Faso and enable the production and export of high volume and high-quality sustainable Shea butter, where the product is facing rapidly increasing demand from cosmetic and food (e.g. chocolate) markets. Importantly, local women's groups are core partners in Serious Shea, and will achieve significantly better access-to-market and 2x-5x higher income.

1.2. Preparation Activities

Serious Shea has secured grant support from the AFT Fund (Agricultural Fast-Track Fund), established by the AfDB (African Development Bank), which will fund key preparation activities for the agribusiness. Serious Shea therefore has grant funding support to complete preparatory planning and assessment, including business planning, forestry mapping, site feasibility assessments, and the technology designs. The key objective in completing this work is to assist Serious Shea in raising funds for implementation.

More specifically, the preparation work includes the following:

Component 1: A Validated and Bankable Business Plan

- Validate key financial assumptions, including capital costs, production volumes, operational costs, etc.
- Engage potential off-takers to estimate purchasing requirements, prices, etc. for financial model
- Develop and finalize financial model, including cashflow, P&L, balance sheet and potential optimal funding structure
- Finalize draft business plan, outlining core strategy, key activities, logistics, team, risks management, financial projections, investment requirements, etc.

Component 2: Site Feasibility and Mapping Assessments

- Identify preferred 60 collection sites and location of 10 processing centres based on assessment of the concentration of women's groups, Shea production areas, REDD+ regions, existing transportation networks, processing volumes, social assessment of communities and women's groups surrounding each collecting site
- Complete optimal design for logistics of Serious Shea's supply chain, which includes the process of collecting the fruit/nuts, storing the fruit/nuts, legal agreements with suppliers, payment processes, processed butter logistics, and export logistics

Component 3: Tree Mapping and Forestry Assessments

- Identify and classify the best areas for collection and processing of Shea fruits
- Characterize each collecting area based on habitat and potential harvest. This objective is to develop a baseline of the existing Shea habitat and associated drivers that are influencing production (e.g. fire, climate, agricultural practices). The objective is to also identify maximum, minimum and most likely volumes of Shea nuts that could be sourced from each of the 60 collection areas
- Develop Shea management plan to be implemented at each collecting centre to ensure sustainable Shea harvesting, inform any tree or habitat improvement. This includes ways to influence mentalities/taboos about Shea trees, training on grafting/pruning/etc., collective action for best Shea management practices (collection dates, fire control, livestock encroachment, etc.), maybe the setup of women's Shea plots with access to land "ownership" through village-level conventions and endorsement from traditional village authorities, etc. This will draw from the vegetation baseline (Component 3) and social baseline (Component 2)
- Establish a Geographical Information System (GIS) as a management tool to track Shea production, harvesting and impact of activities

Component 4: Site Technology Design and Development

- Outline potential technological options for collecting, storing, processing and transporting of Shea butter on a large-scale across Burkina Faso
- Refine potential solutions with key stakeholders (including Serious Shea and the women's groups), rank/prioritize preferred solutions and agree preferred technical design
- Prepare detailed final designs, costings, and implementation plan for 60 collection sites, storage facility, transport logistics and 10 processing centres
- This document refers exclusively to Component 4: Site Technical Development and Planning, which is covered in more detail in the following sections.

Goals & Objectives of Serious Shea Site Feasibility Assessments

Component 4 of the AFT grant focuses on site technology development and planning. The overarching goals are to:

- Outline potential technological options for collecting, storing, processing and transporting of Shea butter on a large-scale across Burkina Faso
- Refine potential solutions with key stakeholders (including Serious Shea and the women's groups), rank/prioritize preferred solutions and agree preferred technical design
- Prepare detailed final designs, costings, and implementation plan for 60 collection sites, storage facility, transport logistics and 10 processing centres

Scope of Services, Tasks, and Expected Deliverables

1.1. Draft Technical Designs and Alternatives

1.1.1. Activities, Components and Tasks

- Provide profiles of equipment (including capacity, suppliers, lifespan, specifications, etc.) within each processing centre including the following:

- *Tools and equipment to collect Shea fruit*
- *Solar cookers and steam boilers to dry and boil the nuts cleanly and efficiently*
- *Warehousing/storage facilities for dried nuts*
- *Solar-powered motors, which enable the women's groups to produce the Shea paste quickly without the strenuous physical labour of traditional manual methods*
- *Large-scale improved cookstoves to boil the mixture and extract the butter*
- *Equipment to assess and monitor quality of final and intermediary products*
- *All the necessary apparatuses and equipment to transport, package and store the final Shea butter product*
- These profiles should itemize the alternatives available to fulfill the task. The typical capacity of equipment, and key strengths and weaknesses of alternatives.
- Outline three to five alternative technical site layouts and preferred choice of technology for the collection sites (this may vary depending on the type of collection sites – e.g., distance, size, number of women's groups, etc.) and the processing centres, ensuring the options utilize clean technology and renewable energy, are modular and scaleable, and cover the full value chain (incl. necessary transport methods and storage facilities), and can achieve the following:
 - *Depulping*
 - *Fruit meat processing (drying/cooking)*
 - *Nut boiling, cracking and drying*
 - *Fat extraction from almonds*
 - *Fat refining and cooling*
 - *Ecological packing for long-haul road transportation*
 - *Renewable power sources*
 - *Water pumping and recycling*
 - *Waste processing and management*
- Refine initial draft designs through interviews with international experts and suppliers
- Engage with local and international suppliers to confirm technical and commercial feasibility for implementation and scale-up
- Prepare details of top two designs, including technical specification, , expected life of equipment, potential source of equipment, capacity, architectural designs, maintenance requirements, skills/training required to operate, estimated costs (capex and operating costs), interoperability with existing technologies used by women's groups, resource flows (energy, water, waste, etc.), scalability, key risks, and product flows (fruit, dried nuts, butter, etc.).

1.1.2. Proposed Method of Execution

- Organize a kick-off meeting with the Serious Shea Team in Ouagadougou
- Organize a weekly update of Serious Shea Team
- Provide a monthly report of progress
- Conduct a workshop in Ouagadougou, Burkina Faso, and present top three high-level technical design alternatives to Serious Shea. The workshop will a) identify the preferred criteria for a successful technical design, b) understand the 3 designs identified; c) consider the alignment of each design to the criteria, d) identify adjustments required to meet stakeholder needs.
- Assist Serious Shea in selecting the preferred design (giving consideration to the stakeholder feedback from the workshop) including making any adjustments.
- Draw on best practice in the vegetable fat processing industry, and prepare designs that are optimal and scalable Burkina Faso
- Ensure designs are commercially viable, and account for local conditions and local equipment supply/import solutions

1.1.3. Expected Deliverables

- Profiles of key items for equipment to be considered within technical designs

- Three high-level technical designs and specifications for the collection sites, storage facilities, transportation process and processing centres. This should include high-level details of the following:
 - *The architecture, equipment, vehicles, etc.,*
 - *Potential contractors and suppliers for the installation and operation*
 - *Maintenance and scalability considerations*
 - *Product flows (fruit, dried nuts, butter, etc.).*
 - *Associated resource flows (e.g., energy use, water consumption)*
- Prepare detailed schematics, 3D representations of the designs for collection sites and processing centres
- Prepare spreadsheets to calculate implementation costs, and reference original quotations from selected equipment suppliers and contractors
- Prepare detailed presentation of resource flows (energy, water, waste, etc.), scalability, and product flows (fruit, dried nuts, butter, etc.).
- Detail operational and maintenance requirements and associated processes
- Prepare detailed quality management processes (including assessing Shea nuts harvested, intermediary products, final products). Summarize results from initial tests with existing producers and compare with potential quality from Serious Shea.
- Prepare summary of technical project description to be used in Serious Shea business plan document (Component 1)

1.1.4. Capacity Building

- NA

1.2. Detailed Designs and Specifications

1.2.1. Activities, Components and Tasks

- Prepare validated technical specifications, architectural designs, and 3D representations, and detail the actual costs (capex and operating costs), resource flows (energy, water, waste, etc.), scalability, and product flows (fruit, dried nuts, butter, etc.) within the schematics
- Prepare detailed implementation and operational plans based on the selected technical design, including the selection of technology suppliers (provide quotes from three potential suppliers of each key item), construction partners (provide quotes from three potential construction partners), construction timelines, operational and maintenance requirements, and import requirements
- Develop quality testing approach and test on existing nuts and Shea butter collected through existing producers. Compare the anticipated results from the Serious Shea processing centres with the quality produced to date

1.2.2. Proposed Method of Execution

- Draw on best practice in the vegetable fat processing industry, and prepare validated designs that are feasible in Burkina Faso
- Ensure designs are commercially viable, and account for local conditions and local equipment supply/import solutions

1.2.3. Expected Deliverables

- Prepare detailed schematics, 3D representations of the designs for collection sites and processing centres
- Prepare spreadsheets to calculate implementation costs, and reference original quotations from selected equipment suppliers and contractors
- Prepare detailed presentation of resource flows (energy, water, waste, etc.), scalability, and product flows (fruit, dried nuts, butter, etc.).
- Detail operational and maintenance requirements and associated processes

- Prepare detailed quality management processes (including assessing Shea nuts harvested, intermediary products, final products). Summarize results from initial tests with existing producers and compare with potential quality from Serious Shea.
- Prepare summary of technical project description to be used in Serious Shea business plan document (Component 1)

1.2.3.1. Capacity Building

- Where possible, identify and engage the local resources in-country that would be suitable to i) supply and install the necessary processing equipment, and ii) provide logistical support (e.g., transporting and storing the Shea)

Team Composition and Qualification Requirements for the Key Experts

The team needs to consist of:

- Shea harvesting specialist (>5 years' experience in the Shea sector). This specialist should have experience in rural Shea fruit collection and demonstrate the requisite expertise to offer guidance on designing an optimal collection site and primary processing facility (e.g., de-husking and drying)
- Food-quality storage and logistics expert (>5 years' technical experience in food storage and logistics and a degree in engineering). This expert should have experience in designing and managing large-scale West African supply chains, particular for cold chains and perishable goods
- Shea butter processing expert (>5 years' experience in vegetable fats processing). This expert should have experience in the processing of vegetable fats. Experience of the process and equipment to produce food-quality Shea butter is preferable.
- Renewable energy engineer (>2 years' experience in renewable energy, with a degree in engineering). This expert should have experience designing small-scale renewable energy systems in rural and remote environments
- Food-quality processing engineer (>2 years' experience in renewable energy, with a degree in engineering). This expert should have a track record in designing and installing efficient large-scale food processing factories and machinery. Experience sourcing and installing equipment in West Africa (and Burkina Faso in particular) is preferable, and ideally past projects involved clean technology and renewable energy components

Reporting Requirements and Time Schedule for Deliverables

The key deliverables will need to be produced before the following deadlines:

- **Task 3.1:** High-level Designs & Alternatives (11th February 2019)
- **Task 3.2:** Workshop (11th March 2019)
- **Task 3.3:** Validated Designs & Specifications (22nd July 2019)

All results need to also include a summary format.